

Senior Fitness Test Manual

Second Edition



**Roberta E. Rikli
C. Jessie Jones**

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Roberta E. Rikli, PhD

California State University, Fullerton

C. Jessie Jones, PhD

California State University, Fullerton



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Human Kinetics

Website: www.HumanKinetics.com

United States: Human Kinetics

P.O. Box 5076

Champaign, IL 61825-5076

800-747-4457

e-mail: humank@hkusa.com

Canada: Human Kinetics

475 Devonshire Road Unit 100

Windsor, ON N8Y 2L5

800-465-7301 (in Canada only)

e-mail: info@hkcanada.com

Europe: Human Kinetics

107 Bradford Road

Stanningley

Leeds LS28 6AT, United Kingdom

+44 (0) 113 255 5665

e-mail: hk@hkeurope.com

Australia: Human Kinetics

57A Price Avenue

Lower Mitcham, South Australia 5062

08 8372 0999

e-mail: info@hkaustralia.com

New Zealand: Human Kinetics

P.O. Box 80

Torrens Park, South Australia 5062

0800 222 062

e-mail: info@hknewzealand.com

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Preface

The purpose of the first edition of this manual was to introduce and describe the newly developed Senior Fitness Test, which had been designed out of a need for simple, easy-to-use tools to assess physical fitness in older adults. Although physical fitness has traditionally been associated more with younger age groups than with older people, it is now recognized as being even more crucial during the later (senior) years. Maintaining adequate strength, endurance, flexibility, agility, and balance is critical whether our later-life interests are playing golf, climbing mountains, or performing simple everyday tasks such as climbing stairs or getting out of a chair or bathtub without assistance. In fact, studies suggest that much of the physical frailty commonly associated with aging could be reduced if we paid more attention to our physical activity level and fitness as we age, especially if evolving weaknesses could be detected and treated early on.

Our goal in developing the Senior Fitness Test (SFT) was to come up with a battery of test items that covered the major components of fitness for older adults (lower- and upper-body strength, aerobic endurance, lower- and upper-body flexibility, agility, and balance) and was capable of measuring older people across a wide range of age and ability levels. The SFT described in this manual provides a simple, economical method for assessing mobility-related fitness parameters in older adults aged 60 to 90-plus. Specifically, the test measures the physical attributes (strength, endurance, flexibility, agility, and balance) needed to perform everyday activities in later life. In addition to being easy to administer and score, the test is safe and enjoyable for older adults, meets scientific standards for reliability and validity, and has accompanying performance standards based on more than 7,000 men and women aged 60 to 94.

What is new in this second edition? Whereas the first edition of the SFT manual included normative percentile tables that made it possible to compare individual scores on each test item with those of other people of the same age and sex, what was not included was information about the clinical relevance, or meaningfulness, of the test scores with respect to the level of fitness needed to support functional mobility and physical

independence. As the SFT has become increasingly popular throughout the United States and a number of other countries, we have received numerous inquiries about the clinical importance of the test scores. Program leaders, clinicians, and older adults themselves are curious about the level of performance (test scores) needed at various ages to be confident of having a sufficient level of fitness to remain physically independent until late in life.

Therefore, important additions to this second edition of the SFT manual are criterion-referenced, clinically relevant fitness standards that indicate the level of fitness needed on a given attribute (such as lower-body strength) in order to maintain functional mobility and physical independence until late in life. Standards of this type that provide a specific criterion, or cut-point score, associated with meeting a specific goal, such as having the physical capacity for independent functioning, are called criterion-referenced standards because they are referenced to a particular goal. Chapters 1 and 3 have been revised to include information about the purpose and process of developing criterion-referenced standards. Chapter 5 has been updated to include a section on interpreting criterion-referenced scores and setting fitness goals for older adults. The performance charts in chapter 5 (figures 5.2 and 5.3) have been modified to reflect the new criterion standards for each age group.

Also in response to requests from users of the SFT, chapter 4 contains expanded information and examples on how to modify test protocols for special populations. The SFT has been used to assess physical capacity in many different special populations, with questions raised about how to best modify the protocols for participants who are not able to do the tests according to published instructions. The SFT has been used, for example, in populations with osteoporosis, obesity, Alzheimer's disease, diabetes, fibromyalgia, heart conditions, multiple sclerosis, hip and knee replacements, chronic renal failure, COPD, and osteoarthritis as well as with stroke patients, amputees, and blind and low-vision participants. Thus, chapter 4 contains additional examples on how test protocols might be modified for special populations.

Another change in the second edition of the manual is the addition of a new chapter, chapter 6, on exercises recommended for older adults. Chapter 5 of the first edition includes a limited amount of information on

recommended exercises; however, at the request of users, this information has been expanded and is presented in a separate chapter. Also included are updated references and statistics throughout much of the book as well as referral information on national physical guidelines that have been recently developed in the United States, in Canada, in the United Kingdom, and by the World Health Organization.

The SFT has proven to be a valuable resource for health and fitness professionals looking to obtain information about the physical status of older adults, either for research purposes or for practical application. The test's minimal requirements with respect to equipment, space, and technical expertise make it feasible for use in common clinical and community settings as well as in the home environment. Also, the test is safe to administer to most older adults without physician approval. The test's multiple uses include providing research data for studies on aging and exercise, assisting health practitioners in identifying weaknesses that may be the cause of mobility problems, and helping fitness and rehabilitation specialists plan appropriate exercise programs for their clients and evaluate their progress over time. The SFT is also suitable for self-administration with the aid of a partner.

The SFT battery has been widely used and referenced¹ in the United States and many other countries, with materials translated into several languages. In fact, as of this printing, the entire manual has been reproduced and published in Danish, Korean, and Portuguese. Clearly, the SFT appears to address an important need in the health and fitness fields—that of providing a valid, reliable, and user-friendly means of evaluating physical capacity in the growing population of older adults. Physical activity and exercise programs are becoming more popular in senior centers, health and wellness clinics, and retirement living complexes throughout the world, and program leaders understand the importance of accountability and of evaluating the progress of their clients and the effectiveness of their programs.

The content of the various chapters in this test manual is discussed in the section How to Use This Manual. The Senior Fitness Test Software 2.0, now web based, can be accessed at <http://sft.humankinetics.com>. If you do not already have a key code to access the software, you can visit this site and click on the link to purchase a subscription. The Senior Fitness

Test Software 2.0 can be used to enter and analyze test scores, to provide personalized reports, and to group statistics showing program outcomes. Reference elements throughout the text refer you to features of the software. The software has been fully upgraded and greatly improved since the first edition; a new mobile version allows users to input test results directly from their smart phones. Data stored in the software are fully downloadable into Microsoft Excel. Also, the personalized reports are much more streamlined and efficient than in the first version and include attractive, user-friendly graphs for helping to interpret performance.

¹The Senior Fitness Test was called the Fullerton Functional Fitness Test when it was originally published with its supporting reliability and validity documentation and normative standards (Rikli & Jones, 1999a, 1999b). As a result, it is sometimes referred to as such in earlier publications.

How to Use This Manual

Senior Fitness Test Manual, Second Edition, presents the information needed to understand the purpose of the Senior Fitness Test (SFT), tells how it was scientifically validated, cites references where the SFT has been used in research and in practice, and explains the procedures for administering the test and interpreting and using the test scores. For a full understanding of the test rationale and procedures, we encourage you to read all the chapters in the order presented. Realizing, however, that some users may have a greater interest in some portions of the material than others, we will summarize the main content of each of the chapters.

Chapters 1 and 2 provide the background information for the test and a brief overview of the test's content. Specifically, chapter 1 introduces the test and explains why fitness is just as important, if not more so, for older people as for younger people. The unique features of the SFT, along with suggested ways of using the test, are also discussed. At the end of chapter 1 is a brief history of how the test has been used and grown in popularity since it was first published.

Chapter 2 establishes the conceptual background for the test by explaining how it relates to traditional theories and models describing physical decline in later years. You will see that the test can be used to assess the major physiological components of functional capacity so that emerging physical weaknesses can be detected and treated before they cause serious functional limitations. Included in the chapter is a discussion of the physical parameters that are important for functional mobility (strength, endurance, flexibility, agility, balance, and body composition) and a list of the criteria used in selecting test items to assess each of these parameters. The overriding goal in developing the test was to select test protocols that meet acceptable scientific standards but at the same time are economical and easy to administer in the community (nonlaboratory) setting. At the end of chapter 2 you will find a brief overview of each of the test items in the SFT battery.

Chapter 3 contains the scientific documentation for the test's validity, reliability, percentile norms, and criterion-referenced fitness standards.

Our preestablished ground rules for including a test item as part of the SFT battery were that it had to meet the criteria for at least two of three types of validity (i.e., content, criterion, or construct) and have a test–retest reliability of .80 or greater. Also described in this chapter is the nationwide study of more than 7,000 older adults (aged 60 to 94) that provided the data for both norm-referenced and criterion-referenced performance standards. Normative standards (percentile tables) provide a basis for comparing a person’s scores with those of others of the same age and sex. Criterion-referenced standards, new to this edition, suggest the fitness (cut-point) scores needed to perform the common everyday activities required for independent living

Whereas chapters 1 through 3 provide the rationale and scientific documentation for the SFT, chapters 4 through 6 contain the essential instructions for test users—how to get ready to administer the tests, how to give the tests, how to interpret and use the test results, guidelines for helping seniors create fitness programs, and what exercises to recommend for improving SFT scores.

Included in chapter 4 is a list of procedures and issues that need to be addressed before test day, along with sample instruction sheets, forms, and equipment lists to use as you plan for the test. Also included are instructions for warming up the participants on test day and descriptions of the official testing and scoring protocols for each of the SFT items, with instructions on how to adapt the protocols for special populations. For those planning to administer the test to classes or groups, the section Guidelines for Group Testing at the end of the chapter should be especially useful. Included are suggestions for test-station setup, tips for group organization and management, and information on selecting and training volunteers to help with testing.

Chapter 5 explains how to interpret the test results and how to use the information to motivate participants to increase their activity level and improve their performance. Included are various performance tables and charts that can help people see how they scored compared with others of their same age and sex and compared with the threshold scores needed to maintain good functional mobility. Also included in chapter 5 is a discussion of ways you can use test results to help your clients set goals and plan effective programs to improve their physical condition.

Chapter 6, an added chapter at the request of SFT users, contains an expanded discussion of exercise recommendations, with a focus on improving SFT performance. Included in the chapter is a discussion of recently published physical activity and exercise guidelines, as well as recommendations for improving fitness levels through both lifestyle exercise and structured exercise programs. Also included are instructions for exercises that can be used to improve SFT scores.

The appendixes in the back of the book contain sample forms, tables, and charts. Conversion charts are also provided that can be used to convert measurements found in this book from English to metric units.

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Members of the program leader and practitioner panel were Jordan Aquino, assistant director of the Pain Management Center at CSUF; Jeana Miller, operations manager of CSUF Center for Successful Aging; Jan Montague, president of Whole Person Wellness Solutions in Cincinnati; and Karen Schlieter, assistant director of FallProof balance and mobility instructor certificate program at CSUF.

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Credits

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Table 3.3, figure 3.1, and table 5.1. Reprinted, by permission, from R.E. Rikli and C.J. Jones, 1999, “Functional fitness normative scores for community-residing adults, ages 60-94,” *Journal of Aging and Physical Activity* 7(2): 162-181.

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Table 6.1. Adapted from NSCA, 2012, *NSCA’s essentials of personal training*, 2nd ed. (Champaign: Human Kinetics), 395.

Chapter 1

Fitness Testing in Later Years

Recognizing Unique Needs of Older Adults



Although physical fitness traditionally has been thought of more as the concern of young people than of older people, this attitude has been changing rapidly. Because average life expectancy is increasing, our ability to enjoy an active and independent lifestyle well into the later years will depend to a large degree on how well we maintain our personal fitness levels. Whereas health promotion and the prevention of

lifestyle diseases (e.g., heart disease, obesity, diabetes) are the major goals of most youth fitness tests, for older adults whose chronic health status generally has already been established, the focus tends to shift from disease prevention to maintaining functional mobility—to being able to continue doing the things one wants and needs to do to stay strong, active, and independent.

The Senior Fitness Test (SFT) described in this manual is a battery of test items that measure the physical capacity of older adults to perform normal everyday activities. The test is considered a functional fitness test because of its purpose in assessing the physical characteristics needed for functional mobility in later years. Specifically, functional fitness is defined as *having the physical capacity to perform normal everyday activities safely and independently without undue fatigue*. As we age, we want to have the strength, endurance, flexibility, and mobility to remain active and independent so we can take care of our own personal and household needs; do our own shopping; and participate in active social, recreational, and sport activities, if that's our choice. The SFT is for professionals in the fields of health, fitness, and aging who need an economical, easy-to-use assessment tool for measuring the fitness of older adults in the clinical or community setting. The test assesses independently living older adults, aged 60 to 90-plus, across a wide range of ability levels, from the borderline frail to the highly fit.

Chapter 1 overviews fitness testing relative to the unique needs of older adults and introduces the SFT. Specific topics include the following:

- Importance of fitness and fitness testing in later years
- Rationale for developing the SFT
- Unique qualities of the SFT
- Uses of the SFT
- History of the SFT's use

Importance of Fitness and Fitness Testing in Later Years

Most of us would agree that quality of life in later years depends to a large degree on being able to do the things we want to do, without pain,

for as long as possible. Because we are living longer, it is becoming increasingly important to pay attention to our physical condition. Ironically, the numerous technological advances in recent years have had mixed benefits for people relative to quantity and quality of life. Whereas medical technology has contributed to a longer life expectancy, computer technology and greater automation are contributing to increasingly sedentary lifestyles and to an increased risk for chronic health and mobility problems. Statistics suggest that in the United States, the health care cost associated with technology-induced inactivity approaches \$1 trillion per year (Booth, Gordon, Carlson, & Hamilton, 2000). Very few jobs or household activities these days provide enough energy expenditure to meet people's physical activity needs. Pushing a button to open the garage door, rolling a trash can out to the curb, or driving through an automated car wash, for example, contributes little to our physical strength, health, and functional mobility.

Several recent publications describing national and international physical activity guidelines (e.g., Canadian Physical Activity Guidelines, 2011; Physical Activity Guidelines for Americans, 2008; Start Active, Stay Active, published by the UK Department of Health, Physical Activity, Health Improvement and Protection, 2011; and the World Health Organization's Global Recommendations on Physical Activity for Health, 2010) provide excellent overviews of the benefits of exercise for older adults as well as the relationship between sedentary lifestyles and the onset of a number of chronic conditions that can lead to frailty and disability in later years. Unfortunately, statistics suggest that most older adults do not get the amount of exercise they need and that 42 percent of those over 65 are experiencing functional limitation in common everyday activities, statistics that have not improved over the past decade (Federal Interagency Forum on Aging-Related Statistics, 2010). As a result, although average life expectancy continues to increase, so too does the possibility of living more years with physical limitations. Many older adults, often because of their sedentary lifestyles, are functioning dangerously close to their maximum ability during normal activities of daily living. Climbing stairs or getting out of a chair, for example, often requires near-maximal efforts for older people who are not very physically active. Any further decline or small physical setback could easily cause them to move from independent to disabled status in which assistance is needed for daily activities.

The good news, though, is that much of the usual age-related decline in physical fitness is preventable and even reversible through proper attention to our physical activity and exercise levels. Especially important is the early detection of physical weaknesses and appropriate changes in physical activity habits. The SFT, therefore, was developed specifically to evaluate and monitor the physical status of older adults so that evolving weaknesses might be identified and treated before resulting in overt functional limitations.

“There is strong scientific evidence that regular physical activity produces major and extensive health benefits in older adults. . . . In addition, physical activity is associated with higher levels of functional health, a lower risk of falling, and better cognitive function.”

World Health Organization, 2010

Rationale for Developing the SFT

With the rapid growth of the older population, finding ways to extend active life expectancy and reduce disability has become the goal of government agencies, gerontology researchers, and health practitioners throughout the world. Physical frailty in later years is costly both in terms of the resources spent on medical care and the diminished quality of life for these people.

Figures show that it costs the United States \$54 billion a year to care for people who have lost their independence, and these figures are expected to increase drastically as the size of the older population continues to grow unless the proportion of those with disabilities is reduced (National Institutes of Health, 2012). The annual health care cost per person jumps drastically as an older adult progresses from independent to dependent status.

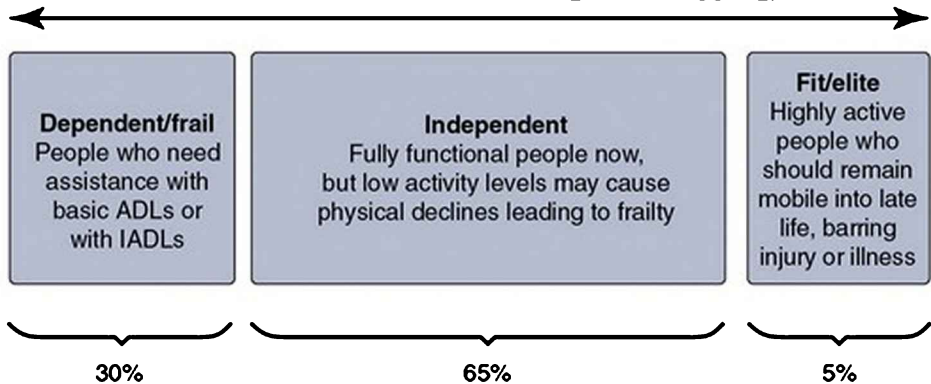
Although a number of conditions (e.g., mental confusion, visual loss) can rob people of their independence, problems with physical mobility rank at the top of the list (U.S. Department of Housing and Urban Development, 1999). Luckily, studies suggest that physical function is

preserved at a much higher level for those who are active and physically fit, with improvements possible at any age, even for many of those with chronic health conditions (American College of Sports Medicine [ACSM], 2009; Physical Activity Guidelines Advisory Committee, 2008). Research clearly shows that it is never too late to improve one's physical fitness and functional ability; even people in their 90s have experienced dramatic benefits from beginning a physical exercise program (Fiatarone Singh, 2002).

In earlier years, because of the lack of available fitness tests for older adults, health professionals were limited in their ability to evaluate clients and make recommendations based on objective data. Instead, program leaders generally had to rely on their own subjective judgment in evaluating older people's physical condition and in planning exercise programs. The SFT was developed to address the need for improved ways of assessing fitness in older adults. Specifically, it was designed to assess the physical capacity of the large proportion of older adults who are still living independently within the community, but because of their declining fitness levels may soon be at risk for losing their functional independence.

As indicated in the physical function continuum presented in [figure 1.1](#), approximately 65 percent of the older adult population fits into this independent but generally low-fit category. At the high end of the continuum are approximately 5 percent of older adults who are considered to be at a high-fit or elite level, such as those athletic older people who continue to engage in strenuous exercise or perhaps still participate in athletic competitions. At the lower end are those people (representing about 30 percent of the older population) who already have progressed into the physically frail and dependent categories and need assistance with common activities of daily living. Although the test items are suitable for use with most frail older adults at the low end of the continuum, they were primarily designed to evaluate fitness in the larger midsection of independent-living adults so that any evolving weaknesses could be detected and treated before causing loss of function and frailty. Cost savings in health care expenses and in diminished quality of life would be substantial if we could prevent, or at least delay, older adults' progression from the independent to the frail and dependent category.

Figure 1.1 Percentage of older adults (65 years of age and up) classified at various points along a continuum of physical ability (Spirduso, Francis, & MacRae, 2005). ADLs refer to the basic activities of daily living such as eating, bathing, and dressing. IADLs are instrumental activities of daily living (i.e., activities required for independent living such as housework, stair climbing, and shopping).



Unique Qualities of the SFT

The SFT was designed to assess physical fitness in older adults across a wide range of age groups and ability levels. Although a limited number of other test batteries have been developed to assess physiological capacity in older adults, the SFT has several unique qualities that distinguish it from others, qualities that have made it popular for use throughout the United States and in many other countries.

- **The SFT is comprehensive.** The test items within the SFT reflect a cross section of the major fitness components associated with independent functioning in later years, whereas other test batteries focus only on selected aspects of fitness. For example, the Short Physical Performance Battery (SPPB) includes tests of lower-body functioning (leg strength, balance, and walking speed) but contains no measures reflecting upper-body function or flexibility (Guralnik et al., 1994, 2000). The SFT, on the other hand, includes measures of all key physiological parameters needed for performing

common everyday activities: upper- and lower-body strength, aerobic endurance, upper- and lower-body flexibility, agility, and dynamic balance. Chapter 2 includes a discussion of relevant fitness parameters for older adults.

- **The SFT provides continuous-scale measures.** Another unique and especially important feature of the SFT is that it produces continuous-scale scores on all test items across a broad range of ability levels, from the borderline frail to the highly fit. A common limitation in other tests is that some items tend to be either too easy or too difficult for a large portion of community-residing older adults, resulting in “ceiling” or “floor” effects in the measurement data. A ceiling effect occurs when a test is too easy for much of the population of interest, resulting in a large number of perfect scores. Conversely, a floor effect occurs when a test is too difficult for the population of interest, meaning that a large number of participants cannot do the test (and therefore score at the floor level). In several large-scale studies using the SPPB, certain items have been found to be too easy (side-by-side balance test) or too difficult (tandem balance and five-times chair stand test) to be good discriminators for 20 to 80 percent of the participants (Giuliani et al., 2008; Guralnik et al., 1994; Seeman et al., 1994). All testing protocols for the SFT were designed to minimize such ceiling and floor effects. The SFT uses a time-based scoring system, for example, for its walking test (i.e., how far a person can walk in 6 minutes), as opposed to a distance-based scoring system that measures how long it takes to walk a prescribed distance, such as half a mile, a quarter of a mile, or 400 meters, distances that cannot be reached by as much as 40 percent of the over-65 population (National Institutes of Health, 2012). Thus, in the SFT protocol it is possible for all participants to obtain a score regardless of how far they can walk. Scores can be obtained for frail people who can walk only a few feet in 6 minutes, as well as for highly fit older adults who can cover several hundred yards within that time. Similarly, on the SFT 30-second chair stand test, participant scores are based on how many stands they can do in 30 seconds rather than on how long it takes them to do a prescribed number of chair stands, such as five stands, as is commonly required in other test protocols. On the SFT chair stand test, where the protocol involves a prescribed number of seconds rather than a prescribed number of stands, all people

can receive a score even though the score may be zero or one in extreme cases.

- **The SFT is usable in the community setting.** Because the items in the SFT have minimal equipment and space requirements, the entire test battery can be administered in most clinical and community (nonlaboratory) settings as well as in people's homes. For the one SFT item that does require a sizable space (the 6-minute walk test), there is an alternative 2-minute step-in-place test that has minimal space requirements.
- **The SFT has accompanying percentile norms and criterion-referenced fitness standards.** Another unique feature of the SFT is that it has accompanying performance standards for use in evaluating test results, both normative standards and criterion-referenced standards. Specifically, the SFT has 5-year age-group percentile norms for men and women, aged 60 to 94, on all test items, thus making it possible for people to compare their scores to others of their same age and sex. Data for the norms were obtained in a nationwide study involving more than 7,000 older Americans from 267 test sites in 21 U.S. states. The tables and charts presented in chapter 5 indicate the percentile scores for each test item, along with scoring ranges that would be considered above normal, normal, or below normal. In addition, criterion-referenced, clinically relevant standards have been developed for the SFT, standards that indicate for each test item the recommended cut-point score associated with the fitness level needed for independent functioning well into later life. See chapter 3 for further discussion of the processes involved in establishing both the normative and the criterion-referenced standards and chapter 5 for the actual performance charts.

Uses of the SFT

The SFT is appropriate for use by health and fitness professionals looking to collect data either for research purposes or for practical application. Specific examples of how the test might be used are discussed here.

- **Conducting research.** Because the SFT has documented

reliability and validity for each test item, it can be used to provide dependable data in a variety of settings: It can provide baseline scores for longitudinal or prospective studies, posttest measures for evaluating intervention effects, and accurate measures for correlation analyses in cross-sectional studies. Also, because the test battery does not require extensive equipment, time, space, or technical expertise, it should be especially useful to researchers needing to collect physical performance data within the community (nonlaboratory) setting or even in people's homes. In addition, the continuous-scale scoring ability of the SFT can provide researchers with a richer set of data than would tests using dichotomous or categorical scoring systems, such as those requiring only yes or no responses or something similar.

- **Evaluating participants and identifying risk factors.** The SFT can tell people whether their physical capacity on each item ranks in the above-normal, normal, or below-normal categories compared with others of their same age and sex. Test results can also tell participants whether they have the recommended fitness levels for their age to remain functionally independent until late in life. Also, by taking the test on multiple occasions and tracking their scores over time, people can monitor changes in performance. They will know if they are improving or declining—and if declining, whether at a slower or faster rate than other people of their age and sex. Individual assessment can help identify specific areas of physical weakness that will need attention if declines leading to functional limitations are to be prevented or reduced.
- **Planning programs.** Group or individualized exercise programs designed to improve functional mobility should be based on as much information as possible to maximize program effectiveness and participant safety. The SFT provides information on physical strengths and weaknesses, thus providing the background information needed to develop fitness programs that target the specific needs of individuals or groups. In fact, program leaders already using the SFT tell us that the inclusion of testing has added quality and substance to their programs as well as helped recruit additional participants.

organization to allow multiple instructors to enter data and assign participants to groups and testing sessions. To learn more, visit <http://sft.humankinetics.com>.

“The SFT is an important part of our StrongWomen Program where we have trained over 2,600 community leaders in 48 states to conduct fitness programs for midlife and older women. Our leaders use the SFT for various outcomes assessment . . . plus the SFT performance standards are helpful in giving participants a sense of where they’re at, what goals to shoot for, and a sense of accomplishment when their scores improve.”

Miriam Nelson, PhD, founder and director, Strong Women Program, and professor of nutrition science, Tufts University

- **Educating and setting goals.** Providing careful interpretation of the test results to participants can help them better understand how their fitness scores relate to their functional mobility. Poor scores on the 30-second chair stand, for example, generally indicate a weakness in lower-body strength that could eventually lead to the inability to climb stairs or to get in and out of a chair, tub, or automobile. Instructors and therapists can explain to their clients that reduced strength in the lower body is associated with functional limitations such as walking and an increased risk of falling. Test results can also provide a meaningful basis for developing a person’s short- and long-term goals. An example of a short-term goal is to improve performance by 20 percent by the end of a 12-week program, or perhaps to move from the 25th percentile in one’s age group to the 50th percentile. An example of a more long-term goal is to be able to comfortably walk a full mile (1.6 km) without stopping by the following summer—a goal that could be especially relevant to someone who is planning an overseas trip and needs more confidence in his ability to keep up with the tour group. Participant goals should always play a major role in planning exercise programs for older people. For most people, setting personal goals tends to increase motivation and improve exercise compliance. See chapter 5 for additional

information on goal setting.

- **Evaluating programs.** Increasingly, exercise leaders are being asked to provide evidence of outcome measures to document a program's effectiveness. The following are examples of possible outcome objectives for a 12-week exercise class.
 1. At least 75 percent of the participants will improve their fitness and mobility.
 2. The average SFT scores of class participants will improve by 20 percent.
 3. More than 50 percent of the participants will report feeling better and having more energy for performing everyday activities.

An example of a more long-term outcome measure might be evidence showing that over a 3-year period, exercise participants either improved or maintained their SFT scores compared with nonexercisers who experienced the usual age-related declines in physical ability.

“We have been using the SFT as the basis of our Senior Fitness Health Check program offered to various older adult groups in Glasgow, including at the 8th World Congress on Active Ageing where over 400 seniors were in attendance—it is a valuable resource when tailoring an action plan of activities to address personal needs.”

Dawn Skelton, PhD, professor, ageing and health, Glasgow Caledonian University, UK

- **Motivating clients.** Most people are inherently curious about their physical abilities and how they compare with those of others of their same age and sex. The performance tables and charts in chapter 5 make it possible for participants to compare their scores with other people their age, to track their physical changes over time, and to compare their scores with the recommended standards of performance for maintaining lifelong physical independence. Other participants, especially the more competitive people, may be motivated to try to score at the top of the scale in their age groups on all test items. Again, the normative tables provide the

information needed for evaluating one's performance compared with others, while the criterion standards provide information about the level of fitness needed to do the kinds of activities associated with being physically independent.

- **Improving public relations.** The U.S. Department of Health and Human Services (2011) has set as a national goal for Healthy People 2020 that of *reducing the proportion of older adults who have moderate and severe functional limitations*. SFT results obtained at regular intervals can be used to evaluate the progress of individuals or programs toward attaining such goals, particularly relative to improving those fitness parameters that support functional ability. Such results can provide the basis for a press release focusing on a program's success in serving local community needs or in addressing a particular political agenda. Positive media coverage is not only good public relations but also very helpful in attracting resources for programs and for recruiting additional participants.

History of the SFT—A Decade of Use

Since the SFT was first described in the 2001 edition of *Senior Fitness Test Manual* (originally published as the Fullerton Functional Fitness Test, Rikli & Jones, 1999a), it has been widely used as an assessment tool in research studies and in numerous clinical and community health and fitness programs for older adults. A review of the literature indicates that SFT test items have been utilized in hundreds of studies focusing on a wide range of topics and participants. Studies utilizing the SFT have evaluated the effects of physical activity and exercise interventions on participants ranging from healthy, community-residing older adults to frail older adults living in nursing homes and long-term care facilities and to both older and younger adults with a variety of conditions including Alzheimer's, diabetes, cardiorespiratory disease, osteoporosis, arthritis, and chronic renal failure as well as cancer survivors, hip-replacement patients, and lower-limb amputees with prostheses.

The SFT is also being used as an integral part of many types of health and fitness programs for older adults, including small programs at local senior centers, state-sponsored programs such as the California Active

Aging community programs (Hooker & Cirill, 2006), and the Center for Positive Aging sponsored by the Georgia Division of Aging Services (centerforpositiveaging.org). In addition, the SFT has also been used as a measurement tool in several nationally funded projects, such as a study looking at exercise effects on colon cancer survivors, funded by the National Cancer Institute of Canada (Courneya et al., 2008); a National Institutes of Health study on the effects of diet and exercise interventions on breast and prostate cancers survivors, led by researchers at Duke University (Demark et al., 2003); and studies funded by the Robert Wood Johnson Foundation, such as the Active for Life initiative looking at various intervention effects on changes in physical function (Baruth et al., 2010).

In addition, several long-standing university-based programs focusing on the study of older adults have incorporated the SFT into their programming (such as the Canadian Centre for Activity and Aging, housed at the University of Western Ontario and the Center for Successful Aging at California State University, Fullerton). The SFT is also being used as part of several large-scale health and well-being software assessment programs, such as Interactive Health Partner (www.interactivehealthpartner.com) and as part of a number of well-known community and clinical programs, such as AREUFIT Health Services (www.areufithealthservices.com), the StrongWomen Program (www.strongwomen.com), ElderFit (www.elderfitpt.com), and the *Sit and Be Fit* award-winning television series (www.sitandbefit.org).

Summary

Personal fitness level is important at any age. As we get older, however, the focus tends to shift from health promotion and disease prevention to maintaining functional mobility and independence. Functional fitness is defined as having the physical capacity to perform normal everyday activities safely and independently without undue fatigue. The SFT provides a simple, economical method for assessing functional fitness in older adults aged 60 to 90-plus. The SFT distinguishes itself from other tests in that it

- is comprehensive,

- provides continuous-scale measures,
- is usable in both the laboratory and community setting, and
- has both normative and criterion-referenced performance standards.

Health and fitness professionals will find many uses for the tests, including the following:

- Conducting research
- Evaluating participants and identifying risk factors
- Planning programs
- Educating and setting goals
- Evaluating programs
- Motivating clients
- Improving public relations

Since the SFT was first published, it has become increasingly popular for use in many settings—research, university outreach and community-based programs, clinical rehabilitation, and various health and wellness software assessment programs. In the next chapter we describe the procedures used to identify the relevant parameters of functional fitness as well as the criteria for selecting the specific test items to assess those parameters. Also included in the chapter is a brief overview of the test items and their scoring protocols.

Chapter 2

The Senior Fitness Test

Defining Functional Fitness Parameters



For a fitness test to be appropriate for older adults, it must reflect the

major physical parameters associated with functional mobility and be safe and feasible for use in the field (nonlaboratory) setting, where most assessment of older adults takes place. To ensure that the Senior Fitness Test (SFT) met these intended criteria, the test was developed based on a thorough review of the scientific literature on functional fitness, a series of pilot studies to test the reliability and validity of the test items, and input from a panel of experts with extensive experience in studying or working with older adults.

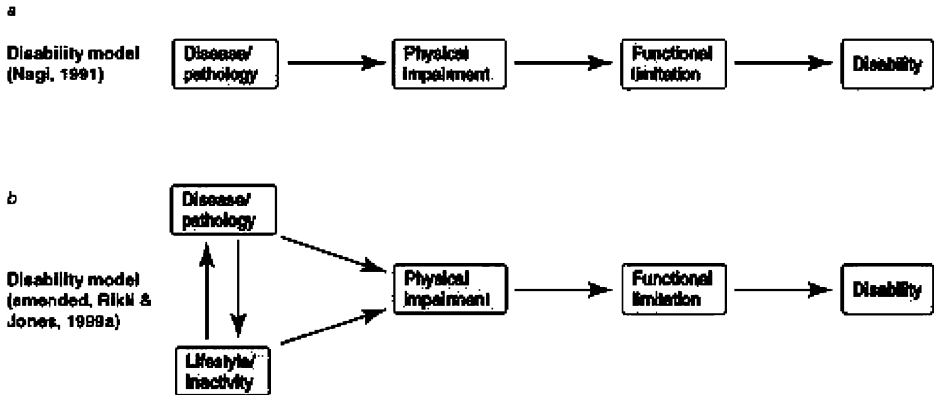
This chapter contains a brief summary of the background knowledge and rationale used in developing the SFT. See Rikli and Jones (1999a) for additional details. Included in the chapter is information on

- the conceptual background for the test development;
- a functional fitness framework illustrating the relationships among fitness parameters, functional behaviors, and activity goals;
- the criteria used in selecting test items to assess each of the fitness parameters; and
- a brief overview of the test items and their scoring protocols.

Conceptual Background

An initial step in developing the SFT was to consider the role of physical activity and fitness within the disability process. The traditional models explaining the process (Nagi, 1965, 1991) describe four main stages in the progression to disability: (1) disease or pathology, (2) physiological impairment, (3) functional limitation, and (4) disability. More specifically, the model (figure 2.1a) suggests that pathology leads to physiological impairment (a decline in body systems, such as muscular, cardiovascular, neurological); physiological impairment leads to functional limitations (restrictions in physical behaviors such as rising from a chair, lifting, or climbing stairs); and functional limitations lead to disability (the inability to perform normal daily activities such as bathing oneself, housework, or shopping).

Figure 2.1 (a) Nagi's 1991 model of the progression leading to disability; (b) an amended version suggesting that an inactive lifestyle can have comparable effects on the disabling process.



Reprinted by permission from Rikli and Jones 1999a.

Although it once was thought that all disability originated from disease or pathology, we now know that a physically inactive lifestyle can also be a primary cause of frailty in later years, especially for people living into their 80s and 90s, and that the model should be amended as shown in figure 2.1b. In fact, data suggest that all older adults, with or without chronic disease, will benefit from an increase in physical activity and experience a higher level of functional mobility in later years (ACSM, 2009; Fiatarone Singh, 2002; Paterson, Jones, & Rice, 2007).

More relevant to the importance of fitness testing, however, is evidence that physical decline, whether due to disease, pathology, or disuse, is modifiable through proper assessment and activity intervention. Research reviews show that increased physical activity, even when begun late in life, results in improved physical fitness (e.g., strength, endurance) as well as improved functional ability (e.g., walking, stair climbing) (ACSM, 2009; Fiatarone Singh, 2002; Paterson, Jones, & Rice, 2007; U.S. Department of Health and Human Services, 2008).

Understanding both the contributing causes of physical decline (e.g., disease, injury, or inactivity) and the subsequent stages leading to frailty is helpful in planning effective prevention and treatment strategies and in

designing appropriate measurement tools. The SFT was designed specifically to assess performance at the physiological stage (e.g., muscular, cardiovascular) so that performance in these critical supporting parameters might be evaluated and monitored, thus curtailing declines that might lead to additional losses, especially losses that progress to the point of affecting everyday functioning.

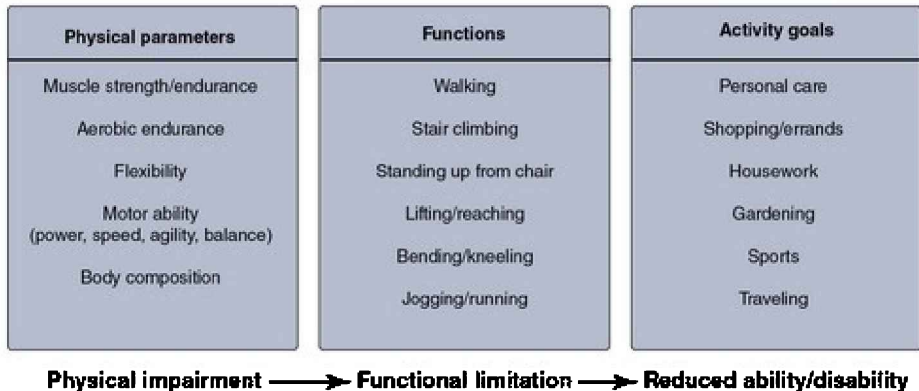
“The ability of older persons to function independently in the community is dependent largely on the maintenance of sufficient aerobic capacity and strength to perform daily activities.”

Findings from the Baltimore Longitudinal Study of Aging (Fleg et al., 2005)

Functional Fitness Parameters

The functional ability framework described in figure 2.2 may be used as a guide for understanding the relationship between the Senior Fitness Test items and functional mobility. Consistent with research on physical activity and fitness in older adults, the framework illustrates the progressive relationships among physical parameters, functional abilities, and activity goals. The common activities in the far-right column in figure 2.2 (e.g., taking care of personal needs, household chores, shopping, and traveling) require the ability to perform the functions listed in column 2 (e.g., walking, stair climbing, lifting, and reaching). These functions, in turn, require an adequate reserve in the physical parameters identified in column 1—muscular strength, aerobic endurance, flexibility, agility, and balance—as well as an optimal, or at least manageable, body mass index (relationship of height to weight).

Figure 2.2 The functional ability framework indicates the physiological parameters associated with functions required for basic and advanced everyday activities.



Reprinted by permission from Rikli and Jones 1999a.

Based on the functional fitness framework in figure 2.2 and the supporting research, the following physical parameters were identified as being the relevant components of functional fitness:

- Muscular strength (lower and upper body)
- Aerobic endurance
- Flexibility (lower and upper body)
- Agility and dynamic balance
- Body mass index

A brief discussion of these parameters and their relevance to functional mobility is presented in the following sections.

Muscular Strength

According to fitness experts, maintaining muscular strength should be a major concern of older adults. A decline in muscular strength, which averages about 15 to 20 percent per decade after the age of 50 (Vandervoort, 2002), can have devastating effects on people's ability to perform normal everyday activities. Lower-body strength is needed for

activities such as climbing stairs, walking distances, and getting out of a chair or bathtub. Upper-body strength is important for carrying groceries, lifting a suitcase, picking up a grandchild or a pet, and many other common tasks. Statistics indicate that many older people, because of their declining strength, begin losing their ability to perform these functions fairly early in the aging process. In a nationwide sample of more than 6,000 community-residing adults over 70, 26 percent could not climb even one set of stairs without stopping, 31 percent had difficulty lifting 10 pounds (5 kg; a bag of groceries), and 36 percent reported having trouble walking several blocks (Stump, Clark, Johnson, & Wolinsky, 1997). Although both lower- and upper-body strength impairments are associated with the inability to perform activities of daily living, a decline in lower-body strength is an especially powerful predictor of the onset of disability in later years (Guralnik et al., 2000).

Maintaining strength and muscle function is also important because of the role strength plays in reducing the risks for falls and fall-related injuries and because of its positive effect on a number of age-related health conditions. Muscular strength can help reduce bone loss, improve glucose utilization, maintain lean body tissue, and prevent obesity (Physical Activity Guidelines Advisory Committee, 2008; Warburton, Gledhill, & Quinney, 2001).

Although the decline in muscle mass and strength can be attributed to multiple factors such as genetics, disease, and nutrition, the most important variable related to muscle loss in older adults is physical inactivity. Fortunately, research now shows that through increased exercise it is possible for people of any age to regain some portion of their lost strength and muscle mass, and as a result, experience improved functional mobility (Fiatarone Singh, 2002; Macaluso & De Vito, 2004). Because of the significance of maintaining muscular strength during aging, its measurement (both lower and upper body) is an important aspect of fitness evaluation and program planning for older adults.

“Not being able to manage the activities of daily living is one of the most common reasons people enter nursing homes. And physical frailty is often what keeps people from these activities and robs them of their independence.”

Aerobic Endurance

An adequate level of aerobic endurance (the ability to sustain large-muscle activity over time) is necessary to perform many everyday activities such as walking, shopping, sightseeing while on vacation, and participating in recreational or sport activities. How much work our bodies can do and how much energy we feel are related to how much oxygen we can take in and use. Although it has been estimated that a $\dot{V}O_2\text{max}$ (a common measure of oxygen consumption, or aerobic capacity) of at least 15 ml/kg/min is necessary to maintain independent living status, declines associated with inactive lifestyles often progress below this point before age 80 (Paterson et al., 2007; Spirduso, Francis, & MacRae, 2005).

Although aerobic capacity tends to decline at the rate of 5 to 15 percent per decade after the age of 30, resulting in as much as a 50 percent loss by the age of 70, studies indicate that physically active people can retain a sufficient reserve of aerobic fitness to maintain functional ability throughout their later years (Jackson et al., 2009; Paterson & Warburton, 2010). Maintaining an adequate level of aerobic endurance has both a direct effect on a person's functional mobility and an indirect effect through its role in reducing the risk for such medical conditions as cardiovascular disease, diabetes, obesity, high blood pressure, and some forms of cancer (Paterson et al., 2007; Physical Activity Guidelines Advisory Committee, 2008).

Clearly, aerobic endurance is an important fitness component for older adults. And as is true of muscular strength, research also shows that increased exercise can lead to substantial improvements in aerobic endurance in older adults. In fact, data show that endurance exercise training in older adults can lead to a similar amount of improvement as that found in young adults (ACSM, 2009).

Flexibility

The importance of flexibility as a component of fitness increases with age. Loss of flexibility (i.e., loss of range of motion around a joint) impairs most functions needed for good mobility, including bending, stooping, lifting, reaching, walking, and stair climbing (Holland, Tanaka, Shigematsu, & Nakagaichi, 2002). Maintaining lower-body flexibility, especially in the hip joint and hamstrings, is also important because of its role in preventing low-back pain, musculoskeletal injury, and gait abnormalities and in reducing the risk of falling (Brown & Rose, 2005).

In the upper body (shoulder area), adequate range of motion is needed for a number of specific functions such as combing one's hair, zipping a back zipper, putting on or removing over-the-head garments, removing a wallet from a back pocket, and reaching for a seat belt. Reduced range of motion in the shoulder girdle can also result in pain and postural instability (Brown & Rose, 2005) and has been found to cause significant disability in as much as 30 percent of the healthy adult population over 65 (Chakravarty & Webley, 1993). Both lower- and upper-body flexibility, both of which decline with age but can be improved through exercise (Brown et al., 2000; Holland et al., 2002), are important aspects of functional fitness for older adults.

Agility and Dynamic Balance

Agility (ability to move the body and change direction quickly) and dynamic balance (maintaining postural stability while moving) are important for a number of common mobility tasks that require quick maneuvering such as getting on and off a bus in a timely manner; moving out of the way to avoid getting hit by a car or other object; and getting up quickly to answer a phone call, go to the bathroom, or tend to something in the kitchen. Also, adequate agility and dynamic balance are needed for safe participation in many recreational games and sports.

Although some might argue that agility and dynamic balance represent two different physical traits and should be evaluated separately, we are treating them as one composite measure since both must work together for the successful performance of many everyday activities, such as the ones just mentioned. Studies indicate that performance on combined agility and dynamic balance tasks is related to gait speed, other measures

of balance, and a composite index reflecting activities of daily living and is a predictor of recurrent falling (Judge 2003; Podsiadlo & Richardson, 1991; Rose, Jones, & Lucchese, 2002). Data from our study to establish norms for the SFT indicate that the rate of decline on the agility and dynamic balance task is similar to that of other fitness variables and that physical exercise is an important factor in maintaining good agility and balance (Rikli & Jones, 1999b).

Body Mass Index

The composition of a person's body, particularly the ratio of fat to lean tissue, can have a marked impact on health and functional mobility. People with excessive body fat relative to their muscle mass are usually unable to function (e.g., climb stairs or walk long distances) as well as people with normal ratios of fat and muscle. Starting around age 30, people on average begin gaining weight at the rate of 1 pound (.5 kg) per year until about age 50 (for men) or 60 (for women), after which there usually is a weight stabilization for a few years and then a gradual decline in weight. Unfortunately, for most people the weight decline in later years usually is due not to a loss in fat but to a loss in lean body tissue (muscle mass and bone). Although body mass index (BMI), which measures the ratio of weight to height, is not a direct reflection of body composition, it is more highly correlated with body weight than with body height and, therefore, has been used for years as a general indication of healthy weight management (ACSM, 2010).

We recommend that BMI be included as part of the SFT because of the role it plays in maintaining functional mobility. Studies indicate that people who are overweight (typically because of excess body fat) are more likely to be disabled in later years than are people with normal body mass ratings (Bouchard, Beliaeff, Dionne, & Brochu, 2007; Sternfeld, Ngo, Satariano, & Tager, 2002). Further, a combination of both excess body weight and low muscular strength puts people at an even greater risk for disability than either of these alone (Bouchard & Janssen, 2010). Researchers are also finding that people with very low weight can be at increased risk for health and mobility problems, possibly due to an associated loss in muscle mass, bone tissue, or both (Arnold, Newman, Cushman, Ding, & Kritchevsky, 2010; Losonczy et

al., 1995).

Body mass index can be determined by multiplying weight in pounds by 703 and dividing by height in inches squared [$BMI = (lb \times 703)/in.^2$] or by dividing weight in kilograms by height in meters squared ($BMI = kg/m^2$). BMI can also be estimated by referring to the [BMI conversion chart](#). Although the optimal BMI range for older adults has not been determined, values between 19 and 25 are generally considered to be in the healthy range, with BMIs higher or lower associated with increased risk for health and mobility problems in later years (ACSM, 2010; Losonczy et al., 1995; Shepard, 1997).

In summary, research suggests that the following physiological parameters are especially important in supporting functional mobility in later years: muscular strength (lower and upper body), aerobic endurance, flexibility (lower and upper body), agility and dynamic balance, and body composition. The importance of these variables relative to the physical health and mobility of older adults has also been documented in a number of major reviews and reports (ACSM, 2009; Fiatarone Singh, 2002; Morey, Pieper, & Cornoni-Huntley, 1998; Paterson & Warburton, 2010; Physical Activity Guidelines Advisory Committee, 2008).

In evaluating fitness in older adults, it is important to select test items that reflect these critical physical components related to functional mobility, as well as tests that meet other criteria such as being valid, reliable, and easy to use in a field (nonlaboratory) setting. Following is a list of the criteria that we established to serve as guidelines in selecting the test items.

Test Selection Criteria

In selecting the test items for the SFT, consideration was given to two overriding goals: (1) the development of test protocols that meet acceptable scientific standards with respect to test reliability and validity; and (2) the development of tests that would be easy to administer and feasible for use in common clinical, community, and home settings where the majority of older adult assessment is likely to take place. More

specifically, the following criteria (the same as those used in developing the SFT) can be used as a checklist in determining appropriate fitness tests for seniors. Test items should have the following characteristics:

- Represent a cross section of the major functional fitness components (i.e., key physiological parameters associated with functional mobility).
- Have acceptable test–retest reliability ($\geq .80$) (Safrit & Wood, 1995).
- Have acceptable validity, with documentation to support at least two of the following: content, criterion, and construct (discriminant) validity. For acceptable criterion validity, correlations between the test item and the criterion measure were to be at least .70, preferably above .80 (Safrit & Wood, 1995). For construct (discriminant) validity, relevant group differences were to be significant beyond the .01 level (Baumgartner, Jackson, Mahar, & Rowe, 2007). Content (logical) validity was to be established through literature review and through the subjective judgment of expert reviewers.
- Reflect usual age-related changes in physical performance.
- Be able to detect physical changes due to training or exercise.
- Be able to assess performance on a continuous scale across a wide range of functional abilities from the low fit and borderline frail to the high fit. The goal, as discussed in chapter 1, is to avoid ceiling and floor effects so that all, or most, participants could receive a score.
- Be easy to administer and score.
- Require minimum equipment and space so they can be administered in typical clinical and community settings as well as in the home.
- Be safe to perform without medical release for the majority of community-residing older adults.
- Be socially acceptable, meaningful, and motivating to older people.
- Be reasonably quick to administer, with individual testing time requiring no more than 40 minutes, and with groups of up to 24 people being able to complete the test within a 90-minute period, using six or seven trained volunteer assistants.

Ensuring that the SFT items met these criteria required extensive trial-and-error pilot testing during development, with the main focus being on adapting and refining test protocols to meet the standards for reliability and validity while at the same time being careful that equipment, space, and time requirements were reasonable. Ultimately, the following seven test items and one alternative were selected for inclusion in the SFT.

Overview of Test Items

Following is an overview of each of the test items included in the SFT. The purpose of each item is given, along with a brief description of the test protocol. Additional rationale for each test item is presented in chapter 3 as part of the discussion on test validity. Detailed descriptions of the test protocols can be found in chapter 4.

30-Second Chair Stand Test



Purpose

To assess lower-body strength needed for numerous tasks such as climbing stairs; walking; and getting out of a chair, tub, or car (increased ability in performing this exercise may reduce the chance of falling)

Description

Number of full stands from a seated position that can be completed in 30 seconds with arms folded across chest

30-Second Arm Curl Test



Purpose

To assess upper-body strength needed for performing household and other activities involving lifting and carrying things such as groceries, suitcases, and grandchildren

Description

Number of biceps curls that can be completed in 30 seconds holding a hand weight—5 pounds (2.3 kg) for women, 8 pounds (3.6 kg) for men

6-Minute Walk Test

Purpose

To assess aerobic endurance—important for walking distances, climbing stairs, shopping, sightseeing while on vacation, and so on

Description

Number of yards (or meters) that can be walked in 6 minutes around a 50-yard (45.7 m) course. See chapter 4 for a diagram of the course layout.

2-Minute Step Test



Purpose

Alternative aerobic endurance test for use when time restraints, space limitations, or weather prohibits giving the 6-minute walk test

Description

Number of full steps completed in 2 minutes, raising each knee to a point midway between the patella (kneecap) and iliac crest (top hip bone); the score is the number of times the right knee reaches the required height

Chair Sit-and-Reach Test



Purpose

To assess lower-body flexibility, which is important for good posture, normal gait patterns, and various mobility tasks such as getting in and out of a bathtub or car

Description

From a sitting position at the front of a chair, with leg extended and hands reaching toward toes, the number of inches (centimeters) (plus or minus) between the extended fingers and the tip of the toe

Back Scratch Test



Purpose

To assess upper-body (shoulder) flexibility, which is important in tasks such as combing one's hair, putting on overhead garments, and reaching for a seat belt

Description

With one hand reaching over the shoulder and one up the middle of the back, the number of inches (centimeters) between the extended middle fingers (plus or minus)

8-Foot Up-and-Go Test



Purpose

To assess the agility and dynamic balance important in tasks that require quick maneuvering such as getting off a bus in time, getting up to attend to something in the kitchen, going to the bathroom, or answering the

phone

Description

Number of seconds required to get up from a seated position, walk 8 feet (2.4 meters), turn, and return to the seated position

Height and Weight



Purpose

To assess body weight relative to body height, because of the importance of weight management for functional mobility

Description

Involves measuring height and weight, then using a conversion table to determine body mass index

Summary

The SFT was developed to reflect the critical physical parameters needed for maintaining functional ability. According to disability models, physical impairment (loss of strength, endurance, and so on) resulting from either pathology or disuse is the initial stage in a progression to disability. Physical impairment, in turn, can lead to functional limitation (restriction in physical behaviors such as rising from a chair or climbing stairs), which eventually can lead to disability (loss of ability to take care of oneself) unless there is an intervention to prevent or reduce further decline.

The following are considered to be the physiological parameters required for maintaining functional ability in older adults and preventing or delaying the progression to disability:

- Muscular strength (lower and upper body)
- Aerobic endurance
- Flexibility (lower and upper body)
- Agility and dynamic balance
- Body mass index

After we identified the general components of functional fitness, the next step was to develop testing protocols to assess each fitness parameter. To meet the goals of the SFT, it was important that the test items

- be reliable and valid;

- be sensitive enough to detect expected changes in performance due to aging or to exercise intervention;
- be able to assess a wide range of performance levels, from borderline frail to highly fit;
- be easy to administer and score and have minimal requirements with respect to equipment, time, space, and training; and
- be socially acceptable and motivating to older people.

After considerable trial-and-error pilot testing to develop protocols to meet the previously cited criteria, we selected the following test items for inclusion in the SFT battery:

- 30-second chair stand test (lower-body strength)
- 30-second arm curl test (upper-body strength)
- 6-minute walk test (aerobic endurance)
- 2-minute step test (an alternative measure of aerobic endurance)
- Chair sit-and-reach test (lower-body flexibility)
- Back scratch test (upper-body flexibility)
- 8-foot up-and-go test (agility and dynamic balance)
- Height and weight (body composition)

In the next chapter we describe the procedures followed to ensure these test items meet the standards of quality required for an effective test. Specifically, we discuss the procedures used for establishing test validity and reliability and for developing norm-referenced and criterion-referenced performance standards for the SFT.

Chapter 3

Test Validity, Reliability, Percentile Norms, and Criterion- Referenced Performance Standards

***Identifying Relevant Fitness Measures
for Older Adults***



For a test to be of value, whether for research purposes or for practical application, it must be valid and reliable (American Psychological Association [APA], 1985). A valid test is one that measures what it is intended to measure. A newly developed field test to measure biceps strength, for example, would be considered valid if it can be shown to correlate well with previously proven measures of biceps strength. A

reliable test is one that results in consistent, dependable, and repeatable test scores free of measurement error. A test is considered reliable if it produces the same scores when given on different occasions, such as from one day to the next, assuming no change in the ability level of the test takers.

According to APA guidelines, published tests should also have accompanying performance standards or scales that can help users interpret test results. Common types of performance standards are norm referenced and criterion referenced. Norm-referenced standards (norms), usually presented as percentile tables, can be used to evaluate people's performance in relation to peers of their same age and sex. Criterion-referenced standards, on the other hand, are those that reflect a level of performance that is needed to achieve a particular goal such as being healthy or being physically independent. The purpose of the criterion standards developed for the SFT is to provide information on the level of fitness needed to perform common everyday activities that are important for independent living, such as rising from a chair, climbing stairs, and being able to walk far enough to do one's own shopping and errands. To address each of these types of assessment, we have developed both normative and criterion standards for the SFT based on data collected nationwide from more than 7,000 older adults aged 60 to 90-plus.

This chapter describes the processes involved in establishing the validity and reliability of the SFT (called the Fullerton Functional Fitness Test in some earlier publications) as well as the procedures followed in developing the performance standards. The topics include the following:

- Validity
 - Types of validity evidence
 - Validity evidence for the SFT items
- Reliability
 - Procedures for estimating reliability
 - Reliability of SFT items
- Percentile norms
 - Normative study procedures
 - Study results and participant characteristics
- Criterion-referenced functional fitness standards

Functional abilities needed for physical independence
Identifying criterion fitness scores
Determining the validity and reliability of the criterion-referenced standards

Validity

The single most important characteristic of any test is its validity. Because a valid test measures what it is intended to measure, its validity must always be determined relative to its purpose. For example, if the purpose of a particular test item is to measure lower-body strength, then one way of evaluating its validity is to compare scores from that test with scores on an already proven test of lower-body strength. A high correlation between the two sets of scores would indicate good test validity relative to the criterion measure. However, if the test's purpose is also to detect change in fitness level as people age or improve after joining an exercise program, then further evidence of the test's validity would be data showing the ability of the test item to detect expected differences between age groups and differences between people who exercise and those who do not. Ideally, tests should be validated using as many sources of evidence as possible.

Types of Validity Evidence

The major sources of evidence for establishing test validity are content related, criterion related, and construct related (APA, 1985). Each of these methods was used in validating the SFT.

Content-Related Validity

Content validity (or logical validity, as it is often called when referring to physical performance measures) is the degree to which a test reflects a defined universe or domain of content, such as functional fitness in the case of the SFT. Support for content validity can come from a literature review showing the relevance of the various test components, as well as

from the subjective judgment of experts in the field. The content relevance (or logical validity) of the SFT was largely addressed in chapter 2 in the section Functional Fitness Parameters. In that section, we document the importance of various fitness parameters relative to maintaining functional mobility in later years, specifically, lower-body strength, upper-body strength, aerobic endurance, lower-body flexibility, upper-body flexibility, agility and dynamic balance, and body composition.

In addition, a 24-person expert advisory panel consisting of noted scientists in the field of gerontology and exercise science, as well as practitioners with extensive hands-on experience in working with older adults, provided further support of the content validity of the SFT. The advisory panel, described in an earlier publication (Rikli & Jones, 1999a), provided feedback and consultation during key phases of the test development.

Criterion-Related Validity

Criterion-related validity is the degree to which a test correlates with a criterion measure that is already known to be valid; this type of measure is sometimes referred to as a gold standard of performance. Criterion validity is typically estimated by calculating the correlation coefficient (Pearson's r) between scores on the test being validated and scores on the comparison measure of performance. Evidence supporting the criterion validity of the SFT items comes from a combination of previously published data on measures similar to the SFT items and from studies designed specifically to look at SFT items relative to appropriate criterion measures. A brief summary of criterion-evidenced validity for each SFT item is presented in this chapter.

Construct-Related Validity

Construct validity, or discriminant validity as it is sometimes called, is the degree to which a test measures a particular construct of interest. A construct is an attribute that exists in theory but cannot be observed

directly such as intelligence, personality, or in this case, functional fitness. The process of establishing construct validity typically begins with the development of a conceptual framework that explains the meaning and relevance of the construct and continues until studies are able to confirm expected predictions or inferences about the construct. Support for the construct validity of the SFT items comes from material discussed in chapter 2 (in the sections Conceptual Background and Functional Fitness Parameters) and from studies showing the SFT's ability to detect expected fitness differences in older adults of different ages, health conditions, and activity levels (Rikli & Jones, 1999a). Subsequent SFT scores of active versus inactive older adults from our normative study are presented in table 3.1.

**TABLE 3.1 SFT Means and Standard Deviations (In Parentheses)
for Active Versus Inactive Older Adults**

	AGE GROUPS							
Activity level	60-64	65-69	70-74	75-79	80-84	85-89	90-94	Combined ages
Group size								
Active	n = 538	n = 986	n = 1,130	n = 847	n = 425	n = 235	n = 101	n = 4,262
Inactive	n = 239	n = 420	n = 504	n = 481	n = 299	n = 200	n = 118	n = 2,261
Chair stand (# stands)								
Active	15.6 (4.3)	14.7 (3.9)	14.0 (3.9)	13.6 (4.1)	12.3 (3.9)	11.3 (3.9)	10.5 (3.9)	13.9 (4.1)
Inactive	13.8 (3.9)	12.8 (3.6)	12.2 (3.6)	11.8 (3.7)	10.5 (4.2)	9.4 (4.0)	6.9 (4.7)	11.7 (4.1)
Arm curl (# reps)								
Active	17.6 (4.7)	16.9 (4.9)	16.0 (4.8)	15.5 (4.5)	14.5 (4.2)	13.3 (3.8)	12.2 (3.5)	16.0 (4.6)
Inactive	15.7 (4.8)	14.9 (4.5)	14.1 (4.3)	13.4 (4.3)	12.9 (4.5)	11.8 (3.9)	10.4 (3.7)	13.7 (4.5)
6-minute walk (# yds)								
Active	638 (91)	607 (102)	588 (94)	551 (106)	524 (97)	485 (108)	427 (120)	576 (110)
Inactive	595 (95)	545 (113)	512 (109)	477 (123)	417 (130)	384 (147)	305 (127)	489 (138)
2-minute step test (# steps)								
Active	100 (23)	98 (26)	92 (25)	92 (25)	85 (24)	78 (22)	78 (21)	93 (25)
Inactive	85 (23)	86 (24)	80 (25)	78 (24)	69 (23)	61 (19)	52 (21)	77 (25)

	AGE GROUPS							
Activity level	60-64	65-69	70-74	75-79	80-84	85-89	90-94	Combined ages
Chair sit-and-reach (in.)								
Active	2.1 (4.2)	1.7 (4.0)	1.1 (4.0)	0.9 (4.1)	-0.1 (4.4)*	-0.3 (4.2)*	-1.6 (3.6)*	1.1 (4.2)
Inactive	0.8 (4.2)	0.6 (4.1)	0.5 (4.0)	-0.1 (4.3)	-0.6 (4.2)	-1.0 (3.6)	-2.9 (4.6)	-0.0 (4.4)
Back scratch (in.)								
Active	-1.2 (3.9)	-1.9 (4.3)	-2.2 (4.2)	-2.7 (4.4)	-3.1 (4.7)	-3.8 (4.8)	-4.6 (4.7)*	-2.3 (4.4)
Inactive	-2.0 (4.3)	-2.6 (4.4)	-3.1 (4.5)	-4.0 (4.9)	-4.2 (4.9)	-5.2 (4.6)	-6.2 (5.4)	-3.6 (4.8)
8-foot up-and-go (sec)								
Active	4.9 (1.1)	5.2 (1.1)	5.5 (1.3)	5.8 (1.4)	6.5 (1.6)	7.2 (2.0)	7.6 (1.9)	5.7 (1.5)
Inactive	5.4 (1.4)	5.8 (1.4)	6.3 (1.9)	6.8 (1.9)	7.6 (2.6)	8.3 (3.1)	10.1 (3.6)	6.7 (2.3)

Data in this table were collected as part of the normative study described in Rikli & Jones, 1999b and reported in Rikli & Jones, 2000. Active older adults were those participating at least 3 times/week in moderate-level physical activity (defined as strenuous enough to cause a noticeable increase in heart rate, breathing, or perspiration). Inactive were those not engaged in regular physical activity. Mean ages: active = 74.8, inactive = 74.6. Gender representation: active = 68% female; 32% male; inactive = 73% female; 27% male.

*Indicates activity group means that are not significantly different ($p > .05$). All other group means are significantly different ($p < .01$).

Validity Evidence for the SFT Items

Presented here is a brief discussion of the purpose and background of each of the SFT items, followed by a summary of the evidence supporting the validity of each item. A full description of each of the SFT items is presented in chapter 4.

30-Second Chair Stand Test

The purpose of the 30-second chair stand test is to assess lower-body strength, an important aspect of fitness in older adults because of its role in common everyday activities such as stair climbing; walking; maintaining balance; and getting out of a chair, bathtub, or car. The test involves counting the number of times, within a 30-second period, that a person can come to a full stand from a seated position with arms folded across the chest.

Background

The 30-second chair stand test is a modification of other versions of chair stand tests that involve recording the time it takes to complete a specified number of stands, such as 10 stands (Csuka & McCarty, 1985) or 5 stands (Guralnik et al., 1994). The reason for changing the protocol from measuring the time it takes to complete a specific number of stands to counting the number of stands that can be completed in a specific amount of time (30 seconds) was to improve the discrimination ability of the test. In the 5- and 10-stand versions of the test, if people are unable to complete the required number of stands, and many cannot, then they are unable to receive a score. In one large-scale study, for example, involving several thousand community-residing older adults, more than 20 percent of the participants could not complete even 5 stands (Guralnik et al., 1994). In another study, 74 percent of assisted-living residents could not complete 5 stands (Giuliani et al., 2008). Using a standardized time protocol (such as 30 seconds) instead of a standardized number protocol (such as 5 or 10 stands) makes it possible for every person to receive a score, even though the score might be only 1 or even 0 for those with very little lower-body strength and 20 or more for higher-functioning older adults. The ability of a test item to discriminate (measure performance) over a wide range of ability levels is especially important when testing older adults across broad age ranges (such as from age 60 to age 90 or 100) or when assessing intervention treatments for older adults who may be very low functioning at the onset of treatment and then progress to higher levels over time.

Validity Evidence

A number of studies show that chair stand performance is a good field-test indicator of lower-body strength in older adults because of its relatively high correlation with proven laboratory measures such as knee extensor and knee flexor strength and one-repetition maximum (1RM) leg press strength (Bohannon, 2002; Csuka & McCarty, 1985; Jones, Rikli, & Beam, 1999), thus supporting the criterion validity of the test. In support of the construct (discriminant) validity, the chair stand test is

effective in detecting anticipated age-related declines (Csuka & McCarty, 1985; Rikli & Jones, 1999a, 1999b; Wiacek & Hagner, 2008); in discriminating between high-active and low-active older adults (Burger & Marincek, 2001; Miotto, Chodzko-Zajko, Reich, & Supler, 1999; Rikli & Jones, 1999a, 1999b, 2000 [scores presented in table 3.1]); and in predicting falling risk (Alexander, Schultz, & Warwick, 1991; MacRae, Lacourse, & Moldavon, 1992; Tinetti, Speechley, & Ginter, 1988).

The 30-second chair stand has been successful in detecting exercise effects in relatively healthy older adults (Cavani, Mier, Musto, & Tummers, 2002; DiBrezzo, Shadden, Raybon, & Powers, 2005; Dobek, White, & Gunter, 2007; Takeshima et al., 2007; Yan, Wilber, Aguirre, & Trejo, 2009); in more frail populations (Beck, Damkjaer, & Beyer, 2008; Hruda, Hicks, & McCartney, 2003; McMurdo & Rennie, 1993); and in populations with special conditions, including cardiovascular disease (Liu, Baiqing, & Shnider, 2010), osteoporosis (Pearson, Burkhart, Pifalo, Palaggo-Toy, & Krohn, 2005), and fibromyalgia (Jones, Rutledge, & Aquino, 2010), as well as in breast and prostate cancer survivors (Damush, Perkins, & Miller, 2005) and patients who have been diagnosed with low- to medium-grade Alzheimer's (Santana-Sosa, Barriopedro, López-Mojares, Pérez, & Lucia, 2008).

30-Second Arm Curl Test

The arm curl test is included in the SFT as a general measure of upper-body strength, which is important for performing many everyday activities such as housework and yardwork and for lifting and carrying things such as groceries, suitcases, grandchildren, and pets. The test involves counting the number of times a person can curl a hand weight, 5 pounds (2.3 kg) for women and 8 pounds (3.6 kg) for men, through the full range of motion in 30 seconds.

Background

The arm curl test in the SFT is similar to many other arm curl tests (Osness et al., 1996) with two exceptions: (1) a change in the prescribed

weight for women from a 4-pound (1.8 kg) to a 5-pound (2.3 kg) weight, and (2) a change in arm position during the curl-up phase of the movement. The 5-pound weight for women was selected for the SFT because upper-body strength in women tends to be about 60 percent of that in men (Sperling, 1980), thus improving the representation of the female–male weight ratio (5 pounds for women versus 8 pounds for men). The change in arm position from the handshake grip to a palm-up protocol during flexion in the SFT test was to more effectively engage the muscles of the upper arm and the biceps tendon relative to muscle action. In the SFT arm curl test, the participant starts the test holding the weight in the handshake position at full extension (arm down at side), then supinates during flexion so that the palm of the hand faces the biceps at full flexion, then returns back to the handshake position during the extension phase. See the arm curl test procedure in chapter 4 for complete details.

Validity Evidence

As an indication of criterion validity, arm curl performance has been positively associated ($r = .82$) with Cybex machine arm curl performance (Osness et al., 1996) and with a composite measure of upper-body strength, which included 1RM biceps, chest press, and upper-back strength scores as measured on weight resistance machines (James, 1999; Rikli & Jones, 1999a). The moderately high correlations ($r = .84$ for men and $.79$ for women) between the SFT arm curl test and the combined biceps, chest, and upper-back composite measure is of particular significance, since this provides support for the validity of the SFT arm curl as a reflection of *overall* upper-body strength.

As with the chair stand test, construct validity of the arm curl test is evidenced by its success in detecting expected age-related declines in upper-body strength (Rikli & Jones, 1999a, 1999b; Wiacek & Hagner, 2008); in discriminating between high-active and low-active older adults (Miotto et al., 1999; Rikli & Jones, 1999a, 1999b, 2000 [see table 3.1]); and in detecting changes due to exercise interventions in both healthy populations (Cavani et al., 2002; DiBrezzo et al., 2005; Dobek et al., 2007; Takeshima et al., 2007; Yan et al., 2009) and special populations

including cancer survivors (Damush et al., 2005), cardiovascular patients (Liu et al., 2010), women with osteoporosis (Pearson et al., 2005), and Alzheimer's patients with low- to moderate-level diagnoses (Santana-Sosa et al., 2008).

Grip strength, measured by a grip dynamometer, is another common field-test measure that is a reasonably valid indicator of upper-body strength in younger people. However, this is not a recommended option for many older adults because of the discomfort associated with performing this test, especially for the numerous older people with arthritis in their hands.

6-Minute Walk Test

The purpose of the 6-minute walk test is to assess aerobic endurance, defined as the capacity to perform large-muscle activity over an extended time. Aerobic endurance is needed to perform a variety of activities including stair climbing, walking, shopping, sightseeing while on vacation, and participating in active sport and recreational activities. The test involves walking continuously around a 50-yard (45.7 m) course, covering as much distance as possible in 6 minutes.

Background

The rationale for standardizing the time (6 minutes) in the SFT instead of a specified distance, such as half a mile, 400 meters, or a mile, as is common in other distance walking tests (Kline et al., 1987; Osness et al., 1996; Simonsick, Fan, & Fleg, 2006), was to improve the discrimination ability of the test. On a timed test, such as the 6-minute walk, scores can be obtained for people of all ability levels—from the borderline frail who can walk only a few feet in 6 minutes, to the highly fit who can cover several hundred yards in the time allowed. Because reports show that a considerable number of community-residing older adults (more than 20 percent of those over 65) have difficulty walking even a few blocks (Federal Interagency Forum on Aging-Related Statistics, 2010; Stump, Clark, Johnson, & Wolinsky, 1997), tests with prescribed distances (such

as half a mile or a mile) are prohibitive to many elderly people.

Validity Evidence

Past studies show that distance walking tests of various types (half mile, 400 meters, 1 mile) are reasonably good indicators of aerobic endurance in both young adults and well-functioning older adults (Bravo et al., 1994; Fenstermaker, Plowman, & Looney, 1992; Kline et al., 1987; Pettee Gabriel et al., 2010; Simonsick et al., 2006; Warren, Dotson, Nieman, & Butterworth, 1993). More specifically, the criterion validity of the 6-minute walk protocol developed for the SFT is supported by a study showing relatively high correlations for older adults ($r = .82$ for men and $.71$ for women) between 6-minute walking scores and treadmill performance using a modified Balke protocol (Rikli & Jones, 1998).

As with most other SFT measures, the 6-minute walk test has been successful in detecting expected performance differences across different age groups and in people with different levels of physical activity (Miotto et al., 1999; Peterson, Crowley, Sullivan, & Morey, 2004; Rikli & Jones, 1998, 1999a, 1999b, 2000). It also has been successful in detecting exercise intervention effects in relative healthy populations (Cavani et al., 2002; DiBrezza et al., 2005; Dobek et al., 2007, Takeshima et al., 2007) as well as in studies involving special populations such as cardiovascular patients (Wilk et al., 2005), people with diabetes (Lambers, Van Laethem, Van Acker, & Calders, 2008), women with fibromyalgia (Jones et al., 2010), and patients with total hip replacements (Hernandez & Franke, 2005; Wang, Gilbey, & Ackland, 2002).

2-Minute Step Test

The 2-minute step test is included in the SFT battery as an alternative measure of aerobic endurance when space limitations (or sometimes weather conditions) prohibit use of the 6-minute walk test. The 2-minute step test protocol involves determining the number of times in 2 minutes that a person can step in place, raising the knees to a height halfway

between the patella (kneecap) and iliac crest (front hip bone). See the 2-minute step test photo in chapter 4.

Background

The 2-minute step test might be considered a self-paced version of several previously published tests such as the Harvard Step Test (Brouha, 1943), the Ohio State Step Test (Cotten, 1971), and the Queens College Step Test (McArdle, Katch, Pechar, Jacobson, & Ruck, 1972), all of which require a specified stepping cadence. Pilot testing during the early stages of the SFT development clearly showed that many older adults cannot or will not maintain a prescribed stepping cadence, thus making these tests inappropriate for this population.

Validity Evidence

Although the validity of step tests involving a prescribed cadence and bench height has been well documented for younger participants, no studies can be found on the validity of these tests for older adults. The criterion validity of the SFT self-paced 2-minute step test has been estimated by comparing it with other published measures of aerobic endurance. Dugas (1996; also reported in Rikli & Jones, 1999a) found a moderate correlation ($r = .73$) between 2-minute step test scores and Rockport 1-mile walk scores in 24 older men and women (mean age = 69.6). In another study involving 25 older men and women (mean age = 70.8), a similar correlation ($r = .74$) was found between 2-minute step test scores and treadmill performance (Johnston, 1999; also reported in Rikli & Jones, 1999a). Treadmill performance consisted of time on the treadmill until the participant reached 85 percent of predicted maximum heart rate using a modified Balke graded exercise protocol (American College of Sports Medicine, 2010).

The 2-minute step test has been successful in detecting expected performance declines across different age groups and physical activity levels (Miotto et al., 1999; Rikli & Jones, 1999a, 1999b; Wiacek &

Hagner, 2008) and has been used to measure exercise intervention effects in various community-residing populations (Yan et al., 2009), nursing home residents (Beck et al., 2008), cancer survivors (Damush et al., 2005), coronary heart patients (Liu et al., 2010), and lower-limb amputee patients with prostheses (Burger & Marincek, 2001). The test's ability to detect significant differences in active versus inactive older adults is seen in [table 3.1](#).

It is also interesting to note that information collected on the rating of perceived exertion (RPE) during the 2-minute step test is fairly comparable to that collected from the same participants during the 6-minute walk test, suggesting that the two aerobic endurance tests are similar with respect to exercise intensity (unpublished data from the Center for Successful Aging at California State University, Fullerton). On an RPE scale of 6 to 20 (Borg, 1998), the average RPE reported during the step test was 13.9 compared with 13.6 during the 6-minute walk test. These RPE scores represent a degree of exertion that is between “somewhat hard” (13.0 on this scale) and “hard” (15.0 on this scale).

Chair Sit-and-Reach Test

The purpose of the chair sit-and-reach test is to assess lower-body flexibility, particularly hamstring flexibility, which is important for good posture and for mobility tasks such as walking, stair climbing, and getting in and out of a car or bathtub. Lower-body flexibility also helps prevent low-back pain and musculoskeletal injuries and reduces the risk of falling. The test involves sitting at the front edge of a stable chair, with one leg extended and the other foot flat on the floor. With hands on top of each other and arms outstretched, the participant reaches as far forward as possible toward the toes. The score is the number of inches (centimeters), either plus or minus, between the tips of the middle fingers and the toes.

Background

The chair sit-and-reach test in the SFT was adapted from earlier versions of floor sit-and-reach tests that have appeared in numerous test batteries including the YMCA battery (Golding, Myers, & Sinning, 1989), the Fitnessgram (Welk and Meredith, 2008), and the AAHPERD Functional Fitness Test for adults over 60 (Osness et al., 1996). Most versions of the sit-and-reach test involve sitting on the floor with both legs extended and reaching as far forward as possible toward (or past) the toes. An exception is the Fitnessgram back-saver sit-and-reach, which involves sitting on the floor but extending only one leg at a time while the other is bent.

We chose to move from a floor to a chair protocol for the SFT flexibility test because many older adults have medical conditions or functional limitations (e.g., obesity, low-back pain, lower-body weakness, hip and knee replacements, severely reduced flexibility) that make it difficult or impossible for them to get down to and up from a floor position. In our pilot studies we also found that some older people, probably due to a combination of weak abdominal muscles and tight hamstrings, could not hold a sitting position on a flat surface, particularly with both legs extended.

In the chair sit-and-reach test, the participant extends only one leg at a time while keeping the other leg bent, with the foot flat on the floor. The rationale for keeping one leg straight and one bent during the testing, as opposed to both legs straight, is based on evidence showing that the simultaneous stretching of both hamstrings causes excessive posterior disc compression, thereby increasing the risk of back injury during the testing (Cailliet, 1988). Another advantage of measuring one leg at a time is the ability to detect any asymmetry in hamstring flexibility between legs, which can cause problems with hip rotation and be a contributing cause of gait abnormalities and low-back pain (Cailliet, 1988).

Validity Evidence

Past studies indicate that sit-and-reach tests, in general, have at least moderate criterion-related validity relative to established measures of hamstring flexibility, with r values ranging from .61 to .89 (Jackson &

Baker, 1986; Jackson & Langford, 1989; Patterson, Wiksten, Ray, Flanders, & Sanphy, 1996). More specifically, the criterion validity of the SFT chair sit-and-reach test has been estimated to be .76 for men and .81 for women (Jones, Rikli, Max, & Noffal, 1998) based on comparisons with goniometer-measured hamstring flexibility, a common gold standard measure of lower-body flexibility (American Academy of Orthopaedic Surgeons, 1966). In fact, data show that scores on the chair version of the sit-and-reach test are better correlated to hamstring flexibility in older people than are scores from the floor sit-and-reach test, even for those who have no trouble getting down and up from the floor position (Jones et al., 1998).

Relative to the construct validity of the chair sit-and-reach test, data indicate it is capable of detecting differences between various age groups (Rikli & Jones, 1999a, 1999b; Wiacek & Hagner, 2008) and physical ability levels (Miotto et al., 1999; Rikli & Jones, 2000), but the test has received mixed reports about its effectiveness in detecting change after an exercise intervention. Improvements in chair sit-and-reach performance have been found in studies involving relatively healthy older adults (Cavani et al., 2002; DiBrezza et al., 2005; Yamauchi et al., 2005; Yan et al., 2009), Alzheimer's patients (Santana-Sosa et al., 2008), and women with osteoporosis (Pearson et al., 2005) but not in other studies involving community-residing older adults (Takeshima et al., 2007), cancer patients (Damush et al., 2005), or patients with heart disease (Liu et al., 2010). Further study will be needed to determine whether the lack of observed differences in exercise groups is due to the inability of the chair sit-and-reach test to detect such change or whether, in fact, it is due to the difficulty of improving flexibility during exercise interventions.

Back Scratch Test

The purpose of the back scratch test is to assess upper-body flexibility, particularly shoulder flexibility, which is important for performing common tasks such as combing one's hair, zipping a dress, putting on an over-the-head garment, or reaching for a seat belt. Reduced range of motion in the shoulder can also result in pain and increased chance of injury and disability in later years (Chakravarty & Webley, 1993; Magee,

1992). The test involves reaching one hand over the shoulder and down the back as far as possible and the other hand around the waist and up the middle of the back as far as possible, trying to bring the fingers of both hands together. The score is the number of inches (centimeters), either plus or minus, between the extended middle fingers.

Background

The SFT back scratch test is a modified version of the Apley scratch test that has been used for years by therapists and orthopedic physicians as a quick way of evaluating overall shoulder range of motion (Gross, Fetto, & Rosen, 1996; Magee, 1992; Starkey & Ryan, 1996; Woodward & Best, 2000). The Apley protocol, which involves reaching behind the head with one hand and behind the back with the other hand toward a specified anatomical point on the opposite scapula, was revised slightly to involve simply trying to touch the middle two fingers together behind the back. The change in assessment protocol was to provide a simpler and more quantifiable method of measuring shoulder range of motion in the field setting.

Validity Evidence

Although no single criterion measure exists for the Apley test, it is considered by experts in the field to be an important indicator of overall shoulder range of motion (Gross et al., 1996; Starkey & Ryan, 1996; Woodward & Best, 2000). The motion of reaching one hand behind the head and over the shoulder involves shoulder flexion, adduction, and external rotation. The behind-the-back position of the other arm involves shoulder extension, adduction, and internal rotation.

Similar versions of this test are included in other published test batteries such as the Fitnessgram (Welk & Meredith, 2008) and the Brockport Physical Fitness Test (Winnick & Short, 1999), but no studies, to our knowledge, have been published describing the criterion validity of these measures. The shoulder stretch test in the Fitnessgram involves the same

motions as the back scratch in the SFT, although the scoring systems are different. In the Fitnessgram, scores are either yes (fingers can touch behind the back) or no (fingers cannot touch), whereas the back scratch test in the SFT involves measuring the distance between the two middle fingers, thus providing a continuous-scale scoring system. The modified Apley test in the Brockport battery involves evaluating only one arm at a time as the person attempts to reach behind the head toward the opposite scapula. Data from an unpublished pilot study indicate some support for the relationship between the modified Apley scores and functional ability in a group of people with cerebral palsy (Winnick & Short, 1999).

The logical (content) validity of the Apley and related shoulder stretch tests has been fairly well established based on the extent of its use by therapists and physicians as a tool in evaluating shoulder range of motion (Gross et al., 1996; Starkey & Ryan, 1996; Woodward & Best, 2000). Support for the construct validity of the back scratch test comes from studies showing its ability to detect expected declines across age groups (60s, 70s, and 80s) as well as expected differences between participants with high and low activity levels (Rikli & Jones, 1999a, 2000; Wiacek & Hagner, 2008). As with the lower-back flexibility measure (chair sit-and-reach), the upper-body back scratch test has been successful in detecting exercise intervention effects in some studies (Dobek et al., 2007; Liu et al., 2010; Peterson et al., 2005; Santana-Sosa et al., 2008; Yan et al., 2009) but not in others (Damush et al., 2005; DiBrezza et al., 2005). Although more studies are needed to verify the criterion-related and construct-related validity of the back scratch test, the evidence supporting its logic (content validity) as an overall measure of shoulder flexibility appears strong.

8-Foot Up-and-Go Test

The purpose of the 8-foot up-and-go test is to assess agility and dynamic balance, attributes that are needed for a number of functions such as getting up and maneuvering quickly enough to get off a bus in time, attend to something in the kitchen, go to the bathroom, or answer the phone or the door in a timely manner. The test involves getting up from a seated position and walking as quickly as possible around a cone (or similar marker) that is 8 feet (2.4 m) away and returning to the seated

position.

Background

The 8-foot up-and-go test is a modified version of a previously published 3-meter timed up-and-go protocol (Podsiadlo & Richardson, 1991). The main purpose in changing the distance from 3 meters (9.8 feet) to 8 feet was to increase the feasibility of administering this test in areas with limited space, particularly in the home setting. In our pilot studies when developing the 8-foot up-and-go test, we found that although it was often difficult to find the needed space in many areas to administer the 3-meter test (which must include room for turning around the cone), we almost always could find space to set up an 8-foot version of the test. Fortunately, reliability studies have shown that little to no accuracy was lost in shortening the distance (Cavani et al., 2002; Miotto et al., 1999; Rikli & Jones, 1999a).

Validity Evidence

Although there is no one gold standard criterion measure to compare performance on a timed up-and-go test, it has been found to be significantly related to the Berg Balance Scale ($r = .81$), gait speed ($r = .61$), and the Barthel Index of ADLs ($r = .78$)—a composite measure involving activities such as getting into and out of a car, walking, and stair climbing (Podsiadlo & Richardson, 1991). Past evidence also indicates that performance on up-and-go tests can discriminate among various functional categories in older adults and is responsive to changes resulting from an increased level of physical activity (Podsiadlo & Richardson, 1991; Tinetti, Williams, & Mayewski, 1986).

Studies conducted specifically on the 8-foot version of the up-and-go test indicate it is an excellent discriminator of performance changes across age groups and that it can detect expected differences between active and inactive older adults as seen in table 3.1 and in Miotto et al. (1999). Study results also show a progressive slowing of average times on the 8-

foot up-and-go test for people in their 60s, 70s, 80s, and 90s (Miotto et al., 1999; Rikli & Jones, 1999b). More recently, the 8-foot up-and-go test has been found to be a good discriminator of fallers and nonfallers, with scores of 8.5 seconds or longer being an indicator of increased risk for falling (Rose, Jones, & Lucchese, 2002). The 8-foot up-and-go has also been used successfully in detecting exercise intervention effects on various types of populations (Beck et al., 2008; Cavani et al., 2002; DiBrezzo et al., 2005; Liu et al., 2010; Pearson et al., 2005; Santana-Sosa et al., 2008; Yan et al., 2009).

Height and Weight

This test item measures participants' body mass index (BMI) and is included in the SFT to assess body weight relative to body height, a measure that is important because of its relationship to body composition (especially the ratio of fat to lean muscle tissue). Technically, BMI is determined by dividing weight in kilograms by height in meters squared ($\text{BMI} = \text{kg}/\text{m}^2$). An alternative formula, using nonmetric units, involves multiplying weight in pounds by 703 and dividing by height in inches squared: $\text{BMI} = (\text{lb} \times 703)/\text{in.}^2$. BMI can also be estimated using a conversion chart such as the [BMI conversion chart](#).

Although body mass index is not a measure that was developed or validated for the SFT, we suggest that it be included as an indicator of functional fitness because of previous evidence showing its role in maintaining functional mobility. Studies show that people with high BMIs (or in some cases very low BMIs) are more likely to be disabled in later years than are people with normal body mass ratings (Arnold, Newman, Cushman, Ding, & Kritchevsky, 2010; Bouchard, Beliaeff, Dionne, & Brochu, 2007; Losonczy et al., 1995; Sternfeld, Ngo, Satariano, & Tager, 2002). High BMI values are also associated with numerous health problems including hypertension, heart disease, and type 2 diabetes (U.S. Department of Health and Human Services, 2008), all of which can have adverse effects on functional mobility.

Although experts have not determined the ideal BMI for older adults, partially because of the unknown changes that occur in muscle and bone

during aging, the following are suggested as general (rough) guidelines for older adults (American College of Sports Medicine, 2010; Losonczy et al., 1995; Ross & Janssen, 2007):

BMI

19-25 Healthy range

≥26 Overweight; associated with increased risk for disease and loss of mobility

≤18 Underweight; could indicate loss of muscle mass and bone tissue

In summary, based on literature review, pilot studies, and feedback from a panel of experts during the development of the SFT, we believe there is sufficiently strong evidence to support using the SFT as a valid measure of physical capacity in older adults. In addition to having acceptable validity, it is important that test items have good test–retest reliability. Procedures and documentation for estimating the test–retest reliability of the SFT items are presented in the following section.

Reliability

Reliability, like validity, is an essential characteristic of a good test. A reliable test produces scores that are relatively free of measurement error and are dependable and consistent from one trial to the next, even one day to the next, assuming there are no changes in ability level or testing conditions. Obtaining stable (reliable) measures is critical, whether tests are to be used for program evaluation, for individual assessment, or for conducting research. A recommended method of estimating the reliability of a physical performance test is to give the test on two different occasions, usually 2 to 5 days apart, and then determine the correlation between the two sets of scores. A high correlation (.80 or above) between scores on day 1 and scores on day 2 indicates the test has acceptable reliability, meaning it produces scores that are relatively consistent from one time to the next (Safrit & Wood, 1995).

Procedures for Estimating Reliability

Because the reliability of a test can vary considerably depending on how and where the test is used, it is important that reliability studies be conducted for the specific population of interest and under conditions that are similar to those where the test is most likely to be given. Therefore, rather than depending on past research to support the reliability of the SFT, we conducted our own test–retest reliability studies using group testing conditions and trained volunteer assistants. Recall that the intent of the test is that it be easy to use in the community setting, with minimal training of testing technicians.

The procedures followed in conducting the reliability have been well described and previously published (Rikli & Jones, 1999a). Briefly, participants for the reliability study (82 men and women, mean age = 71.8 years) were solicited from a nearby retirement housing complex and from enrollees in the senior exercise classes at our university. All participants were living independently, did not require a cane or other assistive device to walk, and had no medical conditions that would prohibit their participation in the testing. All tests were conducted by trained senior volunteers or by graduate student volunteers. Before all testing, participants performed 8 minutes of warm-up and stretching exercises. All tests and retests were administered 2 to 5 days apart. Because our pilot testing indicated that a practice trial was necessary in order to obtain stable scores on both the 6-minute walk test and the 2-minute step test, these items were administered on three different occasions, with the first test treated as a practice trial. The practice trial helps the participants establish an efficient pace for these tests, thus improving their scoring consistency on subsequent trials.

Reliability of the SFT Items

The test–retest reliability value for each test item was established by calculating the intraclass correlation coefficient (R) using a one-way analysis of variance (ANOVA) appropriate for establishing the reliability of a single trial (Baumgartner, Jackson, Mahar, & Rowe, 2007). The one-way ANOVA model treats all sources of measurement variation, including changes in day-to-day performance, as error (lack of reliability), thus resulting in the most accurate estimate of stability reliability. Results indicate that the test–retest correlation values for the

SFT items ranged from .80 to .98, indicating acceptable reliability for all test items. See [table 3.2](#) for specific *R* values and 95 percent confidence intervals for each test item.

Subsequent studies have found comparable reliability results on similar populations of community-residing older adults, with Miotto et al. (1999) reporting reliability (*R*) values ranging from .94 to .98, and Cavani et al. (2002) reporting reliabilities of .95 to .98. High reliability scores have also been reported for the 30-second chair stand and a modified arm curl in a study involving patients with idiopathic inflammatory myositis (Agarwal & Kiely, 2006) and for stroke survivor patients on the 30-second chair stand and 8-foot up-and-go test (Sakai, Tanaka, & Holland, 2002).

TABLE 3.2 Test—Retest Reliability (*R*) and 95% Confidence Intervals (CI) for the Senior Fitness Test

Test item	ALL PARTICIPANTS			MEN			WOMEN		
	<i>R</i>	(CI)	<i>n</i>	<i>R</i>	(CI)	<i>n</i>	<i>R</i>	(CI)	<i>n</i>
Chair stand	.89	(.79-.93)	76	.86	(.77-.90)	34	.92	(.87-.95)	42
Arm curl	.81	(.72-.88)	78	.81	(.66-.90)	36	.80	(.67-.89)	42
6-minute walk	.94	(.90-.96)	66	.97	(.92-.99)	23	.91	(.84-.95)	43
2-minute step	.90	(.84-.93)	78	.90	(.80-.95)	36	.89	(.83-.94)	42
Chair sit-and-reach	.95	(.92-.97)	76	.92	(.85-.96)	34	.96	(.93-.98)	42
Back scratch	.96	(.94-.98)	77	.96	(.93-.98)	35	.92	(.86-.96)	42
8-foot up-and-go	.95	(.92-.97)	76	.98	(.96-.99)	34	.90	(.83-.95)	42

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In addition to being valid and reliable, tests should have performance standards that can be used in evaluating test results. Standards that provide a basis for comparing a person's scores to others within a particular group are referred to as *norm-referenced standards*, or *norms*. Test norms, usually presented as percentile tables, are statistics that summarize the typical performance of a specific population of people. The next section describes the procedures followed in developing age-group percentile norms for the SFT.

Percentile Norms

Normative scores are developed by testing a large number of people in a specifically defined population and then summarizing the data using descriptive statistics. A common method of organizing normative data is through the use of percentile tables. Percentile tables indicate the percentile equivalent (rank) associated with any given raw score. Raw scores are those scores received directly from each test item, such as the number of yards covered during the 6-minute walk test or the number of repetitions completed in 30 seconds on the arm curl test. A raw score falling at the midpoint of a distribution for a specific age group is equivalent to the 50th percentile in that distribution, meaning that 50 percent of those tested scored at or below that particular score and 50 percent scored above it. Similarly, a raw score falling at the 75th

percentile indicates that 75 percent of the population scored at or below that value, with only 25 percent scoring above it. SFT percentile scores are reported separately for men and women in 5-year age groups from 60 to 94. [Percentile tables for the Senior Fitness Test can be seen here.](#) The following section describes the procedures and results of a nationwide study conducted to establish normative scores for the SFT.

Normative Study Procedures

Although the procedures involved in collecting data for the SFT norms have been fully described in a previous publication (Rikli & Jones, 1999b), we provide a summary here because it is important that users of SFT norms understand the nature of the study and the characteristics of the population from which the norms were developed. Technically, normative scores provide accurate comparison data only for participants who are similar to the normative population. Although data collection for the normative study took place in the late 1990s, there is little reason to believe that the norms are not as relevant today as they were at the time they were collected. Statistics suggest that there has been very little change in older adult characteristics in terms of their physical activity patterns, health indicators, and other physical and functional measures (Federal Interagency Forum on Aging-Related Statistics, 2010).

Data for the normative study were collected on 7,183 older Americans (5,048 women and 2,135 men), aged 60 to 94, from 267 test sites in 21 different states. A summary of participant characteristics is presented in [table 3.3](#). The older adult participants at each site were solicited through a call for volunteers made at various senior centers, retirement housing complexes, and similar community locations via posted notices and through local media announcements. To be included in the study, participants had to be residing in the community, living independently, and ambulatory without regular use of assistive devices, as well as have no medical conditions that prohibited them from participating in the tests.

TABLE 3.3 Participant Characteristics: Means and Standard Deviations (in Parentheses)

Characteristic	AGE GROUPS							
	50-59	60-64	65-69	70-74	75-79	80-84	85-89	90-94
Total # of subjects	7,183	861	1,568	1,815	1,451	784	470	234
Women	5,048	620	1,086	1,230	987	543	354	158
Men	2,135	241	482	515	464	241	116	76
Mean age (years)								
Women	73.3 (7.6)	62.1 (1.5)	67.1 (1.4)	72.0 (1.4)	76.8 (1.4)	81.4 (1.2)	86.1 (1.1)	92.0 (2.1)
Men	73.4 (7.4)	62.2 (1.4)	67.1 (1.4)	72.1 (1.5)	76.9 (1.4)	81.4 (1.1)	86.0 (1.0)	92.3 (1.9)
Height (in.)								
Women	63.2 (2.8)	64.0 (2.7)	63.7 (2.8)	63.4 (2.7)	62.9 (2.7)	62.4 (2.7)	62.2 (2.6)	61.1 (3.1)
Men	68.7 (3.1)	69.4 (2.9)	69.3 (3.2)	68.8 (3.1)	68.5 (2.9)	68.1 (3.1)	67.6 (3.0)	66.5 (3.3)
Weight (lb)								
Women	146.5 (28.4)	153.6 (32.4)	153.3 (30.5)	149.7 (27.4)	142.9 (24.6)	136.8 (23.1)	134.0 (22.0)	128.2 (23.7)
Men	178.8 (28.5)	187.6 (31.0)	187.5 (30.1)	178.8 (26.5)	176.4 (26.6)	172.6 (24.9)	161.9 (22.9)	155.5 (20.5)
Education (# of years)	14.5 (3.3)	14.8 (2.9)	14.5 (2.9)	14.4 (3.0)	14.5 (2.9)	14.4 (2.9)	14.6 (3.2)	14.9 (3.9)
Chronic conditions (#)	1.7 (1.4)	1.4 (1.3)	1.5 (1.3)	1.6 (1.3)	1.9 (1.4)	2.0 (1.4)	2.0 (1.4)	2.1 (1.8)
Medications (# of prescriptions)	1.6 (1.6)	1.1 (1.4)	1.4 (1.6)	1.5 (1.6)	1.7 (1.5)	1.8 (1.6)	1.9 (1.6)	1.7 (1.7)
Moderate exercise (%) (≥3 days/week)	65.3	69.2	70.1	69.2	63.8	58.7	54.0	46.0
Self-reported health (%)								
Very good	56.9	65.0	64.7	57.7	53.6	47.2	44.7	49.0
Good	34.1	27.7	28.4	34.8	36.6	40.8	42.3	34.7
Fair	9.0	7.3	6.9	7.5	9.8	12.0	13.0	16.4
Functional level (# per group)^a								
Advanced	3,126	523	838	857	567	220	91	30
Moderate	2,140	144	369	538	515	306	180	88
Low, at risk	1,322	119	216	252	262	206	163	104
Ethnicity (%)								
Caucasian	89.1	80.0	87.5	88.2	92.6	94.2	94.5	90.2
African American	2.5	4.7	2.3	2.1	2.0	1.6	3.0	2.8
Hispanic	4.2	8.0	5.4	5.4	2.6	0.9	0.8	2.1
Other (2.3% were Asian)	4.2	7.3	4.8	4.3	2.7	3.3	1.6	4.9

^aFunctional ratings were based on responses to a 12-item composite physical function (CPF) scale, with age-adjusted scoring. See Park and Kines (2012).

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To maximize consistency of the testing procedures, a large emphasis was placed on training test administrators and assistants. Training workshops were held for most of the lead coordinators from each of the participating states. State coordinators, in turn, held training workshops for local technicians and assistants. A testing manual and an instructional videotape, given to all testing personnel, provided an additional source of

information and training.

Data collection at the various test sites typically took place in a group setting, with up to 24 participants per group. The tests were administered circuit-style in six stations set up around the periphery of a large multipurpose room. After an 8- to 10-minute warm-up, participants were divided evenly among the six stations to begin the tests. To minimize the effects of fatigue, stations were arranged in the following order: (1) 30-second chair stand test, (2) 30-second arm curl test, (3) height and weight and 2-minute step test (if used), (4) chair sit-and-reach test, (5) back scratch test, and (6) 8-foot up-and-go test, with participants beginning at any point in the circuit. A diagram of the circuit setup for group testing can be found in chapter 4, figure 4.3. As indicated, when the 2-minute step test was used to assess aerobic endurance instead of the 6-minute walk test (usually because of lack of space or inclement weather), it was included as part of the height and weight station. If the 6-minute walk test was used as the test of aerobic endurance, it was administered as a group after all other tests had been completed.

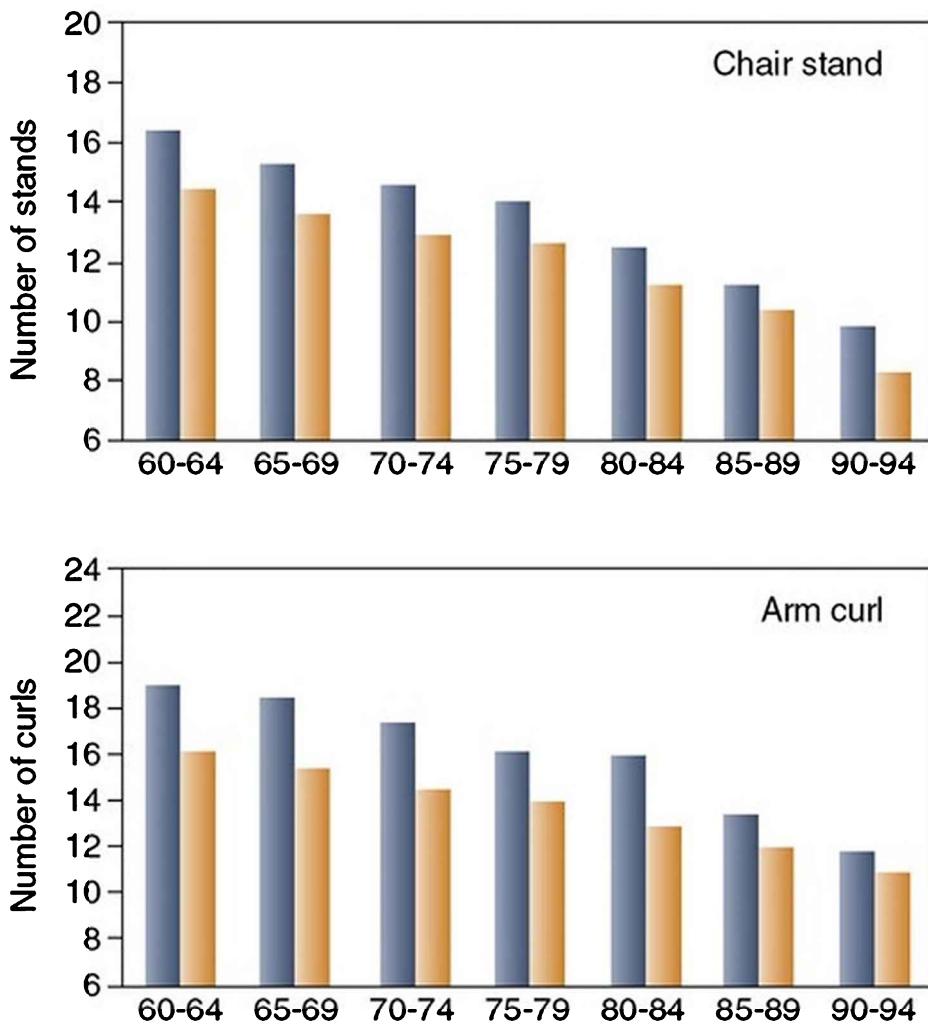
Study Results and Participant Characteristics

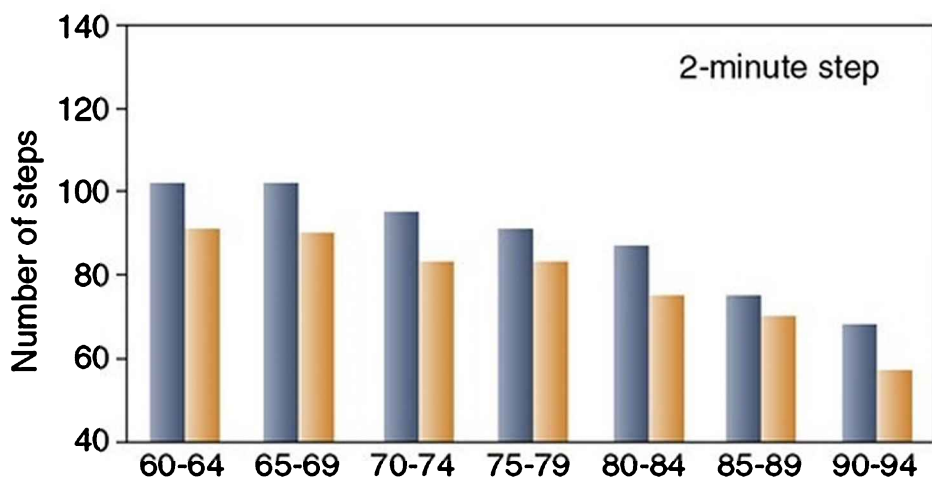
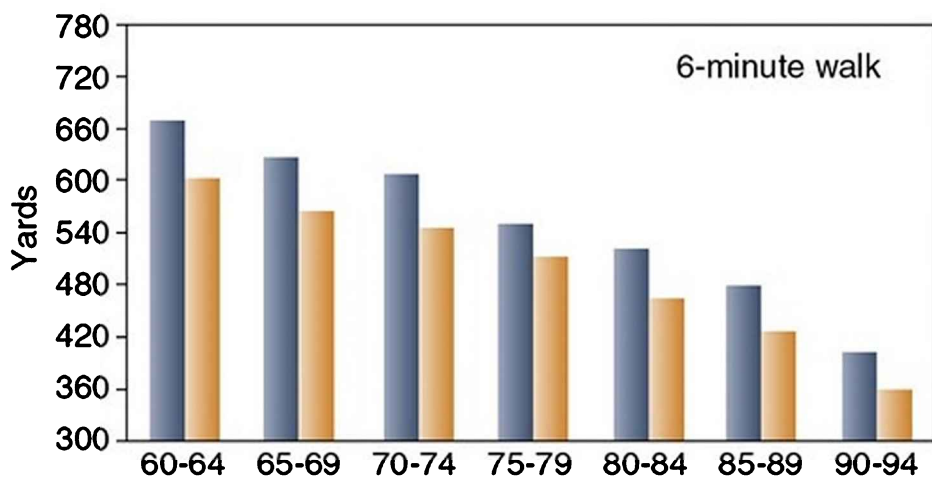
Data from the study provided percentile-based normative performance scores for men and women separately in 5-year age groups from 60 to 94. The percentile score equivalents for raw scores on all test items are presented in the normative percentile tables [here](#). A percentile rank represents that point in a distribution of scores below which that given percentage of scores falls. A raw score falling at the 25th percentile, for example, indicates that 25 percent of the scores in that particular age group would be lower than that score while 75 percent would be higher. As an alternative to the full set of percentile tables presented [here](#), chapter 5 contains tables of the normal range of scores for men and women (see [tables 5.3](#) and [5.4](#)) to provide a simpler, more user-friendly way of looking at the normative data. These abbreviated percentile tables present only the normal ranges of scores, defined as those representing the middle 50 percent for each age and gender group on each test item (i.e., scores that fall between the 25th and 75th percentiles).

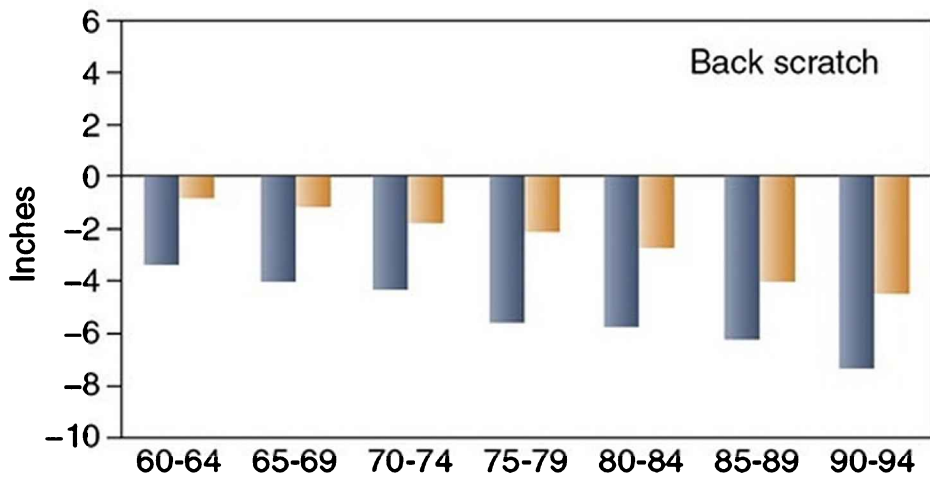
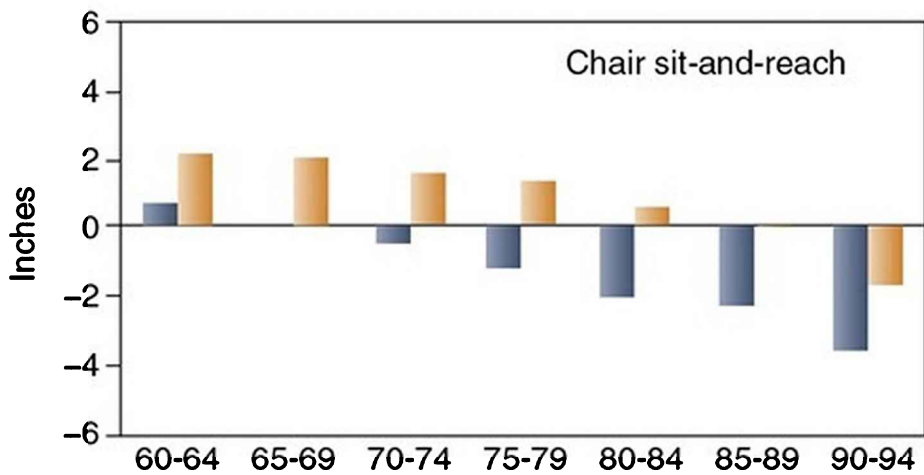
Data from the SFT normative study also provide important information

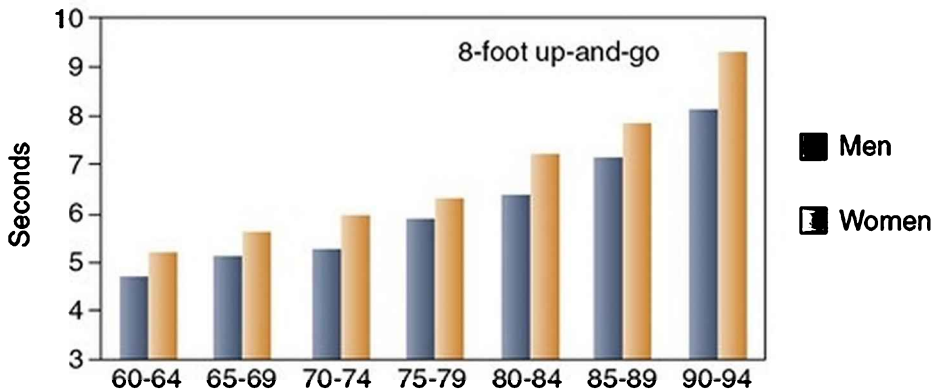
on the typical amounts and rates of declines experienced by this cross section of older adults. As illustrated in figure 3.1, the data reveal remarkably consistent patterns of declines in physical performance scores over most 5-year age groups for both men and women on all test items. Statistical analysis of these data indicate that the average decline across the majority of the adjacent 5-year age groups was statistically significant for both men and women (Rikli & Jones, 1999b). The overall amount of decline across variables from age 60 to age 94 was in the 30 to 45 percent range, resulting in an approximate 1 to 1.5 percent per year or 10 to 15 percent per decade loss in strength, endurance, agility, and balance. The exact percentage of decline was not calculated for the two flexibility tests since the measurement scales involved an arbitrarily determined zero point, thus making it inappropriate to conduct ratio types of calculations. Exact means and standard deviations for all variables can be found in Rikli and Jones, 1999b.

Figure 3.1 Fitness mean scores across 5-year age groups for men and women.







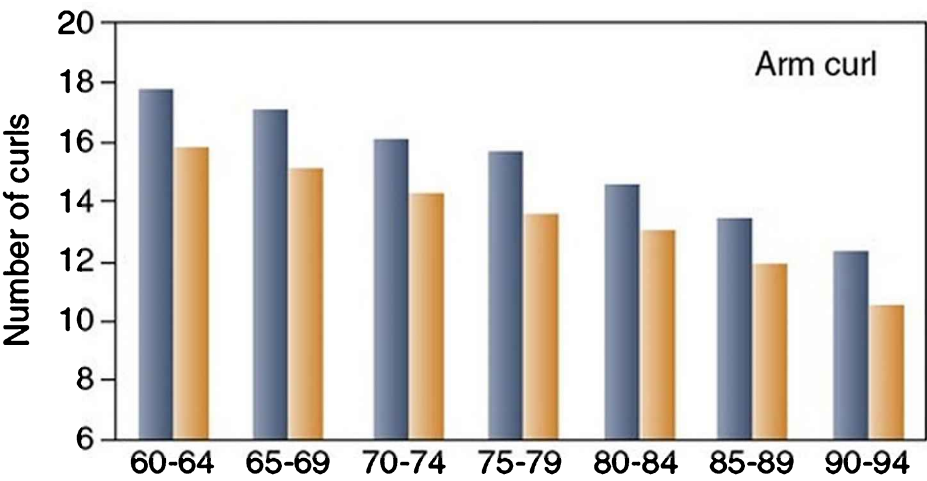
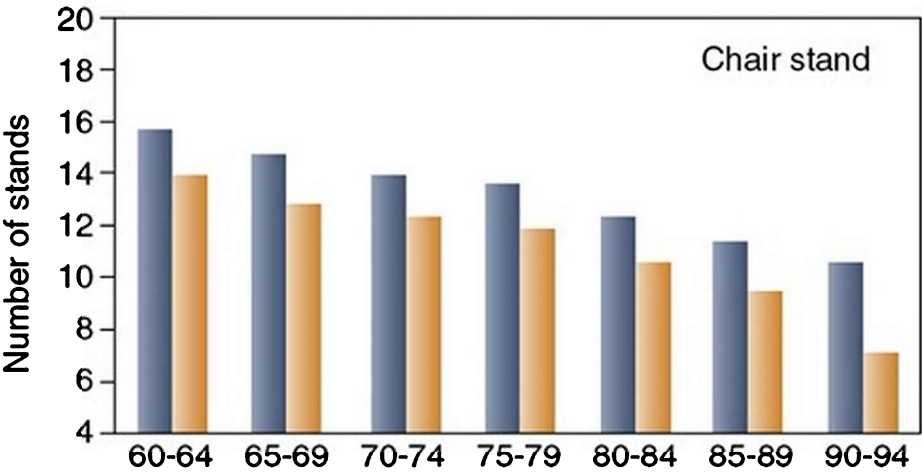


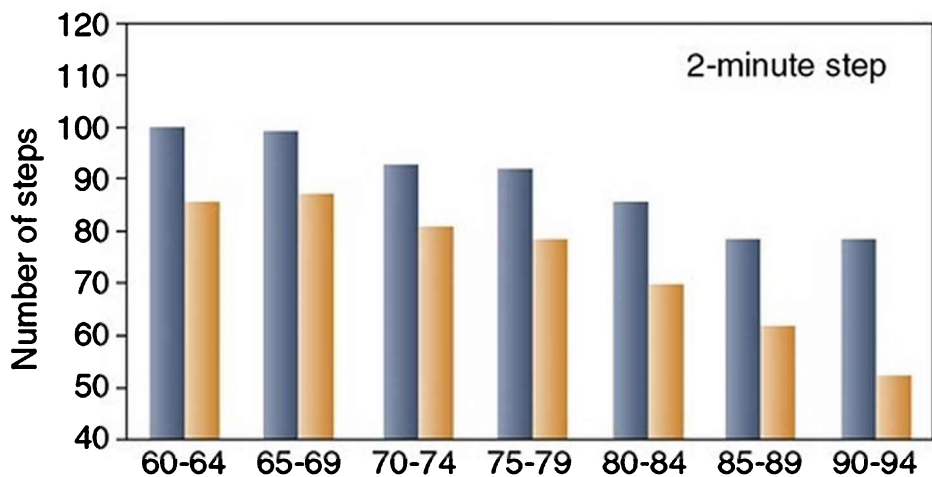
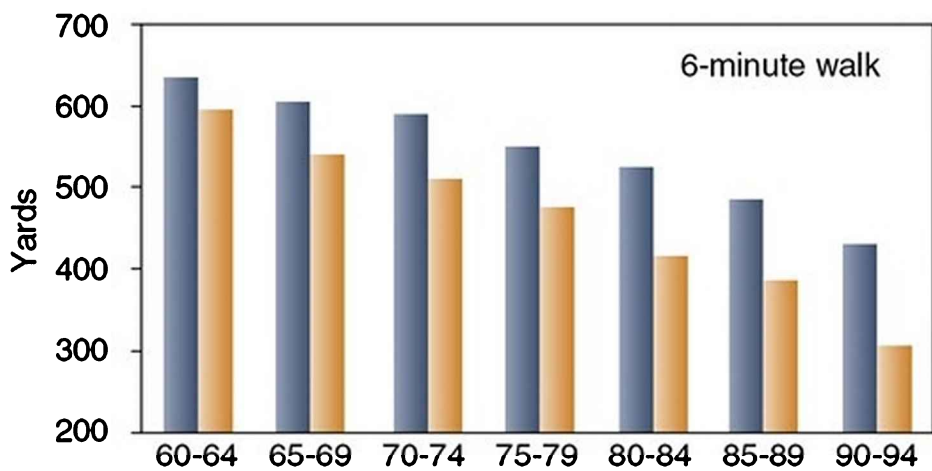
Reprinted by permission from Rikli and Jones 1999b.

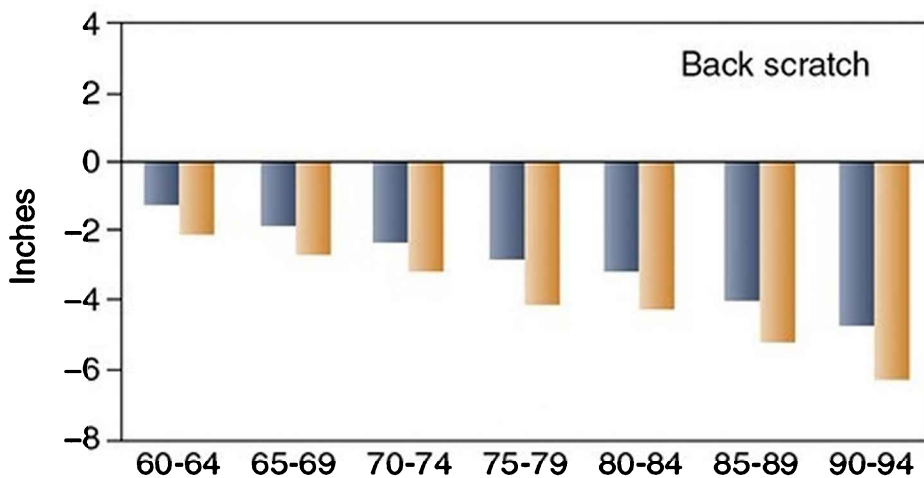
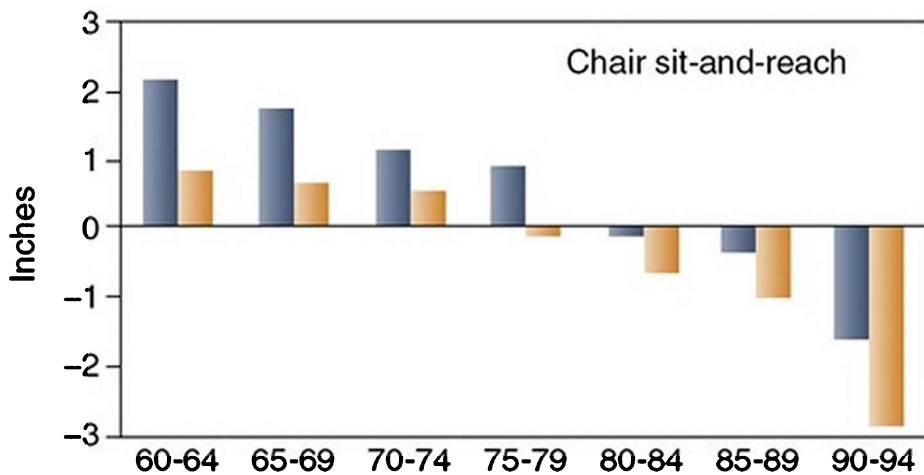
As also can be observed in figure 3.1, there are definite gender differences on the test items, with men consistently scoring better than women on the strength, endurance, agility, and balance tests and women consistently performing better than men on the two flexibility measures. These gender differences were significant across all age groups on all test items.

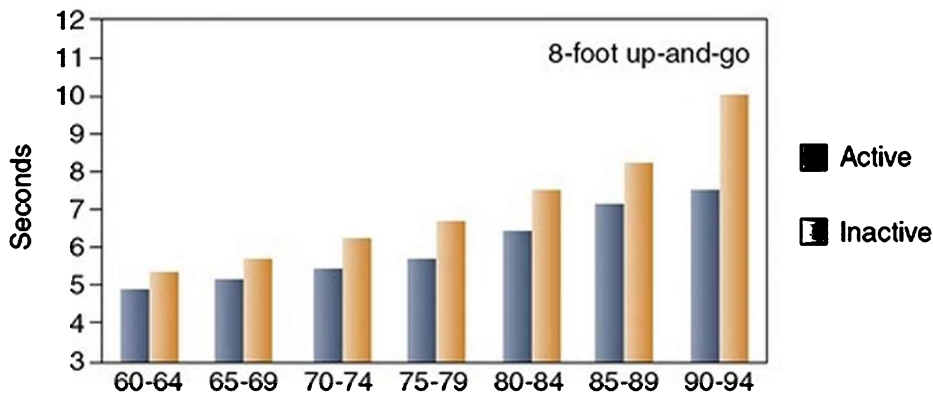
Again, it is important to remember that these normative scores represent average amounts of decline within each age group and can be quite different for different subgroups of people. As shown in figure 3.2, for example, older adults who regularly participate in physical activity (30 minutes, three times a week) have overall better scoring distributions than those who are less active, with almost all comparisons being highly significant (see table 3.1). Further, the scores of active people were not only higher to begin with, but the average rate of decline in performance between ages 60 and 94 was less for the active than for the inactive participants as well (31 percent versus 44 percent). As can be seen in figure 3.2, there tends to be a noticeable acceleration in fitness declines in the late 80s and 90s, especially for inactive older adults.

Figure 3.2 Fitness test scores across 5-year age groups for active and inactive participants.









Data suggest that for sedentary people, approximately 50 percent of the decline in performance might have been offset by being active. To illustrate, look at the chair stand test data in figure 3.2a and the exact mean scores in table 3.1. Over the three decades from 60 to 64 to 90 to 94, the chair stand test performance of inactive persons declines by almost 7 points, from an average score of 13.8 for those aged 60 to 64 to an average score of 6.9 by age 90. Chair stand scores of active people on the other hand are 3.6 points higher at age 90 (10.5 versus 6.9), suggesting that if inactive people had been active they might have improved their performance by approximately 50 percent. In other words, instead of being able to perform only 6.9 chair stands by age 90, if a person had been active instead of inactive, she might be able to perform 10.5 stands, thus improving her performance by 52.2 percent

“Regular physical activity is essential for healthy aging. . . . Even in older adults with existing functional limitations, scientific evidence indicates that regular physical activity is safe and helps improve functional ability.”

U.S. Department of Health and Human Services, 2008

In calculating similar “savings ratios” for the other fitness items, we find that the average percentage of loss that might have been saved by being active is 52 percent. This statistic is especially important when considering that a 50 percent reduction in age-related decline translates,

over a 30-year period, into a 15-year fitness advantage for active compared with inactive people. Such statistics are in line with other reports suggesting that regular participation in moderate physical activity can reduce physical decline by as much as 50 percent and delay age-related physical deterioration by 10 to 20 years (Chandler & Hadley, 1996; Morey, Pieper, & Cornoni-Huntley, 1998; Paterson & Warburton, 2010; Shephard, 1997).

Of special concern in using normative standards to interpret individual scores are the characteristics of the study (normative) population. As indicated in table 3.3, participants in the normative study were 89.1 percent Caucasian, 4.2 percent Hispanic, 2.5 percent African American, 2.3 percent Asian and Pacific Islander, and 1.9 percent from other groups. Although these proportions are reasonably similar to U.S. national statistics, there clearly is an overrepresentation of Caucasian participants and an underrepresentation of other groups, especially African Americans and Hispanics. The current racial and ethnic percentages in the United States for the 65-plus population are 80.1 percent Caucasian, 8.3 percent African American, 7.0 percent Hispanic, 3.4 percent Asian and Pacific Islander, and 1.2 percent other (Administration on Aging, 2010).

Also of interest is that study participants were reasonably well educated (an average 14.5 years of education, equivalent to 2.5 years of college), they were quite active (more than 50 percent reported that they participated at least three times per week for at least 30 minutes a day in moderate activity compared with only about 25 percent nationwide), and most reported being in good to very good health.

Although participants in the SFT study may not vary greatly from other participants of the same type (community-residing older adult volunteers), care must be taken when using SFT norms to interpret scores of people who are different from those represented in this study. Again, it is important to remember that the SFT was designed for independent-living, community-residing older adults and that the test norms are based on a volunteer sample from this population. Using volunteers in studies results in a sample of convenience rather than a sample that is a true representation of the entire population of interest, with convenience samples almost always being healthier and higher functioning than the

population as a whole. In reality, though, the norms probably provide reasonable standards of comparison for the majority of older adults who are most likely to take these tests—that is, those who willingly participate in various community programs or in research studies.

In addition to establishing percentile norms for the SFT, data from the normative study provided a basis for establishing the first ever, as far as we know, criterion-referenced fitness standards for older adults, standards that provide health practitioners and rehabilitation and fitness specialists with clinically relevant information about the level of fitness needed to remain physically mobile and independent. The next section describes the procedures followed in establishing criterion-referenced fitness standards for older adults.

Criterion-Referenced Functional Fitness Standards

Criterion-referenced standards are developed by identifying scores associated with a particular goal or category of interest, such as being healthy or being functionally independent. In many cases, criterion-referenced standards provide more important types of information than do norm-referenced standards. For a 60-year-old man, for example, knowing that his cholesterol level does or does not place him at risk for cardiac problems is generally more important than knowing whether he is average for his age group. Similarly, knowing that one's lower-body strength (as measured by the SFT chair stand test) is at the level needed to remain physically mobile and independent may be of more interest than knowing how one's score compares with others of the same age.

In developing the criterion standards for the SFT (Rikli & Jones, 2012), we were interested in identifying the cut-point score on each test item that is best associated with having the fitness level needed to perform the types of everyday activities required to remain physically independent until late in life, with late in life defined as age 90 and beyond. These cut-point scores (standards), which indicate the level of fitness needed to be physically independent, should be considered appropriate fitness goals for all older adults regardless of their sex, education, or ethnicity.

The Senior Fitness Test Software 2.0 can generate personalized reports that allow test participants to compare their own scores to both normative ranges and the criterion standards associated with functional independence. To learn more, visit <http://sft.humankinetics.com>.

When interpreting results of the SFT, we believe that important information can be gained from both norm-referenced and criterion-referenced standards. Normative standards provide a basis not only for comparing a person's current fitness level with that of peers but also for tracking one's rate of change over the years relative to what is typical of other people the same age. Criterion standards, on the other hand, can be helpful in identifying the fitness level needed to remain functionally independent. With people living longer and the number of older adults increasing throughout much of the world, it is important for both personal and economic reasons that this large segment of the population remain healthy and independent. Although it is well known that preserving mobility and physical independence requires sufficient physical capacity (strength, endurance, agility, and balance), limited information has been available regarding the level of fitness needed to perform the common everyday activities needed for living independently. The criterion standards developed for the SFT represent the first attempt to identify criterion measures for a comprehensive fitness test battery for older adults. Criterion standards are presented for all SFT items except for the two flexibility measures, because there is currently insufficient research concerning the relationship between flexibility and improved functional ability.

The processes followed in setting criterion standards for the SFT were consistent with those recommended by leading measurement experts in exercise science (Baumgartner et al., 2007; Cureton & Warren, 1990; Mahar & Rowe, 2008; Morrow, Jackson, Disch, & Mood, 2011; Safrit & Wood, 1995). Because setting criterion-referenced standards involves referencing a particular score to a particular criterion goal, an important step in the process is determining how the criterion (physical independence, in this case) will be defined and evaluated.

“I use the Senior Fitness Test items routinely in clinical work. The new criterion-referenced standards will enhance their relevance and be a major contribution to the field I look forward to seeing them integrated into our geriatric medicine world.”

Miriam C. Morey, associate director of geriatric research, Durham VA Medical Center; professor of medicine, Duke University School of Medicine

Functional Abilities Needed for Physical Independence

With the goal being developing criterion standards for the SFT that indicate the threshold fitness scores associated with maintaining the functional abilities needed to be physically independent, it was important that there be a way of assessing functional ability. For the study to establish criterion-referenced standards for the SFT (Rikli & Jones, 2012), functional ability was assessed through self-evaluation using the composite physical function (CPF) scale described in figure 3.3, a scale that had previously been evaluated for reliability and validity (Rikli & Jones, 1998). The 12-item CPF scale was designed to assess physical function across a wide range of abilities—basic activities of daily living (ADLs; e.g., dressing and bathing oneself) to intermediate or instrumental activities of daily living (IADLs; e.g., housework and shopping) to advanced activities such as strenuous household, sport, and exercise activities. The CPF scale was developed based on an adaptation and extension of other previously published scales by Siu, Reuben, and Hays (1990) and by Rosow and Breslau (1966), with items also taken from the National Health Interview Survey (National Center for Health Statistics, 1991).

Composite Physical Function (CPF) Scale

Instructions: Indicate your ability to do each of the following by circling appropriate response. Your response should indicate whether you can do these activities, not if you actually do the activities.

	Can do on own without help	Can do with help	Cannot do
a. Take care of own personal needs, such as dressing yourself	2	1	0
b. Bathe yourself, using tub or shower	2	1	0
c. Walk outside (1 or 2 blocks)	2	1	0
d. Do light household chores, such as cooking, dusting, washing dishes, sweeping a walkway	2	1	0
e. Climb up and down a flight of stairs	2	1	0
f. Do own shopping and errands (walk approx. 3-4 blks, 400 yds)	2	1	0
g. Lift and carry 10 pounds (bag of groceries)	2	1	0
h. Walk 1/2 mile (6-7 blocks)	2	1	0
i. Walk 1 mile (12-14 blocks)	2	1	0
j. Lift and carry 25 pounds (medium to large suitcase)	2	1	0
k. Do heavy household activities, such as scrubbing floors, vacuuming, raking leaves	2	1	0
l. Do strenuous activities, such as hiking, digging in garden, moving heavy objects, bicycling, aerobic dance activities, strenuous calisthenics	2	1	0

CPF Rating Scale

High (advanced) functioning: those able to perform all 12 activities without assistance (CPF score of 24).

Moderate functioning: those with a moderate ability to perform at least 7 activities (score of 14-24) without assistance that meet commonly recognized requirements for physical independence—able to take care of personal needs, do light housework, walk 1-4 blocks, do own shopping, etc.

Low functioning at risk: those unable to meet requirements for moderate functioning. This includes a person who is at risk for losing physical independence.

Moderate functioning (age-adjusted score): for use when projecting future rather than current ability.

Age 65 and up: CPF score of 14 (able to perform at least 7 activities without assistance).

Age 55-64: CPF score of 16 (able to perform at least 8 activities without assistance).

Age 45-54: CPF score of 18 (able to perform at least 9 activities without assistance).

Age 35-44: CPF score of 20 (able to perform at least 10 activities without assistance).

(The age-adjusted higher scoring requirements for a moderate rating for those younger than 60 are to allow for an anticipated decline in functional ability level that is similar to the 10-15% age-related rate of physiological decline per decade in older adults. This is roughly a standard 10 moderate functioning level that reflects projected ability for independent functioning in later years 90% (more than) current ability to function independently.)

FIGURE 3.3 Composite Physical Function Scale used for self-evaluation of functional ability.

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CPF scores of study participants in the nationwide database were used to categorize people as high (advanced) functioning, moderate functioning, or low functioning and at risk for loss of physical independence. High-functioning persons indicated they could perform all 12 items with no difficulty or assistance, receiving a perfect CPF score of 24. Moderate-functioning persons scored 14 or above, meaning they were able to

perform at least 7 items on the CPF without assistance, thus meeting commonly recognized requirements for physical independence. Low-functioning persons did not meet the requirements for moderate functioning, thus indicating they may be at risk for loss of physical independence. Defining *moderate functioning* (ability to perform at least 7 CPF activities without assistance) as having the ability to function independently is consistent with data reported by others (Cress, Petrella, Moore, & Schenkman, 2005; Rosow & Breslau, 1996; Siu et al., 1990). It is commonly proposed, for example, that living independently requires such attributes as being able to dress and bathe oneself, do simple housework, do one's own shopping and errands (requiring the ability to walk approximately 400 yards or 366 m, far enough to go to the post office, bank, grocery store, or doctor), lift and carry 10 pounds (4.5 kg), and negotiate stairs. These are the kinds of activities that would have to be checked off on the CPF scale in order to meet the minimum requirement for a rating of moderate functional ability (ability to do at least 7 CPF items without needing assistance) and, thereby, also meeting the requirements for being functionally independent.

The CPF scale also provides for an age-adjusted definition of moderate functioning, which is the definition of interest when the goal is to project into later life (say by age 90) one's ability to do the kinds of activities needed for independent functioning rather than assessing independent functioning at the current time. Since the process of establishing criterion fitness standards for the SFT involved projecting into the future (by age 90) what level of fitness on each test item was associated with retaining at least moderate functional ability and physical independence, age-adjusted scoring was used in defining moderate functioning. As indicated in figure 3.3, an age-adjusted rating of moderate functioning requires that people in their 60s, 70s, and 80s be able to score higher (i.e., be able to perform a proportionately greater number of CPF activities without assistance) than those in their 90s for a rating of moderate.

Identifying Criterion Fitness Scores Associated With Maintaining Functional Mobility and Independence

As is typical in the process of establishing criterion-based standards, a

combination of processes—subjective reasoning, data-based statistics, and literature review—was involved in arriving at the recommended fitness standards (cut-point scores) for the SFT test items that would be associated with the ability (or projected ability) to function independently in later years. More specifically, fitness standards were developed based on the fitness scores achieved by the 2,140 moderate-functioning participants in the previously described normative database (table 3.3), with adjustments made as needed to reflect other relevant information from the literature regarding anticipated age- and gender-related patterns of decline over the 30-year period from the 60s to the 90s. Naturally, the 3,126 study participants who are rated as high functioning in table 3.3 would also meet the criteria for independent functioning in later years. However, fitness scores for this group of study participants were not considered in the analysis when establishing minimum fitness standards for independent functioning because it was assumed their performance, being much higher than what is needed to meet minimum requirements, would have an inflationary effect on the fitness standards being proposed.

The strategy followed in arriving at the recommended fitness standards for older adults as presented in chapter 5 (table 5.5) involved three stages, with the first step being to set fitness standards for 90- to 94-year-olds, utilizing the actual scores obtained by the moderate-functioning (independent) 90- to 94-year-old participants in the normative study database. This was considered a logical first step since the goal of maintaining the level of fitness needed for independent functioning had already been met by this group. Because there is no evidence in the literature suggesting that thresholds for maintaining physical independence should be different for men than for women, the same fitness standards are proposed for those over 90 for both sexes on all test items, standards that are based on the average scores obtained by moderate-functioning 90- to 94-year-old male and female study participants, but with some rounding off and minor smoothing of numbers to make them consistent with SFT scoring procedures and convenient for use by the public.

Previous reports suggest that the proposed standards for the 90- to 94-year-olds are reasonable and in line with other findings concerning the level of fitness needed for independent functioning. The 400-yard (366

m) standard for 90- to 94-year-olds on the 6-minute walk, for example, is within the range of the 360- to 600-meter recommendation that has been previously proposed as the minimum walking distance needed to function independently—that is, to be able to navigate within the community to do one's own shopping and errands (Cohen, Sveen, Walker, & Brummel-Smith, 1987; Lerner-Frankiel, Vargas, Brown, Krusell, & Schoneberger, 1986). Also, the 400-yard standard on the 6-minute walk is close to the quarter-mile (440 yard) criteria used by Medicare as a cut point for defining mobility limitation and disability (U.S. Department of Health and Human Services, 2006). The proposed fitness standard of 8.0 seconds for 90- to 94-year-olds to complete the 8-foot up-and-go test is similar to, but appropriately below, the 8.5-second cut point that has been identified as a predictor for falling in older adults, using the same testing protocol (Rose et al., 2002). Performing below the 8.5 cut point for predicting falls is an especially important consideration, given that falls and fall-related injuries are a major cause of loss of independence and escalating health care costs (Centers for Disease Control and Prevention, 2011).

The second stage in the process involved determining appropriate fitness standards for age groups younger than 90 based on how much anticipated physical decline needed to be planned for over the 30-year period between the ages of 60 and 64 and 90 and 94. It was important that standards be set high enough to allow people to experience normal rates of age-related decline and not progress below that required for independent functioning at age 90. In estimating the rate of decline that needed to be planned for on each of the fitness variables, we first considered the actual scores obtained by the 2,140 independent-functioning participants in the normative database, then we calculated the percent decline seen over the 30-year period from 60 to 64 and 90 to 94. Next, we made adjustments in this figure (percent decline) based on information from the literature showing a greater rate of decline (approximately 1.25 as great) when performance is tracked longitudinally (in the same people from one age to the next) versus cross-sectionally, which occurred in the normative study when different age groups were measured at the same time. Therefore, the standards presented in table 5.5 for 60- to 64-year-olds reflect an anticipated rate of age-related decline that is approximately 1.25 times greater than what was observed in the normative study participants. See Rikli and Jones (2012) for

additional details.

Once fitness standards were developed for the oldest (90 to 94) and youngest (60 to 64) age groups, we established standards for the remaining in-between ages based on the best knowledge available concerning patterns of declines across these age groups, paying particular attention to the fact that the rate of decline tends to be greater for men than women (Doherty, 2003; Hughes et al., 2001); that lower-body strength tends to decline faster than upper-body strength (Paterson, Jones, & Rice, 2007; Vandervoordt, 2002); and that there is a well-recognized curvilinear pattern of age-related decline, with an acceleration in decline in later years (ACSM, 2009; Doherty, 2003; Macaluso & De Vito, 2004; Paterson et al., 2007; Vandervoordt, 2002). Note in [table 5.5](#) that the proposed standards reflect a somewhat slower rate of change during the earlier years compared with the later.

As mentioned previously, the ultimately proposed standards of fitness for all age groups reflect rounded-off numbers to make them consistent with SFT scoring procedures and to make them more user-friendly and convenient for use by practitioners who work with older adults and by older adults themselves. As seen in [table 5.5](#), all standards for the chair stand, arm curl, and step test have been reported in whole numbers to be consistent with the SFT scoring instructions for these tests, with standards for the 6-minute walk reported in 5-yard increments, also consistent with scoring instructions on the SFT.

The criterion performance standards (fitness goals) for each age group are also illustrated by the star symbols (*) in figures 5.2 and 5.3. The low-functioning zones in these performance charts ([figures 5.2 and 5.3](#)) reflect the average fitness scores of older adults who report being unable to perform the types of functional activities required for independent living as described in [table 3.4](#). More specifically, low-functioning people are those who indicate they can perform only 6 or fewer activities on the CPF scale without assistance, thus making them at risk for losing their independence.

TABLE 3.4 SFT Means and Standard Deviations (in Parentheses) for Women and Men With High (Advanced), Moderate, and Low Levels of Functional Ability

Test item	SELF-REPORTED FUNCTIONAL ABILITY		
	High (advanced) n = 3,126	Moderate n = 2,738	Low n = 717
Chair stand (# stands)			
Women	14.1 (3.6)	12.4 (3.5)	8.4 (4.1)
Men	15.5 (4.2)	12.5 (4.1)	8.3 (3.4)
Arm curl (# reps)			
Women	15.7 (4.6)	14.0 (4.0)	10.5 (3.9)
Men	18.0 (4.9)	15.9 (4.5)	10.7 (3.5)
6-minute walk (# of yd)			
Women	589 (84)	515 (101)	363 (135)
Men	636 (96)	521 (126)	369 (138)
2-minute step (# steps)			
Women	94 (24)	81 (23)	58 (25)
Men	100 (24)	87 (26)	59 (25)
Chair sit-and-reach (in.)			
Women	2.4 (3.6)	1.0 (3.6)	-1.9 (4.1)
Men	0.2 (4.5)	-1.9 (4.8)	-3.8 (4.8)
Back scratch (in.)			
Women	-0.1 (3.5)	-2.0 (3.9)	-4.5 (4.7)
Men	-4.0 (4.7)	-5.7 (5.2)	-8.0 (5.7)
8-foot up-and-go (sec)			
Women	5.5 (1.1)	6.2 (1.4)	8.8 (3.2)
Men	5.1 (1.2)	6.2 (1.9)	8.9 (2.9)

Data in this table (previously unpublished) were collected as part of the normative study described in Kell and Jones, 1999b. Functional ability was assessed through self-report via the CFI scale (figure 3.3) using run age-adjusted scoring. High functioning could perform all 12 CFI activities without assistance; for a perfect score of 24; moderate functioning were those scoring 14 or higher (could perform 7 or more activities without assistance); low functioning were those indicating they could perform only 6 or fewer CFI activities, thus being at risk for loss of physical independence. Mean ages: high functioning = 71.1; moderate = 74.8; low = 79.5. Gender representation: high functioning = 60% women; moderate = 77% women; low = 85% women.

To recap, the recommended fitness standards (cut-point scores on the SFT) were developed based on the projected level of fitness needed at each age to maintain mobility and physical independence until late in life, despite normal age-related declines in physical capacity.

Determining the Validity and Reliability of the Criterion-Referenced Standards

Once criterion performance standards had been proposed for the SFT, we assessed their accuracy by testing their validity and reliability on a separate sample of 82 community-residing older adults (48 women and 34 men; mean age = 70.2; $SD = 5.7$) (Rikli & Jones, 2012). In developing criterion standards, it is important to report not only the validity and reliability of the measurement tools (the SFT and CPF, in this case) but also the validity and reliability of the fitness standards as an indication of their accuracy and consistency in identifying those who have or have not met the standards (Baumgartner et al., 2007; Morrow et al., 2011). The validity of a criterion-referenced test addresses the question of consistency in properly classifying people as having met or not met the established standards on each test. In this case, a high validity score would indicate that the established fitness standards were accurate in a large percentage of cases in predicting those who also met (or did not meet) the standard for moderate functional ability as measured by the CPF. As reported in table 5.5, the validity scores were quite good (generally above .80) on all items for both men and women. Per recommended procedures (Baumgartner et al., 2007), phi coefficient correlation values are also presented, providing another indication of classification consistency.

**TABLE 3.5 Validity of Criterion-Referenced Standards
for Senior Fitness Test Items**

Test item	TOTAL			WOMEN			MEN		
	c	(ϕ)	N	c	(ϕ)	n	c	(ϕ)	n
Chair stand	0.87	-0.7	75	0.86	-0.74	45	0.86	-0.71	30
Arm curl	0.83	-0.66	69	0.83	-0.52	40	0.86	-0.72	29
6-minute walk	0.91	-0.67	78	0.88	-0.44	48	0.97	-0.97	30
2-minute step	0.91	-0.79	68	0.92	-0.81	40	0.89	-0.76	28
8-foot up-and-go	0.79	-0.56	73	0.8	-0.6	44	0.79	-0.58	29

Note. c = validity coefficient, the proportion of consistent classifications in having met or not met the standard on both measures—the SFT field measure and the CPF criterion measure for functional independence.

ϕ = phi coefficient; indicates correlation between the classifications on each variable (SFT and CPF), another indication of classification consistency.

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Criterion-referenced reliability is determined by administering the test on two different occasions so that comparisons can be made regarding the accuracy of the classifications on day 1 versus day 2. Criterion tests are considered to have good reliability when a large number of participants who meet (or do not meet) the standard on one day are consistent in meeting (or not meeting) the standard on the retest day. As seen in table 3.6, there was considerable consistency in study participants relative to their meeting (or not meeting) the SFT fitness standards when the test was given on two different days. Modified kappa (k_q) values are also reported to provide a correction for chance in reporting classification consistency.

TABLE 3.6 Test—Retest Reliability of Criterion-Referenced Standards for Senior Fitness Test Items

Test item	TOTAL			WOMEN			MEN		
	Pa	(kq)	N	Pa	(kq)	n	Pa	(kq)	n
Chair stand	.89	(.79)	73	.89	(.78)	42	.90	(.80)	31
Arm curl	.80	(.60)	71	.79	(.58)	39	.81	(.62)	32
6-minute walk	.93	(.86)	73	.91	(.82)	44	.97	(.94)	29
2-minute step	.88	(.76)	69	.90	(.80)	39	.87	(.74)	30
8-foot up-and-go	.90	(.80)	71	.88	(.76)	40	.94	(.88)	31

Note. Pa = proportion of agreement in participants being consistently classified as having met or not met the SFT performance standard on both day 1 and day 2.

kq = modified kappa, providing a correction for chance in reporting classification consistency (Looney, 1989). Reprinted by permission from Rikli and Jones 2012.

Although further study over longer periods is required to fully assess the accuracy of the recommended criterion standards, with the true test being tracking participants over time to determine whether the proposed standards for the younger age groups are good predictors of functional ability and physical independence in later years. In the meantime, however, we believe the rationale and empirical evidence supporting the standards are sufficiently solid to make them useful as guidelines for evaluating fitness capacity in older adults relative to what is needed to remain physically independent until late life. Most important, the standards can provide health and fitness professionals with previously unavailable reference points for planning exercise interventions that target specific areas of weakness, thus reducing the risk for eventual loss of independence.

Summary

Published tests should be valid and reliable and should have accompanying performance standards to aid in the interpretation of test scores. Ideally, tests should be validated using multiple sources of evidence, including content-related, criterion-related, and construct-

related evidence.

In developing the SFT, content (logical) evidence regarding the relevance of the selected fitness categories was provided through literature review and through expert opinion, as was discussed in chapter 2 in the sections Conceptual Background and Functional Fitness Parameters. Criterion-related evidence was documented by showing the correlation (r values) between performance on each test item and performance on a recognized criterion measure, when an appropriate criterion could be identified. Construct-related validity of test items was evidenced by their ability to detect expected performance differences from one age group to another, and between people with high versus low levels of physical activity, and by their ability to detect exercise intervention effects.

The test–retest reliability of the SFT was assessed using intraclass correlation (R) procedures to compare test scores on day 1 with retest scores on day 2. Studies to assess SFT reliability were designed specifically to reflect conditions similar to those where most testing is likely to take place (i.e., in a group setting within the community, using trained volunteer assistants). As shown in table 3.2, test–retest R values for the test items range from .80 to .96, indicating acceptable reliability for all items.

Percentile norms for the SFT are based on a nationwide study involving more than 7,000 older Americans from 267 testing sites in 21 states. Data from the study provide age-group norms (reported in percentiles) for men and women aged 60 to 94. Test users are reminded that the SFT norms are based on a volunteer sample of independent-living older adults who are relatively healthy and tend to be more active than the population as a whole, which included a slight overrepresentation of Caucasian versus minority participants. The characteristics of the SFT participants are, however, similar to other large-scale samples of community-residing older adults and therefore appear to provide relevant standards of comparison for the majority of people most likely to take the test—that is, willing participants residing within the community.

Test users are further reminded that the norms represent average scores of broad ranges of ability levels and that the performance of specific subgroups of older adults might be expected to vary considerably.

Finally, the data from the normative study also provided a basis for developing the first-ever criterion-referenced standards for a comprehensive fitness test battery for older adults. Specifically, the average SFT fitness scores of moderate-functioning older adult study participants, with adjustments made as needed to better reflect anticipated patterns of age-related decline, were used in establishing the criterion fitness standards as presented in chapter 5 (table 5.5 and figures 5.2 and 5.3). The proposed fitness standards provide previously unavailable reference points for evaluating fitness levels in older adults relative to those needed to perform the kinds of everyday activities required for physical independence. Chapter 5 contains additional information on using the SFT performance tables and charts to interpret individual test scores.

In the next chapter we describe the procedures for administering and scoring the SFT items, including information needed to prepare participants properly for testing; checklists for gathering supplies and equipment; descriptions of the specific testing and scoring protocols, with examples of how to adapt test protocols for special populations; and guidelines for conducting group testing and training volunteer assistants.

Chapter 4

Test Administration

Establishing Consistent Testing Protocols



The Senior Fitness Test (SFT) was designed to be easy to administer in common community settings without extensive time, equipment, technical expertise, or space requirements. The complete test battery can be given in approximately 20 to 30 minutes to an individual or to pairs, using partners to assist with scoring. However, we also designed the test to be easily administered to a group of participants (up to 24 at a time) within a 60- to 90-minute period with the help of trained assistants.

In previous chapters we describe the rationale for the SFT, possible uses of the SFT, and why each physical parameter is important for keeping people active and independent in later years. In this chapter we provide detailed instructions for planning and administering the test items and discuss guidelines for group testing. More specifically, the topics include the following:

- Pretest procedures and considerations
- Administering the tests
- Guidelines for group testing

Pretest Procedures and Considerations

Although the SFT is comparatively simple to administer and score, as with any test, careful planning is required to ensure reliable testing protocols and meaningful data. The following is a list of pretesting procedures and conditions that need to be addressed before test day. Attention to these points is important to ensure participant safety, testing efficiency, and accurate measurements.

The Senior Fitness Test Software 2.0 allows you to enter participant information individually or import information for a group of participants from your own spreadsheet. To learn more, visit <http://sft.humankinetics.com>.

1. **Technician training.** Those administering the test items should be properly trained and well practiced regarding the testing procedures—that is, they should understand and be able to follow the exact protocols for administering the tests and recording the scores. Strict adherence to the established test procedures is essential if meaningful comparisons are to be made to the normative standards, or if comparisons are to be made from one testing time to another, such as before and after initiating an exercise program. After becoming familiar with the test instructions contained in this chapter, technicians should engage in ample practice testing before actually giving the tests to clients.
2. **Informed consent and assumption of liability.** If test results are being used for research purposes, the examiner must obtain informed

consent from the participants before testing. The purpose of obtaining informed consent is to protect the rights of human subjects (i.e., to ensure they have been informed of the test's purposes and risks and are aware of their right to discontinue testing at any time). Although a signed informed consent form may not be required in your program, we recommend that it (or something similar) be used anyway as a way of explaining the risks and responsibilities associated with testing. Here is a sample informed consent and assumption of liability form describing testing purposes, risks, protocols, and individual rights and responsibilities.

3. **Screening of participants.** Although the test items are safe to administer to most community-residing older adults without medical screening, there are some exceptions. People who should *not* participate in testing without physician approval are those who
 - have been advised by their doctors not to exercise because of a medical condition;
 - have had congestive heart failure;
 - are currently experiencing joint pain, chest pain, or dizziness or have exertional angina (chest tightness, pressure, pain, heaviness) during exercise; or
 - have uncontrolled high blood pressure (greater than 160/100).
4. When gathering the information you normally obtain from your clients (name, address, emergency contact, health background, and so on), you should explain the previously listed conditions and ask clients to obtain medical clearance from their doctors if needed. Asking clients to sign an informed consent and assumption of liability form that describes these conditions is one way of communicating this information. Here is a sample medical clearance form, should it be required.
5. **Pretest instructions to participants.** To ensure maximum safety and performance, participants should also be given information before test day concerning the best way to prepare for testing. Specifically, participants should be asked to
 - avoid strenuous physical activity for a day or two before assessment;
 - avoid excess alcohol use for 24 hours before testing;
 - eat a light meal an hour before testing;
 - wear clothing and shoes appropriate for participating in physical activity;
 - bring a hat and sunglasses for walking outside and reading glasses (if needed) for completing forms;

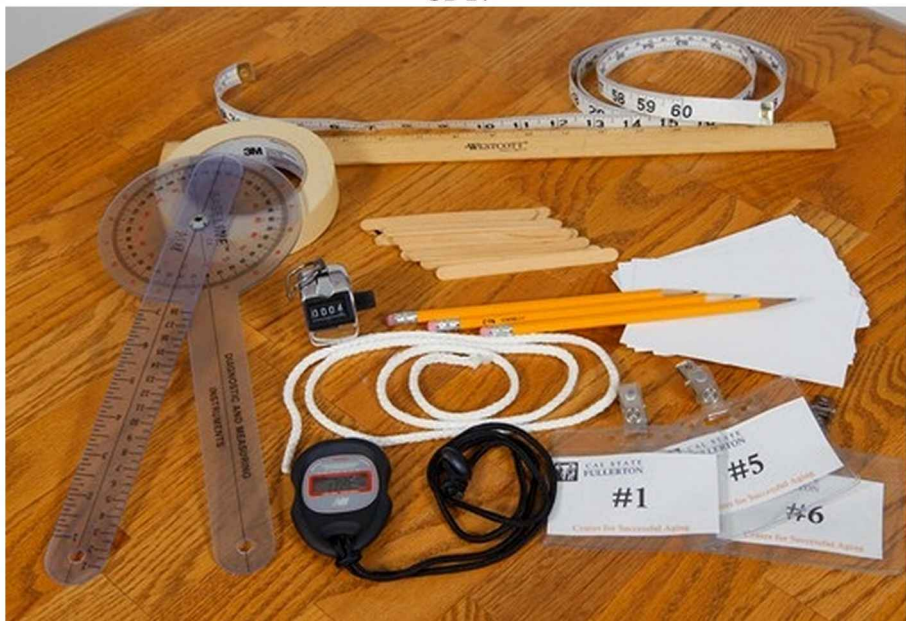
- bring the informed consent and medical clearance forms, if required; and
 - inform the test administrator of any medical conditions or medications that could affect performance.
6. Also, to improve their scoring accuracy, participants need to practice taking the aerobic endurance test before test day. They should time themselves either walking for 6 minutes or stepping for 2 minutes. Practicing the test will help them determine the pace that works best for them. The informed consent and medical clearance forms, if needed, should be distributed to clients at the same time as the list of participant instructions. Here is an example of a participant instructions before assessment form. You may copy or adapt this form, as needed, for use in your program.
7. **Testing equipment and supplies.** All testing equipment and supplies should be gathered and readily available before testing. Most of the materials required for the SFT are common items that can be found in most exercise facilities, brought from home, or obtained easily at local stores. Table 4.1 contains a list of all equipment and supplies needed to conduct the SFT, along with possible vendors or sources where they can be obtained, and these supplies are shown in figure 4.1. Specific uses of each equipment and supply item are indicated in the test item descriptions in the next section.

TABLE 4.1 Vendors and Sources for SFT Equipment and Supplies

Equipment and supplies*	Possible vendor or source
Folding chairs	Generally available at major discount stores, sporting goods stores, and some department stores.
Stopwatches	
Hand weights (5 & 8 lb [2.3 & 3.6 kg] dumbbells)	
Scale	
Masking tape	Usually available in craft and fabric sections of discount stores or in specialty craft and fabric stores.
30 in. (76 cm) cord	
Long tape measure (>20 yd or m)	
4 cones (or similar markers)	
Popsicle sticks (for counting laps on 6-minute walk)	Available in office supply stores.
60 in. (150 cm) tape measure	
18 in. (45 cm) ruler (half a yardstick)	
Tally counter (for 2-minute step test)	
Small pencils	Information in appendix D can be copied onto 8.5 × 11 (22 × 28 cm) card stock, then cut in half.
Name tags	
3 × 5 in. (7.5 × 12.5 cm) cards (may be used as an alternative way of marking laps in the 6-minute walk)	
Individual score cards (approx. 5 × 8 in. [13 × 20 cm])	
Station signs (for use in group testing)	Information in appendix G can be copied and enlarged onto heavy paper. Could also be mounted and laminated if so desired.
12-in. plastic goniometer	Available at health/medical supply stores or online.

* For group testing, multiple numbers of some items will be required. See table 4.2

Figure 4.1 Equipment and supplies you will need to administer the SFT.





8. **Data recording forms (scorecards).** Forms for recording the test scores, similar to the sample shown in figure 4.2, should be prepared before testing. The “comments” line on the scorecard is for indicating any deviations from proper protocol. Scores obtained from adjusted protocols, of course, should not be compared with the normative standards but can be used for individual comparisons from one testing time to the next. Note that the information appears twice on the same page, thus allowing it to be copied onto heavy paper and then cut in

half to form two approximately 5- by 8-inch scoring cards.

Scorecard: Senior Fitness Test

Date: _____

Name: Mary Jacob M: _____ F: _____ Age: _____ Ht: _____ Wt: _____

Test Item	Trial 1	Trial 2	Comments	
1. Chair stand test (# in 30 sec)	15	N/A		
2. Arm curl test (# in 30 sec)	10	N/A		
3. 2-minute step test (# of steps) or 6-minute walk test (# of yds)	520 yd	N/A		
4. Chair sit-and-reach test (nearest 1/2 in.: +/-)	+1.5	+2.0	Right	or Left (extended leg)
5. Back scratch test (nearest 1/2 in.: +/-)	-4.5	-3.0	Right	or Left (over shoulder)
6. 8-ft up-and-go test (nearest 1/10 sec)	6.1	6.3		

FIGURE 4.2 Sample scorecard.

The Senior Fitness Test Software 2.0 allows you to enter scores directly into the system using your smart phone or other mobile device. To learn more, visit <http://sft.humankinetics.com>.

9. **Testing order.** In planning testing order, it is important to consider whether you will be using the 6-minute walk test or the 2-minute step test as the aerobic endurance measure. If the 2-minute step test is used, the test items should be scheduled in the following order to minimize fatigue: chair stand test, arm curl test, 2-minute step test, chair sit-and-reach test, back scratch test, and 8-foot up-and-go test. When test stations are set up circuit-style to assess large groups,

participants can begin at any point in the rotation, but should then follow the recommended order so that the same muscle groups are not involved in back-to-back tests. If the 6-minute walk test is used, it should be administered last, after all other tests have been completed. Normally, if the 6-minute test is given, the 2-minute step test is omitted from the rotation. If you choose to give both the 2-minute step test and the 6-minute walk, we recommend that the 6-minute walk be administered on a separate day to avoid overfatiguing the participants. The height and weight measurements can be taken at any time, since they involve no exertion. During group testing we recommend that the height and weight measures be scheduled at station 3, along with the 2-minute step test if it is given or instead of the 2-minute step test if it is omitted.

10. **Environmental conditions.** The tests should not be administered if the temperature or humidity conditions are uncomfortable or appear unsafe for the participants. Because people differ substantially with respect to their tolerance of temperature and humidity, the best guide is each person's own comfort level. Always watch for signs of overheating or overexertion, and stop the test immediately if symptoms occur or if the participant requests to stop.
11. **Signs of overexertion.** The following are common physiological signs associated with overheating or overexertion. Testing should be stopped immediately if any of the following conditions appear:
 - Unusual fatigue or shortness of breath
 - Dizziness or light-headedness
 - Tightness or pain in chest
 - Irregular heartbeat
 - Pain of any kind
 - Numbness
 - Loss of muscle control or balance
 - Nausea or vomiting
 - Confusion or disorientation
 - Blurred vision
12. **Emergency procedures and accident reporting.** Before you begin testing, be sure to plan procedures that will be followed in case of emergency. Know where the closest phone is, and be sure that emergency procedures and phone numbers are posted clearly. It is also a good idea to have typed directions to your facility posted nearby in case you need to call for emergency assistance. If any kind of injury or illness occurs during testing, you should complete an accident report describing the situation and the procedures that were

followed. Here is a sample accident report form.

Administering the Tests

This section describes the procedures you should follow on test day. Included are guidelines for a proper warm-up and the instructions you should give to participants before they begin the tests. Also presented are detailed descriptions of each of the official SFT protocols.

“Normally, at this stage in life, taking a test is the last thing I want to do. But the Senior Fitness Test is an exception. It’s mostly fun to take and confirms the value of exercise classes I attend.”

Ken Higdon, age 86, Lakeview Village, Lenexa, Kansas

Warm-Up Exercises and Participant Instructions

Before testing begins, participants should engage in 5 to 8 minutes of warm-

up and stretching activities. It doesn't really matter what specific activities are performed during the warm-up, as long they involve the large-muscle groups and are not too strenuous. Marching in place, swinging the arms, and up-and-back and side-to-side walking steps are good ways to warm up the muscles. Performing these activities to music can make them more enjoyable and provide a fun way to begin the testing experience. After the warm-up, participants should do some simple stretches, paying special attention to the areas that will be stretched during the tests, especially the lower-body (hamstring) muscles and the upper-body (shoulder) area. The following are some sample exercises that could be used before the SFT to stretch the major muscles and joints involved in the testing.

Sample Stretching Exercises

Head Turns

Slowly turn your head to the right until you feel gentle tension on the side of your neck. Hold 5 seconds, and then slowly turn your head to center. Repeat the stretch to the left side.



Head Half Circles

Slowly tilt your head over to the right side until you feel gentle tension, and then slowly rotate your head forward and to the left side. Repeat the stretch

on the opposite side.



Single-Arm Crossover

Using your left hand, grab your right arm and slowly pull it across your chest until you feel gentle tension. Hold 5 seconds. Repeat with your left arm.



Chest Stretch

Grasp your hands behind your back, and slowly raise your arms until you feel gentle tension in your chest, shoulders, and arms. Hold 5 seconds.



Calf Stretch

Step forward with your left foot, keeping your feet parallel to each other. Shift your body weight forward by bending your left knee, keeping your right leg straight with your heel on the floor. Hold 10 seconds. Repeat with your right foot forward.



Hamstring Stretch

Extend your left leg forward with your foot lightly flexed. Bend your right knee and lean forward at the hips, using your hands for support. Keep leaning forward until you feel gentle tension in the back of your left leg. Hold 10 seconds. Repeat with your right leg forward. Be sure to keep your back straight, not rounded.



Guidelines for Stretching

Do

- Perform some type of warm-up activity before stretching (to increase circulation and body temperature).
- Gradually ease into each stretch and hold for 5 to 10 seconds.
- Stretch to the point of gentle tension but not pain.
- Repeat each stretch at least two times.

Don't

- Bounce, jerk, or force a stretch.
- Stretch to the point of pain.
- Hold your breath.

Just before beginning the tests, tell all participants to *do the best they can on all tests but never to push themselves to a point of overexertion or beyond what they think is safe for them*. Such a statement not only standardizes the testing instructions for all participants (these were the same directions given during the normative testing) but also helps clarify for the participants that the objective on all test items is to try as hard as they *comfortably* can, while staying within their own safety limits.

Instructions for Participants

Encourage participants to do the best they can on all tests but never to push themselves to a point of overexertion or beyond what they think is safe for them.

Strict adherence to the points mentioned, as well as to the test protocols that follow, contributes not only to participant safety but also to consistent and reliable testing procedures. Again, scores obtained from protocols that have been altered in any way or adapted to meet individual needs should *not* be compared with the performance standards, but they can be used for individual evaluation from one testing time to another. The specific variation from the recommended protocols should be described in the comment section of the scorecard.

Safety Precautions and Adaptations for Special Populations

Although the SFT was originally designed for generally healthy community-dwelling older adults, because of the ease and safety of test administration, the test items also have been used with various clinical and frail populations, including people with metabolic disorders (e.g., cardiovascular conditions, pulmonary disorders, diabetes), musculoskeletal conditions (e.g., osteoporosis, arthritis, hip or knee replacement), neurological and cognitive disorders (e.g., Alzheimer's disease and other forms of dementia, Parkinson's), sensory impairments (e.g., vision, auditory, sensorimotor), chronic pain conditions (e.g., fibromyalgia, arthritis, complex regional pain syndrome), and mobility limitations (e.g., balance disorders, sarcopenia, morbid obesity). Although some elements in the test item protocols may need to be adjusted, there is no need to change the time of the 30-second chair stand, 30-second arm curl, 2-minute step, and 6-minute walk.

Although some people may not be able to perform the task for that long, you just need to write down how much they could do within that period. The extended time was deliberately set to measure a wide range of physical abilities and provide an opportunity for people to demonstrate improvements over time. If the time of the 30-second chair stand was reduced to 10 or 15 seconds, there would be less opportunity to demonstrate improvement.

The safety precautions and adaptations for special populations at the end of each SFT test item protocol are based on recommendations from a panel of expert clinicians and practitioners who have experience using the SFT with various clinical populations. In addition, when administering all test items, keep in mind the following general guidelines:

- Stop testing immediately if participants are using improper technique or form (especially if they are in danger of hurting themselves).

- Stop testing if there are any signs of overexertion, as described in the section Pretest Procedures and Considerations.
- For people with visual impairments, use bright objects, clear any clutter, avoid glaring lights, and always ask how you can assist them.
- For participants with auditory impairments, use more visual demonstrations; have them read the directions; speak clearly and slowly, using low tones; and remember that facial expressions communicate a lot.
- For people with chronic pain, assess level of pain before testing. If pain level is greater than 5 on a 0 to 10 scale, delay testing until the level is lower. Also, spreading out the assessments over multiple days would be helpful.
- For participants with cognitive impairments, use simple words; speak slowly and distinctly; be calm, reassuring, and positive; maintain eye contact; don't give too many verbal instructions; use nonverbal gestures and demonstrations to help with communication; and allow participants adequate time to respond.

“It was found that this test battery (SFT), with minimal adaptations such as helping with one or both hands while standing up and the use of additional support for walking, is suitable for subjects after lower-limb amputation.”

Burger & Marincek, 2001

Official SFT Protocols

This section offers a full description of each of the SFT protocols. Included under each test item are its purpose, equipment requirements, test procedures, scoring instructions, safety precautions, and adaptations for special populations. For each test item, the instructor should first demonstrate the proper procedures at a reduced pace to ensure that participants understand what is expected. Then, on each of the timed tests (chair stand, arm curl, 2-minute step, 8-foot up-and-go, and 6-minute walk), the demonstration should be repeated at a faster pace to illustrate that the objective is to do the best one can within safety limits. As previously described, to minimize fatigue, we recommend giving the test items in the order they are presented here: chair stand test, arm curl test, 2-minute step test (if 6-minute walk test is not given), chair sit-and-reach test, back scratch test, and 8-foot up-and-go test. Again, when testing a large group using

circuit-style stations, participants can begin at any point in the rotation, then follow the recommended order from that point on. If the 6-minute walk test is used instead of the 2-minute step test, it should always be administered after all other tests are completed. The height and weight measures can be taken at any time because they don't involve any exertion.

30-Second Chair Stand Test

Purpose

To assess lower-body strength

Equipment

Stopwatch and straight-back or folding chair with a seat height of 17 inches

(43 cm); chair is placed against a wall to prevent slipping

Procedure

Instruct the participant to sit in the middle of the chair with back straight, feet flat on the floor, and arms crossed at the wrists and held against the chest. On the signal “go,” the participant rises to a full stand, then returns to a fully seated position. Before testing, have the participant practice one or two stands. Demonstrate the test slowly to show proper form, then at a faster pace to show that the objective is to do the best one can within safety limits. Encourage the participant to complete as many full stands as possible in the 30 seconds.

Scoring

The score is the total number of stands completed in 30 seconds. If a person is more than halfway up at the end of 30 seconds, it counts as a full stand. Administer only one test trial.

Safety Precautions

- Brace the chair against the wall, or have someone hold it steady. Ideally, use a carpeted surface for the chair to further keep it from slipping away from its position.
- Watch that the chair is under the participants when they sit, especially for people who are visually impaired or physically and cognitively frail.
- Watch for balance problems; quick movement could especially increase instability for people with sensory impairments (e.g., vision or inner ear problems).

- This test item may be contraindicated for people with chronic pain disorders and tall people who have had a knee or hip replacement (using a 17-inch chair may cause an angle greater than 90 degrees at the hips and knees, causing additional strain). Adapt the test to reduce pain or improve the angle, or do not use the test.

Adaptations for Special Populations

- If participants cannot perform even one stand with arms crossed on the chest, allow them to use their hands to push off their legs or the chair, or use a cane or walker, or use a higher or lower seat height. Describe the exact adaptation on the scorecard (e.g., pushed off thighs, used chair to push off, adjusted seat height up or down [note inches]). Although the recorded test score is zero for purposes of comparing to normative standards, also indicate the *adapted score* (i.e., 0/14) so that personal performance can be evaluated from one test time to the next. Of course the goal is to eventually use the test protocol as written, which doesn't allow using hands to press off.
- Remember, although the test protocol is the number of stands in 30 seconds, the test time can stop once you observe the person is no longer able to perform additional stands.
- For the cognitively impaired, you may want to repeat the test item demonstration.
- If the person is rather frail, she does not have to perform a practice stand.

30-Second Arm Curl Test

Purpose

To assess upper-body strength

Equipment

Stopwatch, straight-back or folding chair with no arms, 5-pound (2.3 kg) dumbbell for women, and 8-pound (3.6 kg) dumbbell for men

Procedure

Have the participant sit on a chair with back straight and feet flat on the floor, with the dominant side of the body close to the edge of the seat. The

weight is held down at the side, perpendicular to the floor, in the dominant hand with a handshake grip. From the down position, as the elbow bends the weight is curled up, with the palm gradually rotating to a facing-up position during flexion of the elbow. The weight is returned as the elbow is fully extended down, with the hand returning to a handshake grip. The wrist should not move—the bending is at the elbow. Demonstrate the test slowly to illustrate the form, then at a faster speed to illustrate the pace. Have the participant practice one or two repetitions without the weight to ensure proper form.

On the signal “go,” the participant curls the weight through the full range of motion (from full extension to full flexion of the lower arm) as many times as possible in 30 seconds. The upper arm must remain still throughout the test. Bracing the elbow against the body helps stabilize the upper arm.

Scoring

The score is the total number of arm curls executed in 30 seconds. If the arm is more than halfway up at the end of 30 seconds, it counts as a curl. Administer only one trial.

Safety Precautions

- Make sure participants do not have extensive flexion at the wrist when performing the test. The flexion and extension are from the elbow—not the wrist. The wrist should not bend forward and backward.
- Ask participants if they have any elbow, wrist, or hand pain; adapt the test to reduce pain, or do not use the test.

Adaptations for Special Populations

- If a participant cannot hold the hand weight because of a health condition such as arthritis, a Velcro wrist weight can be used.
- If the weight is too heavy for the participant to complete even one repetition using the correct form, a lighter weight can be substituted.
- If some participants are confused by the motion described in the test protocol or have difficulty switching from the handshake grip to the palm-up position, have them hold the weight in only one position during the arm curl.
- Allow the person to perform the test while standing if it is easier.
- For any adaptations in the test protocol, record both the official test score (zero) and the adapted test score (e.g., 0/13). Note the type of adaptation used to complete the test on the comment section of the scorecard (e.g., 2 lb weight, used green resistance band, did not return to handshake grip).
- Remember, although the test protocol is the number of curls in 30 seconds, the test time can stop once you observe the person is no longer able to perform additional curls.
- For the cognitively impaired, you may need to repeat the test item demonstration.

Height and Weight

Purpose

To assess body mass index (BMI)

Equipment

Scale, 60-inch (150 cm) tape measure, masking tape, and ruler (or other flat object for marking the top of the head)

Procedure

- **Shoes.** For the sake of time, shoes can be left on during height and weight measurements, with adjustments made as described later.

- **Height.** Tape a 60-inch tape measure vertically on the wall with the zero end positioned exactly 20 inches up from the floor. Have the participant stand with the back of the head against the wall (the middle of the head is lined up with the tape measure) and the eyes looking straight ahead. Place a ruler (or similar object) on top of the participant's head, and while keeping it level, extend it straight back to the tape measure. The person's height is the score in inches indicated on the tape measure plus 20 inches (the distance from the floor to the zero point on the tape measure). If shoes were worn, subtract .5 to 1 inch (or more) from the measured height, using your best judgment. Record height to the nearest half inch. (Note: A 60-inch tape measure is approximately equivalent to a metric tape measure of 150 cm. If a metric tape measure is used, for ease in calculating height, position it exactly 50 cm [equivalent to 19.7 in.] up from the floor. To adjust for shoe height, subtract 1 to 3 cm, using your best judgment. Record height to the nearest centimeter.)
- **Weight.** Have the participant remove any heavy articles of clothing (e.g., jackets, heavy sweaters). Measure the person's weight and record it to the nearest pound (or kilogram), with adjustments made for the weight of the person's shoes. In general, subtract 1 pound (approximately .5 kg) for lightweight shoes and 2 pounds (approximately 1 kg) for heavier shoes, using your best judgment.

Scoring

Record the person's height and weight on the scorecard. You can estimate body mass index later using the [BMI conversion chart](#). More precisely, you can determine BMI by dividing weight in kilograms by height in meters squared:

$$\text{BMI} = \text{kg/m}^2$$

BMI can also be calculated by multiplying the weight in pounds by 703, then dividing by the height in inches squared:

$$\text{BMI} = (\text{lb} \times 703)/\text{in.}^2$$

2-Minute Step Test

Purpose

To provide an alternative test of aerobic endurance

Equipment

Stopwatch, tape measure or piece of cord about 30 inches (76 cm) long, masking tape, and a tally counter to help count steps

Setup

For maximum scoring accuracy, have participants practice the test (stepping in place for 2 minutes) on a day before the test. On test day, begin by setting the minimum knee-stepping height for each participant, which is at a level even with the midway point between the kneecap and the front hip bone (iliac crest). It can be determined using a tape measure or by simply stretching a piece of cord from the middle of the patella (kneecap) to the iliac crest, then folding it over and marking this point on the thigh with a piece of tape.

Monitoring Step Height

You can monitor the correct knee height (stepping height) by moving the participant to the wall, to a doorway, or next to a high-back chair and transferring the tape from the thigh to a spot at the same level on the wall or the chair. If the person is tall enough, step height can also be marked by stacking books on a nearby short table.

Procedure

On the signal “go,” the participant begins stepping (not running) in place as many times as possible in the 2-minute period. Although both knees must be raised to the correct height, use your tally counter to count only the number of times the right knee reaches the target. When the proper knee height cannot be maintained, ask the participant to slow down or to stop until she can regain the proper form, but keep the time running.

Scoring

The score is the number of full steps completed in 2 minutes (i.e., the number of times the right knee reaches the proper height). Administer only one trial on test day.

Safety Precautions

- Participants with balance problems should stand next to a wall, next to or in a doorway, or between rails or chairs (for support in case of lost balance) and should be spotted carefully.
- Monitor participants for signs of overexertion.
- At the end of the test, ask participants to walk slowly for a minute to cool down.
- If any participants stomp their feet on the floor during the test, encourage them to bring their feet down lightly to prevent sore knees.

Adaptations for Special Populations

- If participants are unstable or visually impaired, have them hold on to a table, or stand next to a wall, or stand between rails or chairs to complete the test; closely monitor their progress.
- If participants are unable to lift their knees to the proper height or can lift only one to the proper height, allow them to complete the test to the best of their ability.
- If participants are unable to perform the steps in place, allow them to do a moving march.
- For any adaptations in the test protocol, note the type of adaptation used to complete the test on the comment section of the scorecard (e.g., unable to lift one knee to proper height, held on to rails with both hands, held on to the wall with one hand).
- Remind the frail or less aerobically fit participants that they can rest at any time (score is the number of steps completed in the 2 minutes).

Chair Sit-and-Reach Test

Purpose

To assess lower-body (primarily hamstring) flexibility

Equipment

Folding chair with a seat height of 17 inches (43 cm) and with legs that angle forward to prevent tipping, and an 18-inch (46 cm) ruler (half a

yardstick); chair is placed against a wall to prevent slipping

Procedure

The participant sits on the edge of the chair. The crease between the top of the leg and the buttocks should be even with the front edge of the chair seat. One leg is bent and slightly off to one side with the foot flat on the floor. The other leg is extended as straight as possible in front of the hip. The heel is placed on the floor, with the foot flexed at approximately 90 degrees.

With arms outstretched, hands overlapping, and middle fingers even, the participant slowly bends forward at the hip joint, reaching as far as possible toward or past the toes. If the extended knee starts to bend, ask the participant to move slowly back until the knee is straight. The maximum reach must be held for 2 seconds.

The participant should practice the test on both legs to see which is preferred (the one resulting in the better score). Only the preferred leg is used for scoring purposes (for comparison with norms). Once the preferred leg is determined, have the participant practice a couple more times for warm-up.

Scoring

After the participant has had two practice trials on the preferred leg, administer two test trials and record the better test score. Measure the distance from the tips of the middle fingers to the toe end of the shoe to the nearest half inch (centimeter). The midpoint at the toe end of the shoe represents the zero point. If the reach is short of this point, record the distance as a minus (–) score; if the middle fingers touch the toes, record a score of zero; and if the reach is past the midpoint of the toes, record the distance as a plus (+) score.

Safety Precautions

- Place the chair securely against a wall so it doesn't slip during testing.
- Remind participants to exhale as they bend slowly forward and to avoid bouncing.
- Participants should stretch only to a point of slight discomfort, never to the point of pain.
- Remind participants not to hold their breath—just continue breathing throughout the test.
- Do not administer the test to people with severe osteoporosis, with recent knee or hip replacements, or who have pain when flexing forward.
- Tester should get down beside the participant to the outside of the extended leg and place one hand on the knee (gently) so that if the tester feels the knee start to bend, she can have the participant stop or pull back if necessary.

Adaptations for Special Populations

- For people who cannot fully extend the knee, note approximate flexion on the scorecard, using a goniometer (if available) or using best judgment. The goniometer, if used, is positioned on the outside of the extended leg with the center axis at the midpoint of the knee joint, with one arm of the goniometer placed in line with the femur and one arm in line with the middle of the lower leg.
- If a participant is visually impaired, ask if you can touch him to help direct him.
- Repeat the demonstration for people who have a difficult time following directions.
- Allow participants to perform the test from a wheelchair (with wheels locked) or a walker with a seat.

Back Scratch Test

Purpose

To assess upper-body (shoulder) flexibility

Equipment

18-inch (46 cm) ruler

Procedure

Have the participant stand and place the preferred hand over the same shoulder, palm down and fingers extended, reaching down the middle of the back as far as possible. Note that the elbow is pointed up. Ask the participant to place the other arm around the back of the waist with the palm up, reaching up the middle of the back as far as possible in an attempt to touch or overlap the extended middle fingers of both hands. The participant should practice the test to determine the preferred position (the hand over the shoulder that produces the best score). Two practice trials are given before scoring the test.

Check to see if the middle fingers are directed toward each other as best as possible. Without moving the participant's hands, direct the middle fingers to the best alignment. Do not allow participants to grab their fingers together and pull.

Scoring

After giving the participant two warm-up practice trials in the preferred position, administer two test trials and record the better test score to the nearest half inch (cm), measuring the distance of overlap, or distance between, the tips of the middle fingers. Give a minus (–) score if the middle fingers do not touch, a zero score if the middle fingers just barely touch, and a plus (+) score if the middle fingers overlap. Always measure the distance from the tip of one middle finger to the tip of the other, regardless of their alignment behind the back.

Safety Precautions

- Stop the test if the participant experiences pain.
- Remind participants to continue breathing as they stretch.
- Remind participants to avoid any bouncing or rapid movements.
- Try to take the measurement as quickly as possible so participants don't have to hold an uncomfortable position.
- Have participants shake and roll their shoulders between trials.

Adaptations for Special Populations

- This test is contraindicated for people with neck and shoulder injuries or problems (e.g., frozen shoulder, rotator cuff problems, pinched nerves).

8-Foot Up-and-Go Test

Purpose

To assess agility and dynamic balance

Equipment

Stopwatch, folding chair with 17-inch (43 cm) seat height, tape measure, and cone (or similar marker)

Setup

Place the chair against the wall, facing a cone marker exactly 8 feet (2.4 m) away (*a*), measured from the back of the cone to a point on the floor even with the front edge of the chair.

Procedure

Instruct the participant to sit in the middle of the chair with back straight, feet flat on the floor, and hands on the thighs. One foot should be slightly in front of the other foot, with the torso slightly leaning forward. On the signal “go,” the participant gets up from the chair, walks as quickly as possible around either side of the cone (*b*), and sits back down in the chair. Be sure to start the timer on the signal “go” whether or not the participant has started to move, and stop the timer at the exact instant the person sits back down on the chair.

Scoring

After you have demonstrated the proper form and desired pace, have the participant practice the test once, and then administer two test trials. Record the best (fastest) time to the nearest tenth of a second.

Safety Precautions

- When administering the 8-foot up-and-go test, stand between the chair and cone in order to assist participants in case they lose their balance. For the frail, you may need to spot them more closely, especially as they stand, turn around the cone, and sit down. If at any time you believe a person is at risk for falling, do not administer the test.
- With a frail or very obese person, watch that he stands up and sits down safely; you may have to direct the person's bottom to the chair as he sits down. Also, you may need to use a larger and sturdier chair and possibly get assistance from a strong person.

Adaptations for Special Populations

- If needed, participants can use a cane or walker for this test.
- For the visually impaired, use a brightly colored or larger cone to prevent tripping; provide verbal guiding cues; and, if needed, physically guide the participant.
- For the cognitively impaired, mark the walking path with markers or arrows.
- Allow people who are unable to get up from a chair to start and stop the test from a standing position.

*Note: When administering tests, the 6-minute walk test should always be given last.

Purpose

To assess aerobic endurance

Equipment

Long measuring tape, two stopwatches, four cones (or similar markers), masking tape, felt-tip marker, 12 to 15 popsicle sticks per person (or index cards and pencils to keep track of laps walked), chairs for waiting partners and for walkers who need to rest, and name tags

Setup

For improved pacing and maximum scoring accuracy, have participants practice a 6-minute walk before test day. To set up on test day, mark off in 5-yard segments a flat 50-yard rectangular area (20 yards by 5 yards; see diagram). The inside corners of the measured distance should be marked with cones, and the 5-yard lines marked with masking tape or chalk. (In metric units, this can be a 50 m rectangular course marked off in 5 m segments, with scores then translated to yards for comparison with the norms in this text).

Procedure

Two or more participants should be tested at a time to standardize motivation. A skilled instructor can test up to 12 people at once, using partners to assist with scoring, but 6 at a time is more manageable. Starting (and stopping) times are staggered 10 seconds apart to encourage participants to walk at their own pace and not in clusters or pairs. Participants wear numbers (using name tags) to indicate the order for starting and stopping. On the signal “go,” the participant begins walking as fast as possible (not running) around the course, covering as much distance as possible in the 6-minute time limit. We recommend using two stopwatches to time the test, just in case one stops working. To keep track of the distance walked, partners give popsicle sticks (or similar objects) to participants each time they complete a lap. Or partners can mark a scorecard each time a lap is completed, using the “picket fence” system (/ / / and so on).

The timer should move to the inside of the marked area after everyone has started. To assist with pacing, the remaining time should be called out when walkers are about half finished and when about 2 minutes is left. Participants can stop and rest on the chairs provided, but the time keeps running. The tester should encourage participants a few times by saying, “You’re doing well,” and “Keep up the good work.” When a participant’s 6 minutes has elapsed, the tester asks her to stop, move to the right (across from the nearest 5-yard marker), and slowly step in place for a minute to

cool down.

Scoring

Record the scores when all the walkers have been stopped. Each popsicle stick (or mark on a card) represents 50 yards (or meters). For example, if a person has eight sticks (representing eight laps) and was stopped next to the 45-yard (or 45-meter) marker, the score is a total of 445 yards or 445 meters, with meters needing to be converted to yards for comparison with norms. Administer only one trial on test day.

Safety Precautions

- Select a well-lit walking area with a level, nonslip surface.
- Position chairs at several points along the outside of the walking area for participants who want to rest during or after the test and for emergency situations (e.g., dizziness or overexertion).
- Monitor participants for signs of overexertion.
- For people with diabetes, watch for signs of hyperglycemia or hypoglycemia. Have fruit juice available for them to drink.
- At the end of the test, ask participants to continue walking slowly for a minute to cool down, followed by a lower-calf stretch.
- Have water available for participants to drink before, during, and after testing.
- Remind participants that they can slow down; or they can stop, rest, and start again until the 6 minutes are up.

Adaptations for Special Populations

- If needed, participants can use a cane or walker for this test to help

with balance.

- For the frail or visually impaired participant, offer to have a “buddy” (testing assistant) walk with the participant (or even allow the participant to hold the assistant’s arm if needed).
- For any adaptations in the test protocol, note the type of adaptation used to complete the test on the comment section of the scorecard (e.g., walked with cane, held arm of tester).
- Remind the frail or less aerobically fit participants that they can rest at any time (score is the number of yards or meters completed in the 6 minutes).

Guidelines for Group Testing

It has been our experience that both practitioners and researchers generally find it preferable to administer the SFT to several people at a time. In fact, group testing is preferred when the goal is to compare performance to the normative data, because all normative scores were collected in a group setting where there tends to be more social interaction and encouragement than when tests are given in an isolated environment. Although the SFT is especially suitable for group administration, careful planning and organization are needed for the testing to run smoothly and efficiently. With the help of six or seven trained assistants, it is possible to test up to 24 participants at a time in a 60- to 90-minute period using a circuit-style setup. If all tests, including the 6-minute walk test, are to be administered indoors, a large community center or gymnasium (approximately 50 feet by 100 feet, or 15 m by 30 m) will be needed. However, a much smaller area will work if the 6-minute walk test can be administered outdoors or if the 2-minute step test is substituted for the 6-minute walk test as the measure of aerobic endurance. Included in this section are instructions for setting up the stations for testing, a list of equipment and supplies that will be needed at each station, guidelines for selecting and training testing assistants, and step-by-step procedures for administering the test items on test day.

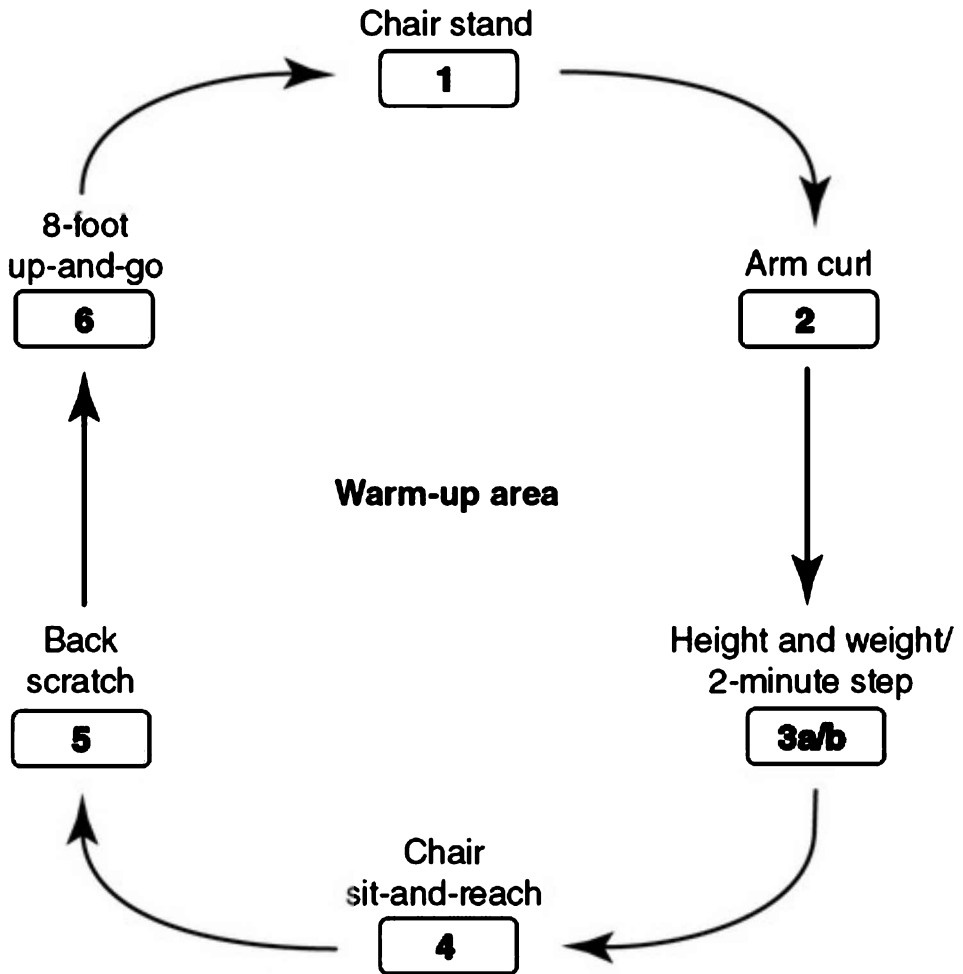
The Senior Fitness Test Software 2.0 allows you to schedule testing sessions and assign participants to groups and to test

sessions. To learn more, visit <http://sft.humankinetics.com>.

Station Setup

For the most efficient use of time and to minimize the fatigue effect for participants, testing stations should be set up circuit-style in the following order: (1) chair stand test, (2) arm curl test, (3) height and weight and 2-minute step test (if 6-minute walk test is not used as the aerobic endurance test), (4) chair sit-and-reach test, (5) back scratch test, and (6) 8-foot up-and-go test. As indicated in the diagram in figure 4.3, the stations should be set up around the periphery of the room, allowing space in the center for the pretest warm-up exercises and for the 6-minute walk if there is room. This type of station setup allows participants to begin their testing at any point in the circuit and then rotate in order to the next station.

Figure 4.3 Order of station setup for group testing.



*If the 6-minute walk test is substituted for the 2-minute step test, it should always be administered after all other tests are completed.

When the 6-minute walk test is used, it should always be administered after all other tests are completed. If it is not possible to give the 6-minute walk test (e.g., because of space limitations, bad weather), then the 2-minute step test is administered at station 3 along with the height and weight

measurements. If you want to administer both the 2-minute step test and the 6-minute walk test, we recommend including the 2-minute step test as part of the regular circuit and giving the 6-minute walk test on a separate day. For many older adults, it is too exhausting to complete both aerobic tests on the same day. Doing so could result in unsafe conditions as well as inaccurate scoring.

For group testing, the specific procedures for administering each of the test items within the circuit is the same as those described in the preceding section on official SFT protocols. However, additional equipment and supplies will be needed, as well as trained assistants to help at each of the stations.

Testing Equipment, Supplies, and Assistants

For group testing to run smoothly, each station should be properly set up in advance with all the required equipment and supplies. Although a special feature of the SFT is that it does not require extensive equipment, you should not underestimate how long it takes to gather the needed items. [Table 4.2](#) lists the specific equipment and supplies, as well as the number of assistants, needed at each station. Brief organizational instructions are also provided. Recall that [table 4.1](#) provides suggested vendors and sources for obtaining the equipment and supplies.

TABLE 4.2 Station Setup and Instructions for Testing Up to 24 Participants at Once*

Station	Equipment	# of testers	Brief instructions
1. Chair stand test	Stopwatch, folding chair, 17 in. seat height (43 cm)	1	Demonstrate test, have all participants at station practice at once, test one at a time.
2. Arm curl test	Stopwatch, 5 lb & 8 lb (2.3 kg & 3.6 kg) dumbbells (1 each), folding chair, 17 in. seat height (43 cm)	1	Same instructions as station 1
3. Height and weight; 2-minute step test	Scale, 60 in. (150 cm) tape measure (mounted on wall), two stopwatches, two tally counters, two 30 in. (76 cm) pieces of cord, masking tape	2 ^b	Demonstrate step test to all, have all practice at once; each tester measures two people's height and weight and administers their step test.
4. Chair sit-and-reach test	18 in. (45 cm) ruler, folding chair, 17 in. seat height (43 cm), 12-in goniometer	1	Same instructions as station 1.
5. Back scratch test	18 in. (45 cm) ruler	1	Same instructions as station 1
6. 8-foot up-and-go test	Stopwatch, cone, measuring tape, folding chair, 17 in. seat height (43 cm)	1	Demonstrate test, participants line up and take turns for practice test, trial 1, and trial 2.
All stations (1-6)	Each station needs enough chairs (a total of 4) for resting while waiting for turn, four scorecards, and four pencils		
*6-minute walk test	This test is given after all others, with all assistants helping. Equipment needs: Long measuring tape, two stopwatches, four cones, masking tape, felt-tip marker, 12-15 popsicle sticks per person (or index cards and pencils to keep track of laps walked), chairs for walkers who need to rest, and name tags.		
	See 6-minute walk test protocol. An experienced tester can test 12 people at a time, with another 12 used as scoring partners; however, if time permits, testing 6 at a time works better.		

*With 24 total participants, 4 are assigned to each station to begin testing; then they are rotated clockwise through the 6 stations in turn.

^bOnly one tester is needed at this station if the 6-minute walk test is given instead of the 2-minute step test.

Although not absolutely necessary, it is convenient to have a small table (such as a card table) set up at each testing station, or, if possible, to arrange the testing stations near a counter, a table, a bench, or a ledge of some type that can provide a place to lay out the testing supplies (scorecards, pencils, stopwatches). To facilitate assigning and rotating participants, it is also helpful to post station signs indicating the test name and number of each station. The signs can be copied, mounted, and laminated to make them more durable for repeated use.

As an aid in organizing and keeping track of all supplies and equipment, and in ensuring they are readily available at each station on test day, we have found it helpful to use numbered containers (e.g., gift bags or shopping bags) to collect and transport the items for each station. For example, a small bag labeled *Station 1* would contain all the items needed to conduct the chair stand test: a stopwatch, scorecards, and pencils. Similarly, a somewhat larger bag labeled *Station 6* would contain the items needed for the 8-foot up-and-go: a stopwatch, tape measure, cone (or similar marker), scorecards, and pencils. In fact, if you plan to conduct the tests on multiple occasions, it is helpful to list the contents on the outside of each bag, making it easy to double-check to see that all equipment and supplies are available each time they are needed.

Selecting and Training Testing Assistants

For assessing groups of approximately 24 people at a time, resulting in 4 participants per station, 6 to 8 testing assistants are required. If only 3 or 4 assistants are available, we suggest limiting your testing group to 12 instead of 24. Generally, 1 assistant is assigned to each station, with one exception. If the 2-minute step test is included at station 3 (meaning it is given instead of the 6-minute walk test), then at least 3, preferably 3, assistants are needed at this station to maintain good timing and prevent a backlog in the rotation of groups from one station to the next. If possible, the lead instructor or researcher should not be assigned to a testing station but should be free to walk around to monitor the testing and help rotate groups and to be available in an emergency situation. If additional assistants happen to be available to help with testing, by all means use them. Working in partnership during testing is more enjoyable for the assistants and has a positive effect on testing accuracy and efficiency.

Testing assistants can be older adult volunteers, coworkers, friends, or student volunteers from a nearby university. Active older adult volunteers are an especially ideal source of assistance provided they have good communication skills as well as the physical and mental capabilities needed to demonstrate and conduct the tests properly. In fact, we have found older adults to be enthusiastic about helping with the tests, reliable, accurate, and even willing to go on the road to assist in testing at other locations, if needed. Regardless of the type of assistants you have, it is essential that they be properly trained and given ample time to practice the tests before test day. All testers need to understand the exact procedures (protocols) for each

test item and the importance of strictly following such procedures, or if necessary, of describing any deviation in proper protocol in the comment section of the scorecard. The following is a summary of the procedures that should be followed in training testing assistants.

- Provide all assistants with written descriptions and visual diagrams of the test protocols several days before testing, and assign each assistant only one or two test items to learn and administer. The test descriptions in this chapter can be copied and distributed to test assistants.
- Ask the testing assistants to study their assigned test protocols and practice giving the tests to a friend or a family member.
- Schedule a testing practice day (rehearsal day) before the official test day for all assistants; demonstrate and review procedures for each of the test items.
- During the rehearsal, have the assistants practice giving their assigned tests to each other, and check carefully for accuracy in testing and scoring.
- If you are including the 6-minute walk test as one of the test items, explain the protocol fully to your assistants during the practice day, emphasizing their role in helping to get all participants to the proper location for testing, helping to assign partners, watching for signs of overexertion during the test, and assisting with scoring at the end of the test. If time permits, assistants should partner up and practice the 6-minute walk test exactly as it will be given on test day.

Test-Day Procedures

To help with planning, we have compiled the following list of test-day procedures based on our experience in administering the SFT to many different groups of older adults at numerous test sites. Following these steps should help your test day run smoothly.

1. You and all your assistants should arrive at the testing site at least 30 minutes before testing to set up the test stations and the 6-minute walk test course, if it is to be included.
2. Before participants arrive, call all test assistants together to review procedures and answer any last-minute questions.
3. Assign someone to collect all informed consent and medical clearance forms (if applicable) from participants.

4. After welcoming participants, you or an assistant should conduct a 5- to 8-minute warm-up for the participants, including gentle stretching exercises. If possible, use lively music for the warm-up to promote enjoyment and a positive mood for the day.
5. Before sending participants to their stations, explain that the goal is to do the best they can on all tests but never to push themselves to a point of overexertion or beyond what they think is safe for them. Remind participants they should discontinue any activity that causes them pain or discomfort.
6. After the warm-up and special instructions, direct the participants (in equal numbers) to one of the six stations. Participants will start at different stations; however, they should rotate from station to station in numerical order. For example, if a participant starts at station 5, she rotates to station 6, then station 1, and so on until she completes all stations. Have the participants remain at their respective test stations until everyone has completed the tests. Then ask participants to take their scorecards and rotate in a clockwise direction to the next station.

Sample testing scenario 1: Assuming 24 participants and 6 to 8 assistants, 4 participants would be assigned to each of the six stations to begin the testing. Each assistant remains at the same station throughout the testing, while participants rotate from one station to the next.

Sample testing scenario 2: Assuming 12 participants and 3 or 4 assistants, 4 participants are assigned to stations 1, 3, and 5 to begin the testing. Each assistant is responsible for conducting tests at two stations in a row. For example, when the assistant at station 1 completes the chair stand tests, he moves with these participants to station 2 to conduct the arm curl. Meanwhile, the assistant at station 3 also conducts the test at station 4, and so on. The participants eventually rotate to all six stations, while the assistants go back and forth between their two assigned stations.

7. If you have scheduled the 6-minute walk (which is always given after all other tests are completed), have participants bring their scorecards and walk together as a group to the test location. Refer to the 6-minute walk protocol described earlier in this chapter for setup and instructions. With a skilled instructor, it is possible to test up to 12 people at a time, using partners to help count laps. However, if time permits, testing 6 at a time is more manageable.

8. After testing, collect all scorecards and thank participants for their cooperation. Provide them with information about when and how they will receive their test results.

Summary

The SFT can be administered easily within the community setting. The complete test battery can be given to one or two people in 20 to 30 minutes and, with the help of trained assistants, can be given to a group of up to 24 older adults in a 60- to 90-minute period. Important pretest considerations include the following:

- Properly training test technicians
- Obtaining informed consent
- Properly screening participants
- Providing pretest instructions to participants
- Gathering testing equipment and supplies
- Preparing scorecards
- Planning proper testing order
- Considering environmental conditions and signs of overexertion

On testing day it is important to include a proper warm-up, provide standardized instructions to participants, and conduct the tests according to the official SFT protocols described in this chapter. When deviations from proper protocols are necessary, they should be described in the comment section of the scorecard.

The SFT is especially conducive to group testing, but this requires careful planning for the testing to run smoothly on test day. Specifically, special attention is needed with respect to (1) planning the testing stations, (2) gathering and organizing the equipment and supplies, (3) selecting and training testing assistants, and (4) planning the step-by-step test-day procedures.

Now that you know how to administer the SFT, the next chapter provides information on interpreting the test results and providing feedback to clients. Suggestions are made for using feedback to motivate clients and improve their performances.

Chapter 5

Test Results

***Interpreting and Using Feedback to
Motivate and Improve Performance***



We have found that after taking the Senior Fitness Test (SFT), most of the participants immediately want to know three things: (1) what their scores are, (2) what their scores mean, and (3) how they can improve their scores. In this chapter we explain how to interpret the results of the SFT items to your clients, including how to read the performance tables and charts that were developed as part of the national study to establish

the fitness standards. We also discuss ways of using the feedback from the test items to motivate your clients to increase their level of physical activity and improve their fitness level. Specifically, we will provide information on

- interpreting test scores,
- providing feedback to participants, and
- using test results to motivate participants.

Interpreting Test Scores

An important feature of the SFT is its accompanying performance standards that can be used in interpreting test results. As you will recall from chapter 3, performance standards can be either norm referenced (standards that make it possible to compare one's scores with those of others of the same age and sex) or criterion referenced (standards that represent a criterion behavior or goal, such as having the fitness level needed to perform everyday activities). Both normative and criterion-referenced standards have been developed for the SFT based on a national study involving more than 7,000 older adults, aged 60 to 94, from 267 different test sites throughout the United States. Data from the study were analyzed and then organized into various tables and charts that can be used in interpreting test scores.

Normative Tables

A common method of presenting normative data is through the use of percentile tables. Percentile norms indicate how a person's test scores rank relative to her peers. A percentile rank indicates the point in a distribution of scores below which that percentage of scores falls. For example, a chair stand test score of 15 for a 62-year-old woman would fall at the 50th percentile (as seen in the sample percentile norms in table 5.1), meaning half (50 percent) of the women her age typically score below her and half score above her. However, another woman of the same age scoring 20 on the chair stand test would have a percentile rank of 90, indicating she was better than 90 percent in her age group and that only 10 percent scored above her. Percentile tables for the Senior Fitness

Test contain the full distribution of percentile scores for women and men on each of the SFT items.

TABLE 5.1 Sample Chart of Age-Group Percentile Norms: Chair Stand Test (Women)

Percentile rank	60-64	65-69	70-74	75-79	80-84	85-89	90-94
95	21	19	19	19	18	17	16
90	20	18	18	17	17	15	15
85	19	17	17	16	16	14	13
80	18	16	16	16	15	14	12
75	17	16	15	15	14	13	11
70	17	15	15	14	13	12	11
65	16	15	14	14	13	12	10
60	16	14	14	13	12	11	9
55	15	14	13	13	12	11	9
50	15	14	13	12	11	10	8
45	14	13	12	12	11	10	7
40	14	13	12	12	10	9	7
35	13	12	11	11	10	9	6
30	12	12	11	11	9	8	5
25	12	11	10	10	9	8	4
20	11	11	10	9	8	7	4
15	10	10	9	9	7	6	3
10	9	9	8	8	6	5	1
5	8	8	7	6	4	4	0

Note: Percentile norms for all SFT test items are presented in appendix H.

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Percentile tables can also be used to compare a person's own scores across the different fitness categories. Determining the corresponding percentile rank for various test scores as illustrated in table 5.2 indicates a person's relative strengths and weaknesses. For example, if a 73-year-old man obtained the raw fitness scores presented in table 5.2, his corresponding percentile ranks (determined from the percentile tables for the Senior Fitness Test) would tell us he is above average on the first three tests (chair stand, arm curl, and 6-minute walk, scoring at the 75th,

70th, and 90th percentiles, respectively) but below average (at the 20th percentile) in lower-body flexibility.

TABLE 5.2 Example of Raw Scores and Percentile Equivalents

Test item	Raw score	Percentile rank (approx.) (for men aged 70-74)
Chair stand test (lower-body strength)	17	75th
Arm curl test (upper-body strength)	20	70th
6-minute walk test (aerobic endurance)	740	90th
Chair sit-and-reach test (lower-body flexibility)	-4.0	20th

Test results, such as those presented in table 5.2, can help in planning programs that are specifically targeted toward the needs of clients. Based on this gentleman's test scores, a recommended exercise program probably should include extra activities for improving his lower-body flexibility. Recording your client's percentile scores on a personal profile form, such as the one in figure 5.1, can help interpret personal strengths and weaknesses and track progress from one testing time to the next. Also, for those who have access to a computer, the Senior Fitness Test Software 2.0 can be used to prepare personalized, professional-looking records for clients and can help maintain class records and document program outcomes.

Reports generated from the Senior Fitness Test Software 2.0 show an individual participant's scores placed on a continuum of normative values to help participants interpret the scores. To learn more, visit <http://sft.humankinetics.com>.

Personal Profile Form

Name John Doe Test Date 2-12-13

Age 73 M X F

Test item	Score	Percentile category*			Fitness standard met? Yes / No	Comments
		Below average	Normal range	Above average		
		25th%	75th%			
Chair stand (# of stands)	17	_____	_____	<u>X</u>	<u>Yes</u>	Keep up the good work!
Arm curl (# of repetitions)	20	_____	_____	<u>X</u>	<u>Yes</u>	Also good!
2-minute step (# of steps) or 6-minute walk (# of yds)	740 yd	_____	_____	<u>X</u>	<u>Yes</u>	Excellent! Keep up your walking program.
Chair sit-and-reach (# of inches +/-)	-4.0	<u>X</u>	_____	_____	<u>N/A</u>	Flexibility needs work. Add stretches for calf and hamstring muscles.
Back scratch (# of inches +/-)	-8.5	<u>X</u>	_____	_____	<u>N/A</u>	Should add exercises for shoulder flexibility.
8-foot up-and-go (# of sec.)	4.2	_____	_____	<u>X</u>	<u>Yes</u>	Very good mobility.
Body mass index (see BMI chart)	Ht <u>67 in.</u>	≤18 Underweight may signify loss of muscle or bone 19-25 Healthy range ≥26 Overweight; may cause increased risk of disability or disease				
	Wt <u>154 lb</u> BMI <u>24</u>					

*Percentile categories and criterion fitness standards can be determined from tables 5.3, 5.4, and 5.5 and are illustrated in the SFT performance charts (see figures 5.2 and 5.3).

FIGURE 5.1 Sample personal profile form.

Tables 5.3 and 5.4 provide an alternative version of the normative performance standards on each of the SFT items. In these simplified tables, only the normal range of scores is given for each age group, with *normal* defined as the middle 50 percent of the scores—that is, those falling between the 25th and 75th percentiles. To use these tables, you simply check to see whether a particular score falls within the normal

range or if it is above or below the normal range. For a 72-year-old woman, a chair stand test score of 12 falls within the 10 to 15 range indicated for 70- to 74-year-olds and therefore would be evaluated as normal or typical for this age group. A chair stand test score of 9 or lower, on the other hand, would be considered below normal for 70- to 74-year-olds, while scores falling above the indicated range (i.e., 16 or higher) would be considered above normal.

TABLE 5.3 Normal Range of Scores for Women*

Test	60-64	65-69	70-74	75-79	80-84	85-89	90-94
Chair stand (# of stands)	12-17	11-16	10-15	10-15	9-14	8-13	4-11
Arm curl (# of reps)	13-19	12-18	12-17	11-17	10-16	10-15	8-13
6-minute walk** (# of yd)	545-660	500-635	480-615	435-585	385-540	340-510	275-440
2-minute step (# of steps)	75-107	73-107	68-101	68-100	60-90	55-85	44-72
Chair sit- and-reach† (in. +/-)	-0.5-+5.0	-0.5-+4.5	-1.0-+4.0	-1.5-+3.5	-2.0-+3.0	-2.5-+2.5	-4.5-+1.0
Back scratch† (in. +/-)	-3.0-+1.5	-3.5-+1.5	-4.0-+1.0	-5.0-+0.5	-5.5-+0.0	-7.0-+1.0	-8.0-+1.0
8-foot up- and-go (sec)	6.0-4.4	6.4-4.8	7.1-4.9	7.4-5.2	8.7-5.7	9.6-6.2	11.5-7.3

* Normal range of scores is defined as the middle 50% for each age group (between the 25th and 75th percentiles). Scores above the range would be considered above normal for the age group, and those below the range would be considered below normal.

** Scores are rounded to the nearest 5 yd.

† Scores are rounded to the nearest half inch.

TABLE 5.4 Normal Range of Scores for Men*

Test	60-64	65-69	70-74	75-79	80-84	85-89	90-94
Chair stand (# of stands)	14-19	12-18	12-17	11-17	10-15	8-14	7-12
Arm curl (# of reps)	16-22	15-21	14-21	13-19	13-19	11-17	10-14
6-minute walk** (# of yd)	610-735	560-700	545-680	470-640	445-605	380-570	305-500
2-minute step (# of steps)	87-115	86-116	80-110	73-109	71-103	59-91	52-86
Chair sit-and-reach† (in. +/-)	-2.5-+4.0	-3.0-+3.0	-3.0-+3.0	-4.0-+2.0	-5.5-+1.5	-5.5-+0.5	-6.5-+0.5
Back scratch† (in. +/-)	-6.5-+0.0	-7.5-+1.0	-8.0-+1.0	-9.0-+2.0	-9.5-+2.0	-9.5-+3.0	-10.5-+4.0
8-foot up-and-go (sec)	5.6-3.8	5.9-4.3	6.2-4.4	7.2-4.6	7.6-5.2	8.9-5.5	10.0-6.2

* Normal range of scores is defined as the middle 50% of each age group (between the 25th and 75th percentiles). Scores above the range would be considered above normal for the age group, and those below the range would be considered below normal.

** Scores are rounded to the nearest 5 yd.

† Scores are rounded to the nearest half inch.

“The SFT is the best tool I know of that allows for national benchmarking of program effectiveness. With only a small space, some simple equipment, and the test’s easy-to-follow instructions, our program leaders have a powerful tool for tracking the progress of their clients and programs.”

Jan Montague, president, Whole-Person Wellness Solutions,
www.wpws.net

Criterion Performance Scores

Another perhaps more important way to evaluate test scores is to compare them with criterion standards or reference points, such as the fitness cut-point scores presented in table 5.5, which indicate the level of fitness needed to remain physically mobile and independent until late in life. The data collected in our national study described in chapter 3, in addition to providing information about what is typical for women and

men at various ages, also provided information on fitness levels associated with maintaining functional mobility and physical independence. The average fitness scores for moderate-functioning women and men (i.e., those in the study who reported having no difficulty performing normal everyday activities as they aged, such as climbing stairs, walking a half mile, or carrying groceries, as indicated on the CPF scale presented in figure 3.3) provide a type of threshold value or criterion reference point that indicates having a sufficient level of fitness to maintain functional mobility and independence. See chapter 3 for additional details concerning the definition of moderate-functioning (independent) older adults and procedures for establishing the recommended fitness standards.

TABLE 5.5 SFT Scores (Criterion Fitness Standards) Associated With Maintaining Functional Mobility and Physical Independence

Test	AGE GROUPS						
	60-64	65-69	70-74	75-79	80-84	85-89	90-94
Chair stand (# in 30 sec)							
Women	15	15	14	13	12	11	9
Men	17	16	15	14	13	11	9
Arm curl (# in 30 sec)							
Women	17	17	16	15	14	13	11
Men	19	18	17	16	15	13	11
6-minute walk (# of yd)							
Women	625	605	580	550	510	460	400
Men	680	650	620	580	530	470	400
2-minute step (# of steps)							
Women	97	93	89	84	78	70	60
Men	106	101	95	88	80	71	60
8-foot up-and-go (sec)							
Women	5.0	5.3	5.6	6.0	6.5	7.1	8.0
Men	4.8	5.1	5.5	5.9	6.4	7.1	8.0

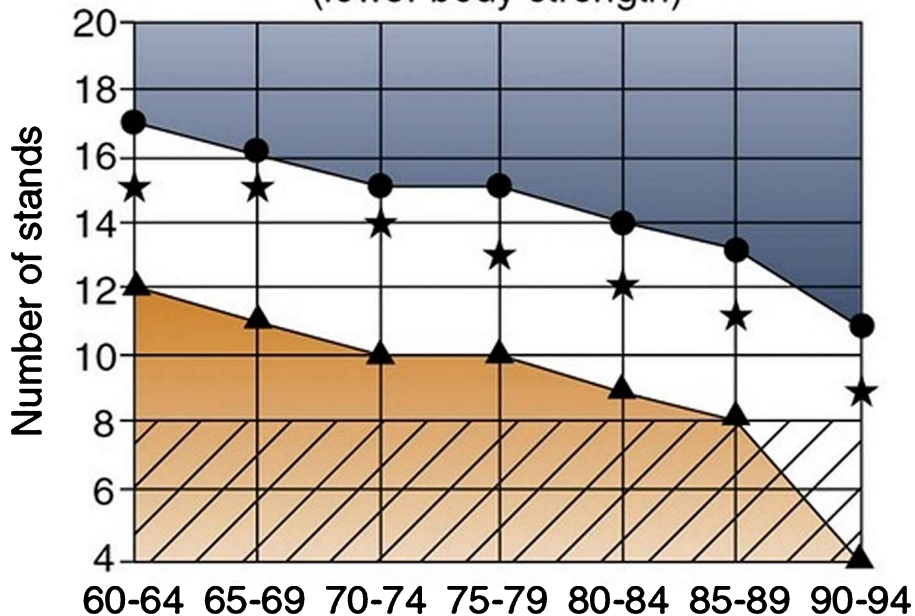
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Figures 5.2 and 5.3 provide graphical information showing above average, normal, below average, and low functioning scoring zones for each of the test items for women and men, as well as the criterion fitness scores (signified by the star symbol) that are the recommended fitness goals for maintaining functional mobility and independence. Scores falling at or above the fitness standards (star symbols) indicate that a person has met the recommended standard for his age group, a standard that was set high enough to allow for normal age-related declines in fitness level without progressing below that associated with independent functioning in later life (by the age of 90). The low functioning zone reflects the average scores of people in the normative database who were unable to perform without assistance at least 7 activities on the CPF scale, the number required for maintaining physical independence (see

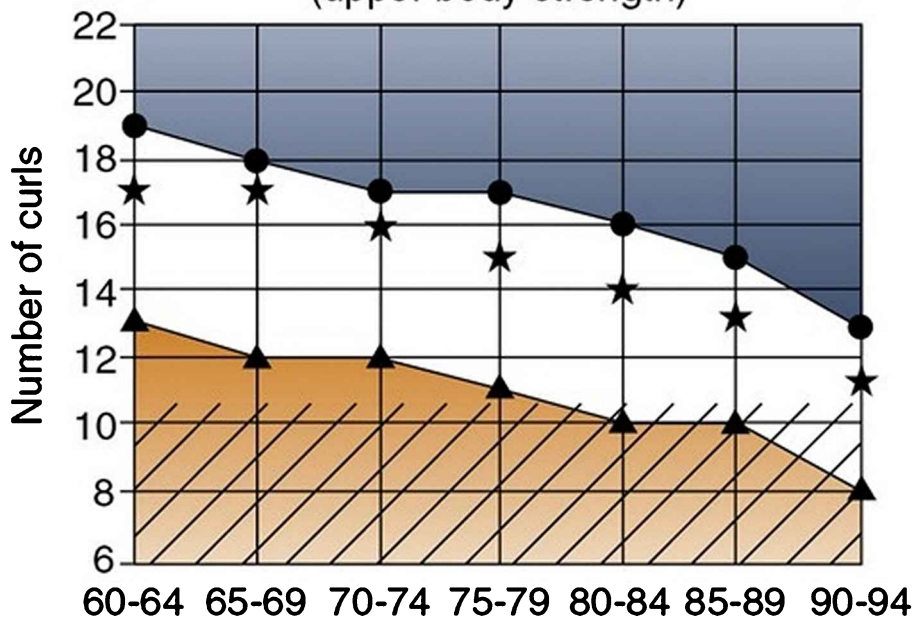
table 3.4).

Figure 5.2 SFT performance charts for women. Exact figures are reported in table 5.5.

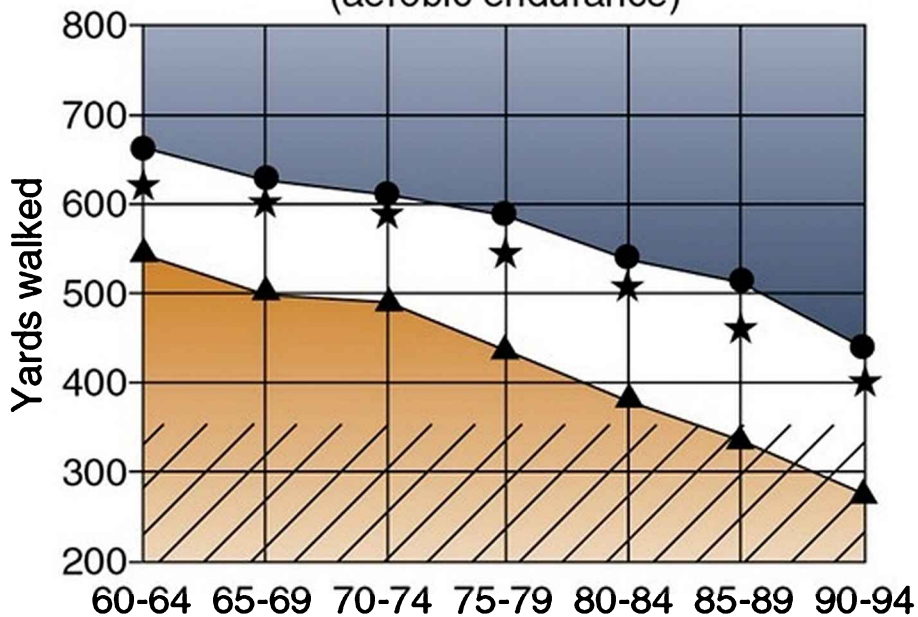
30-second chair stand—Women
(lower body strength)



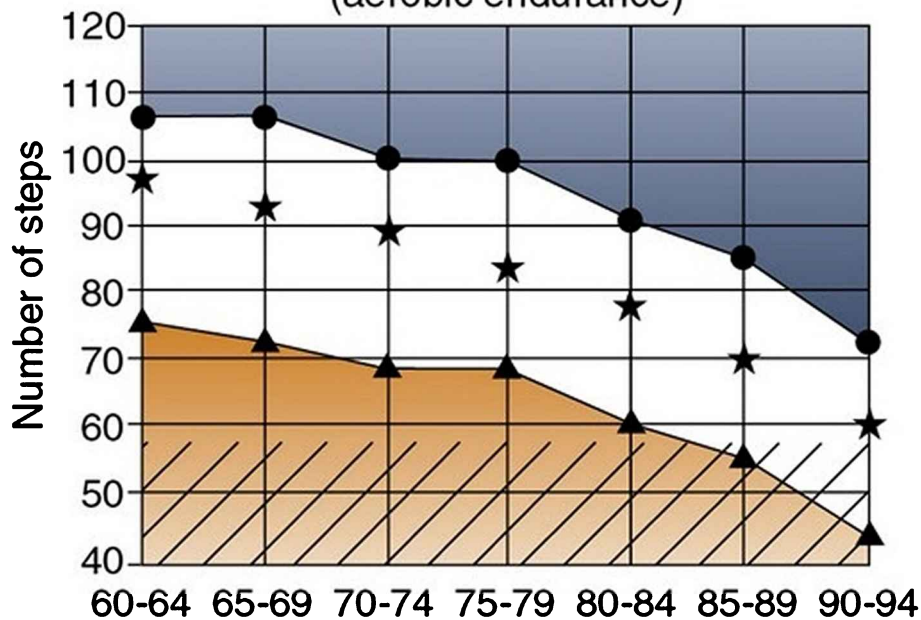
30-second arm curl—Women (upper body strength)



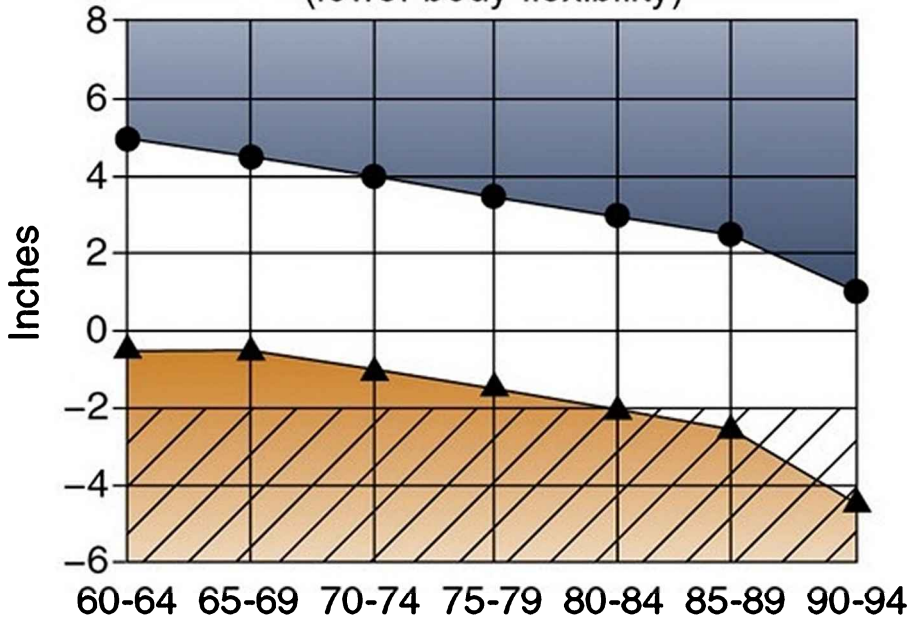
6-minute walk—Women (aerobic endurance)



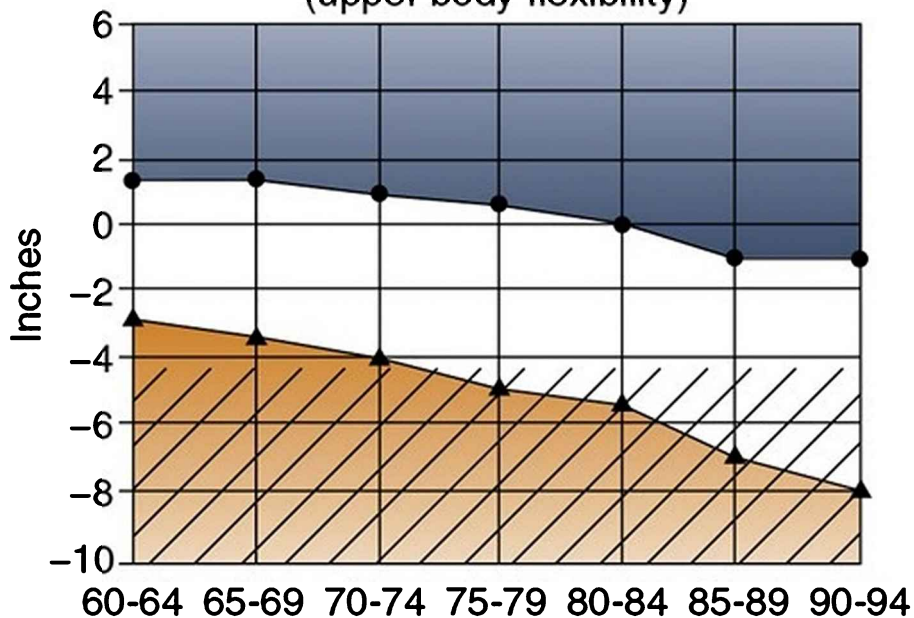
2-minute step—Women (aerobic endurance)

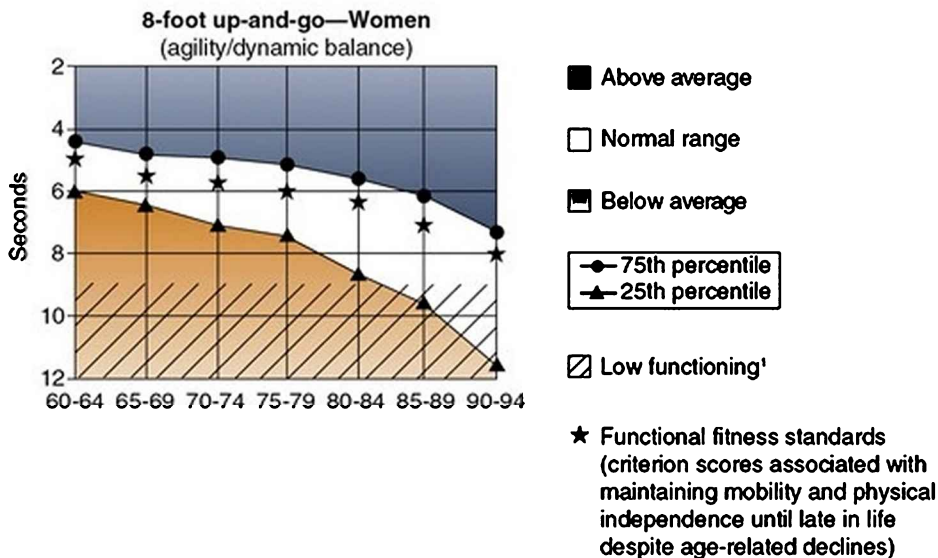


Chair sit-and-reach—Women (lower body flexibility)



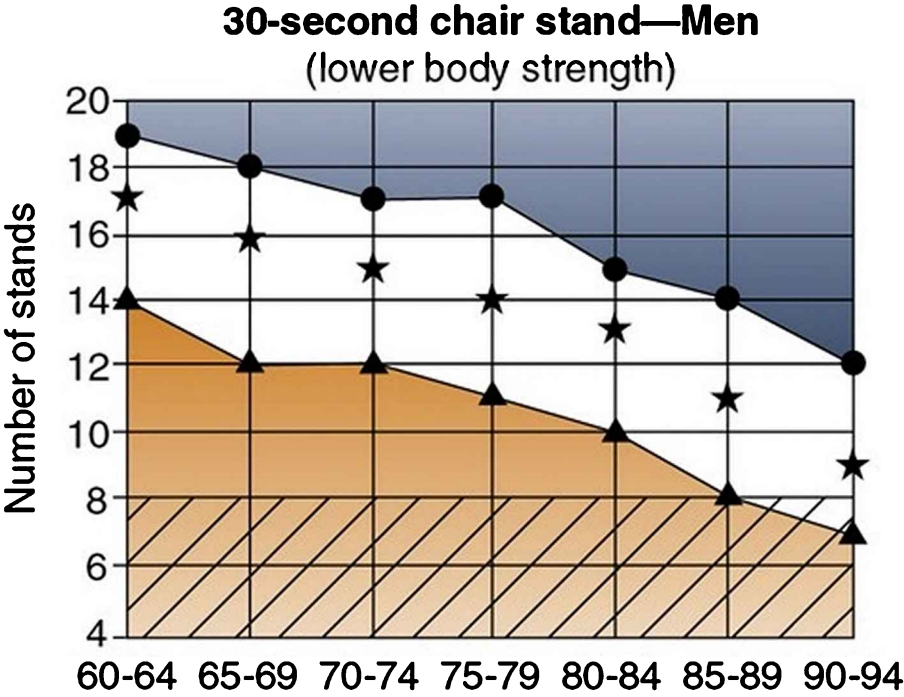
Back scratch—Women (upper body flexibility)



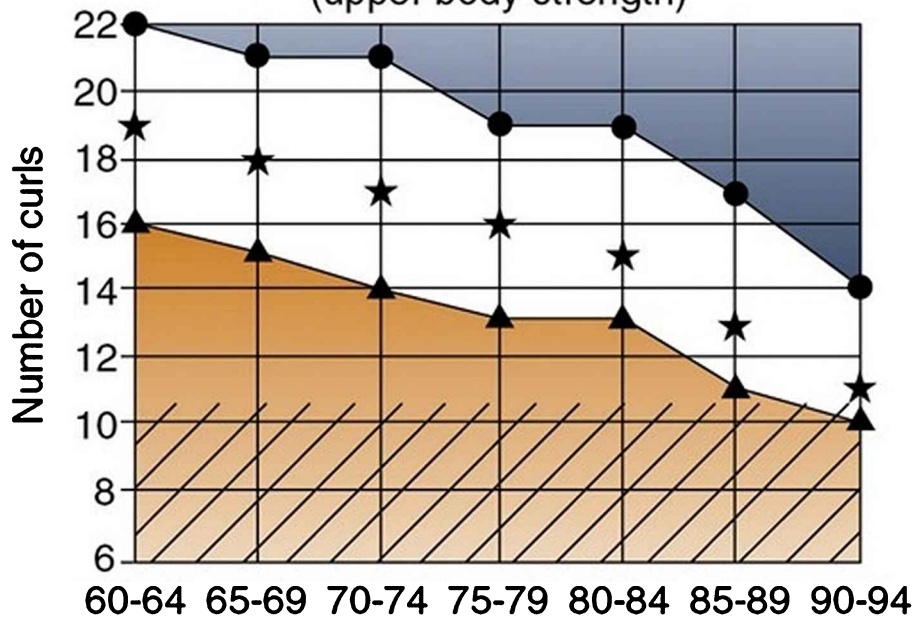


¹Reflects average fitness scores of men and women who can perform only 6 or fewer activities on the CPF scale without assistance (see table 3.4).

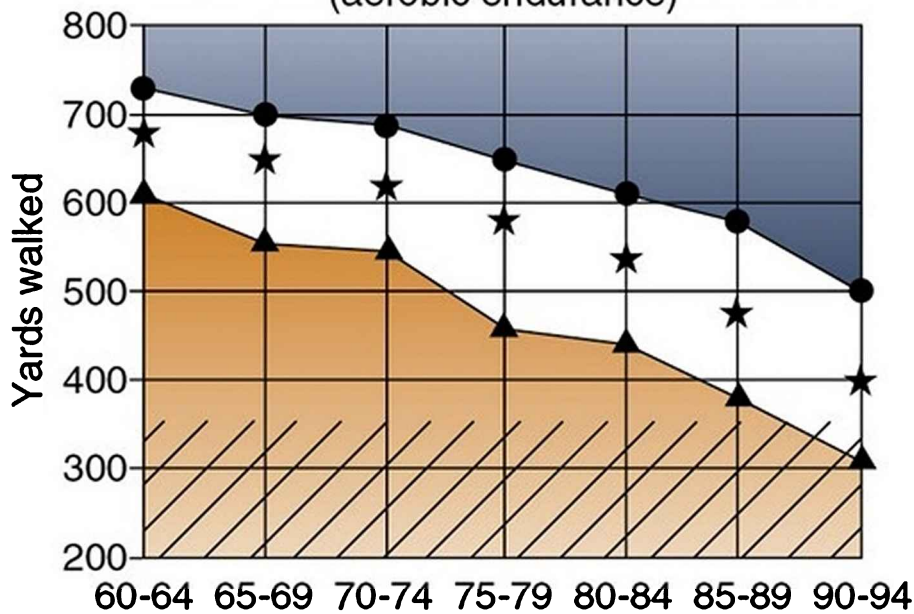
Figure 5.3 SFT performance charts for men. Exact figures are reported in table 5.5.



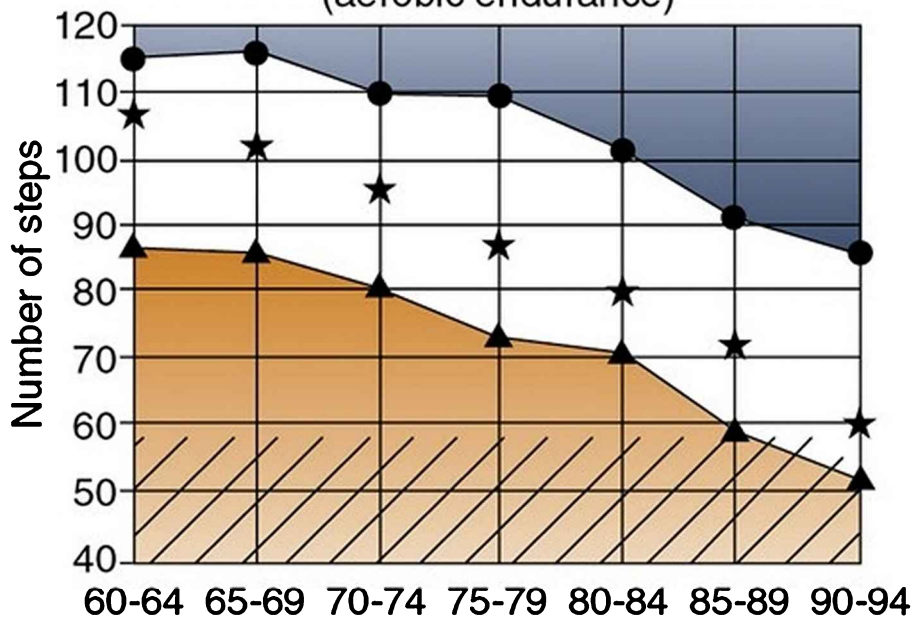
30-second arm curl—Men (upper body strength)



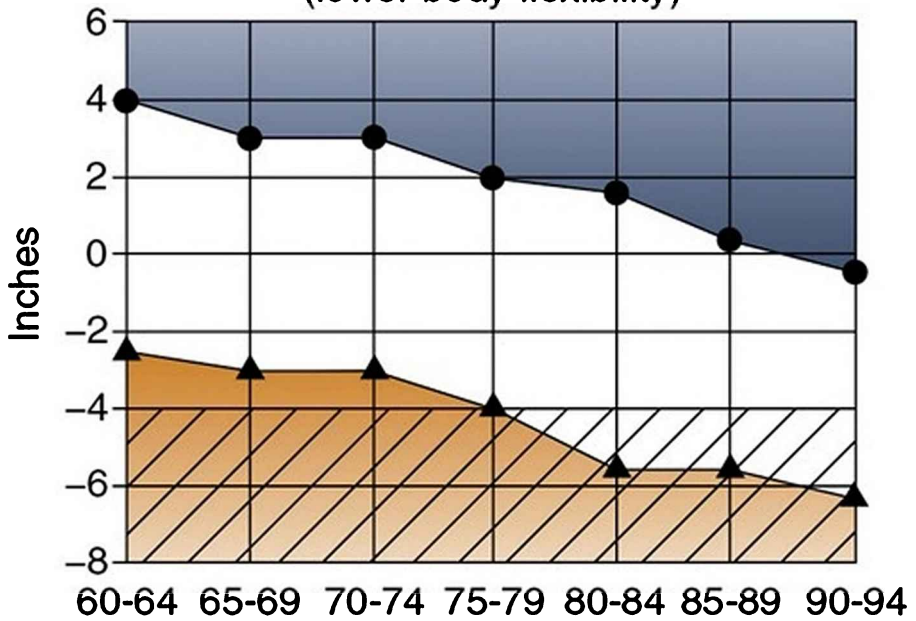
6-minute walk—Men (aerobic endurance)



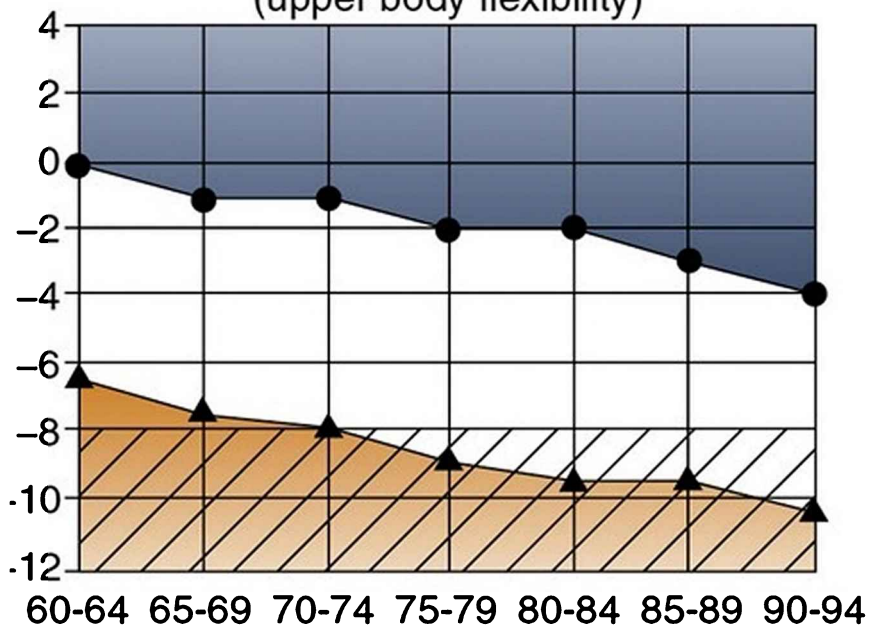
2-minute step—Men (aerobic endurance)

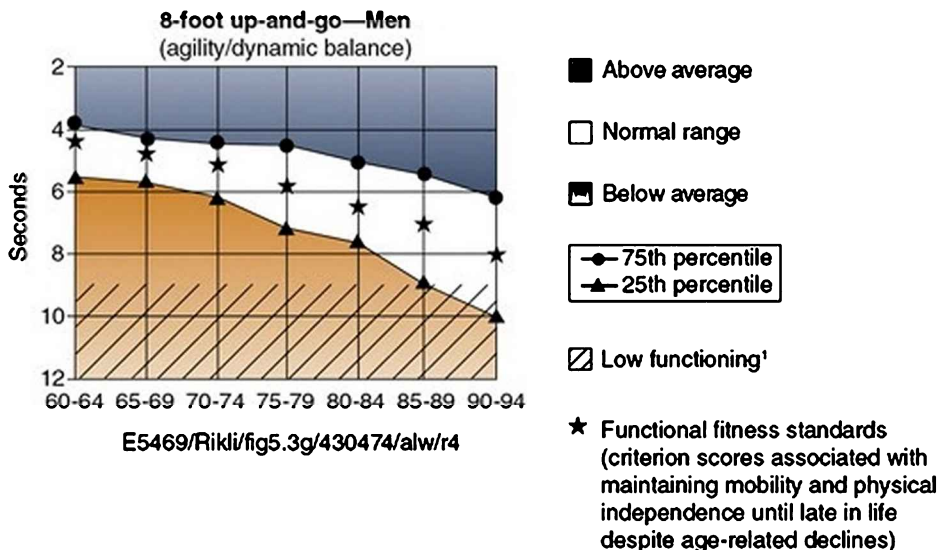


Chair sit-and-reach—Men (lower body flexibility)



Back scratch—Men (upper body flexibility)





¹Reflects average fitness scores of men and women who can perform only 6 or fewer activities on the CPF scale without assistance (see table 3.4).

Age, of course, is an important factor in interpreting one's score relative to having a sufficient level of fitness to support long-term functional mobility and independence. A chair stand test score of 11, for example, would be interpreted quite differently for a woman who is 82 years old than for one who is 62. For an 82-year-old woman, a score of 11 falls within the normal range of lower-body strength for her age group and meets the recommended fitness standard for those aged 80 to 84. For a 62-year-old woman, however, a score of 11 indicates she is below average for her age group and below the recommended fitness standard associated with having good functional mobility for her age. A 62-year-old woman who scores 15 or more chair stands, on the other hand, meets the recommended fitness standard for her age group, meaning she should have a sufficient level of lower-body strength to withstand normal age-related declines and not progress below that which is associated with independent functioning at age 90. The 25th and 75th percentile lines in figures 5.2 and 5.3 provide a general indication of the typical, anticipated

aging curves for each of the various test items.

Body mass index (BMI), another important variable in maintaining good health and mobility, can be evaluated through use of the BMI conversion chart. Although an optimal BMI range for older adults has not been determined, various sources suggest that BMI values above 25 or below 19 may be associated with increased risk for disease and disability in later years (ACSM, 2010; Shephard, 1997). Although BMI values provide useful body weight guidelines for most people, they should be interpreted with caution in older adults since unknown changes in bone and muscle loss can result in an increased chance of misclassification in this age group.

Precautions in Interpreting Performance Standards

The recommended fitness standards presented in table 5.5 and in the performance charts (figures 5.2 and 5.3) should be thought of as general performance guidelines rather than as absolute predictors of functional mobility level. Certainly, additional research is needed to validate these data. However, because of the large number of people tested in our study, we do think these data provide useful, previously unavailable reference points (standards) for evaluating fitness levels in older adults relative to their ability to remain physically mobile and independent until late in life.

Test users are reminded that the cut-point scores used to identify the recommended fitness standards represent group averages and may not apply equally to all people. For example, people who are very short (or perhaps very tall) may not have scored well on some of the test items because of their size, but they may in fact be maintaining their fitness and functional ability quite well. With test directions standardized for all participants, such as that requiring a 17-inch (43 cm) seat height for the chair stand, it could be that a very short or very tall person, for example, may not score as well as those of more average height because of the extra difficulty of getting up and down from the chair seat.

Also, when using the tables and charts to assist with interpretation of

scores, keep in mind the overall characteristics of the study population. As indicated in chapter 3, those participating in the study were independent-living volunteers who were fairly active, 89 percent Caucasian, and fairly well educated. Additionally, it is important to note that the data in the tables represent *overall means* and may not be representative of individual subgroups of older adults. It is not clear, for example, how these scores would compare with those of different ethnic and racial groups, groups from other countries, less motivated (e.g., nonvolunteer) participants, or people with multiple chronic health conditions.

Methods of Providing Feedback to Participants

Of the mechanisms discussed earlier (tables, charts, personal profile forms, and computerized printouts), the ones that work best for explaining test results to your participants will depend on each specific situation. If you have given the SFT to a large group and have little time to provide individualized feedback, it will probably work best for you to return the individual scorecards (after you've made copies or entered the scores into the Senior Fitness Test Software 2.0). Participants can then compare their scores against posted standards, using either the normal range of scores tables (tables 5.3 and 5.4), the criterion fitness standards in table 5.5, or the SFT performance charts. By referring to the charts, participants can compare their scores with those of other people their age to find out whether their performance on each test is considered normal, above normal, or below normal and whether or not they have met the recommended fitness standards for their age. Before giving back the scorecards, we recommend that you meet with your group to explain what the performance standards represent and how to use the tables and charts. It is also important at this time to assure participants, particularly those with low scores, that fitness can be improved regardless of a person's age or current condition.

If time and resources permit, more meaningful feedback can be provided to clients through individualized personal profile forms with the reports generated from the Senior Fitness Test Software 2.0. These forms provide clients with the information they need to interpret their scores.

Raw scores are listed for each test item along with the corresponding percentile ranks, the evaluation category (normal, above normal, or below normal), and whether or not the recommended fitness scores have been met. It is important to point out that scoring within the normal range does not necessarily mean that a person's score is good. Considering that many older adults in the normative data base were quite inactive, the average scores in this distribution are often below the fitness goal that is recommended in order to maintain physical independence.

Using these forms makes it easy to evaluate a person's relative strengths and weaknesses within the fitness categories (i.e., what the client is best or worst at) and makes it easy to compare scores from one testing time to the next. In the SFT Software 2.0, the program evaluates each person's scores relative to the established standards and provides clients with easy-to-read charts explaining their scores. The software also facilitates ongoing record keeping of your clients and provides you with handy group statistics for evaluating the effectiveness of your total program. See the sidebar for more information on the reporting options included with this software.

Again, regardless of the feedback method used, remember it is critically important for you to help clients understand that *no matter what their age or current condition, improvement is always possible!* Never just assess clients and give them feedback about their fitness, especially if their scores are low, without giving them hope and encouragement at the same time. Instead, explain to them that numerous studies show it is *never too late* and that people are *never too old* to improve their fitness, and you will help them plan exercise activities to improve their performance if they so desire. We also suggest that you help clients understand that although the performance standards (charts) make it possible for them to compare their scores with those of others their age, the more important concern is how their own scores change over time. This is especially important for people with low performance scores so they won't become discouraged. Suggestions for motivating participants to increase their physical activity level and improve their performance are discussed in the next section.

Reporting With the Senior Fitness Test Software 2.0

The Senior Fitness Test Software 2.0 provides several reporting functions to help you organize and interpret test results as well as provide individualized feedback to participants. The software allows you to do the following:

- Display basic information for each participant and export participant lists.
- Generate individual test result reports that compare a participant's scores to the normative ranges for his or her age and sex, as well as to the criterion-referenced standard scores he or she should target to remain functionally independent.
- Graphically display the participant's scores on a continuum of normative values for his or her age.
- Graphically display a participant's scores from several testing sessions in a report that tracks individual progress over time.
- Generate group performance reports for your own use that display the number of participants in a testing session, participants' individual scores on a given test, and the range of scores within the group. The report also calculates average age of participants, average score, and standard deviation to help you interpret the results and understand the needs of your participant group.
- Print information sheets on important fitness topics to distribute to participants.
- Transfer data to or from Microsoft Excel.

To access the Senior Fitness Test Software 2.0, go to <http://sft.humankinetics.com>. If you have already purchased a subscription, you can follow the link on this page to enter your key code. If you have not purchased the software, follow the links on this page to purchase a subscription.

Using Test Results to Motivate Participants

Almost everyone recognizes that adequate physical activity is important for optimal health and functioning, yet few people (less than 20 percent of older adults) get the amount of activity they need. According to exercise experts, an important first step in motivating people to become more active is to assess their fitness level and provide individualized feedback, presuming, of course, that people are at the stage of at least contemplating becoming more active. In fact, Dr. Kenneth Cooper, the founder and medical director of the world-renowned Cooper Institute in Dallas, designed his entire program on the premise that evaluation is the most powerful motivator for getting people to improve their fitness level. According to Cooper, and according to our own experiences in working with older adults, the steps to success in getting people to change their activity behaviors are as follows:

- *Evaluation*—assessing people’s current level of fitness and identifying strengths and weaknesses
- *Education and motivation*—helping people understand why fitness is important in their lives
- *Goal setting and program planning*—planning and carrying out a relevant program based on individual goals designed to meet individual needs and interests
- *Monitoring progress and reevaluation*—checking participants’ progress and adjusting the program as needed

If you have given the SFT to your clients, you already have completed step 1 of the steps to success, that of evaluation. For many people, simply having their fitness level assessed motivates them to increase their activity level and improve their fitness. The remainder of the chapter describes procedures for addressing the steps subsequent to evaluation.

Educating and Motivating Participants

After administering the SFT to your clients or program participants, we suggest that you meet with them, either individually or as a group, to

give them their feedback. This provides an ideal time to educate and motivate about the importance of being physically active and fit in later life. You especially should point out the strong relationship between people's activity level and their fitness scores (surprisingly, many people are not aware of this) and the fact that as much as half of the typical decline associated with aging may not be due to aging at all, but rather to disuse—that is, people's tendency to become less and less active as they age.

Point out to your clients that as people are living longer, it is becoming increasingly important that they pay attention to their fitness level if they want to remain healthy and independent during their later years. Learning that participation in physical activity can help delay the onset of physical frailty and extend their physical independence can be very motivating. Statistics indicate that many older adults by their late 70s or early 80s, particularly those with sedentary lifestyles, will lose the strength and endurance they need to perform common everyday activities, such as climbing stairs, walking to the store, and taking care of their own personal and household activities, unless steps are taken to increase their activity level. For many, maintaining a physically active lifestyle can easily add 10 years or more to their functional or active life span, thus significantly delaying the onset of physical frailty.

It is especially critical for older people to understand the importance of preventing the vicious cycle that occurs for so many—the cycle that begins with becoming less active as they age, which in turn leads to lower energy levels (less desire to be active), which then results in even further reductions in activity, then further declines in energy levels, and on and on. Eventually, this downward spiral can result in people's strength and endurance declining to the point that it is hazardous to their health and their ability to carry out normal everyday activities. Scores obtained on the SFT items can provide important feedback about the strengths and weaknesses of older adults and whether or not their physical declines are placing them at risk for losing their functional mobility.

The good news that you can share with your clients, however, is that no matter what their age or current physical condition, it is always possible to improve their level of fitness by increasing their activity level. You

can tell them, citing research reported in national and international physical activity guidelines, that people of all ages—even into their late 80s and 90s—have experienced significant gains in fitness after beginning exercise programs, gains that have led to improved functional performance (e.g., walking and balance) and, for some, the ability to discard their canes and walk without assistance. Such guidelines have recently been developed by the World Health Organization and by researchers in the United States, Canada, the United Kingdom, and elsewhere (ACSM, 2009; Canadian Society for Exercise Physiology, 2011; U.S. Department of Health and Human Services, 2008; UK Department of Health, Physical Activity, Health Improvement and Protection, 2011; World Health Organization, 2010).

You can also motivate your clients by sharing success stories from your own program. Older people are interested in the activities of their peers and are often motivated by their successes. In our program, for example, a 79-year-old woman who had several chronic health conditions and was on 13 medications scored below average on most test items when she first came in to be tested. She was so surprised at how poorly she scored compared with others her age that she immediately hired a personal trainer. Six months later, when she came back for a reevaluation, she not only scored in the normal range on most tests but was also down to taking only 4 medications and saying she felt like a new person. She also looked like a new person—full of energy and smiles. Certainly we are not suggesting that exercise can be used to cure medical problems, but we do know it can help in managing numerous conditions.

“The Senior Fitness Test (SFT) battery is quick, simple, noninvasive and very functional—all important considerations for our senior participants. It also proves to be a great tool for motivation as it offers a thorough report highlighting how participants’ scores compare to others of their same age and gender and also to their own previous test scores.”

Clara Fitzgerald, director, Canadian Centre for Activity and Aging,
London, Ontario

Even relatively small success stories can be interesting and meaningful.

One woman, after 8 weeks in our program, was thrilled when she found she had gained enough upper-body strength to be able, for the first time, to pull down the back door (hatch) in her minivan without having to ask for help. Another woman reported a tremendous gain in self-confidence when her lower-body strength (and balance) improved to the point she could get up from the floor easily without help, something she had been unable to do for some time. We could go on and on with stories about how increased physical activity level has improved people's quality of life.

Beginning or modifying an exercise program, however, generally takes more than a motivational talk about the value of exercise and a few motivational stories. Changing behavior in people is difficult and almost always requires applying some type of behavior modification strategy. One behavior modification technique that has been successful in changing exercise behavior, especially for those who have already expressed a desire to change, is goal setting.

Goal Setting and Program Planning

Goal setting is an essential step for anyone wanting to make a change in behavior. Establishing goals, especially written goals, tends to move people from a good-intentions or wishful-thinking level to an action-plan level. As an exercise leader, you should encourage your clients to participate in goal setting, but you should *not* set their goals for them. People are much more likely to work toward goals *they* think are important rather than goals that have been established by others, unless, of course, these can be personalized and seen as important to them as individuals. In helping your clients with goal setting, the following procedures and considerations may be helpful. Specifically, you should encourage your clients to do the following:

1. **Identify long-term goals or major exercise objectives.** You can expect great variation in what people hope to accomplish in their exercise programs. Some may be interested in addressing a specific weakness identified in their SFT results, while others may just want to maintain their current mobility and energy level. Others may be interested in managing a health condition (e.g., reducing

their blood pressure, controlling their diabetes). Some may want to lose weight, improve their appearance, or reduce their risk of falling. Still others may want to improve their performance in a particular sport, such as golf or running. Also, some people's goals may be very specific, such as losing 10 pounds (4.5 kg) before a granddaughter's wedding the next summer, whereas others may be more general. For the goal-setting process to be most effective, goals should be recorded in writing, perhaps using a form similar to the Personal Goals and Activity Plans.

2. **Identify short-term goals (1- or 2-week activity plans).** Long-term goals are more likely to be attained if broken down into more immediate short-term goals. Short-term goals should be realistic, measurable, and time specific, and they should not extend beyond 2 weeks. They should be expressed as specific activities clients think will move them toward their long-term goals. Some suggest a goal is realistic only if it meets the 90 percent rule—that is, it is a goal the client is 90 percent sure of being able to achieve within the time stated. A measurable goal (or activity) is one that is observable. A goal of exercising more is not observable, but a goal of increasing walking time to at least 30 minutes a day at least four times a week is observable. Being time specific relative to one's goals means planning the days and times that the activity will take place, such as walking every Monday, Wednesday, and Friday morning at 9:00 a.m., or performing 10 minutes of stretching exercises each evening during the 6 o'clock news. Scheduling activity into one's day or week (i.e., putting it on the calendar) increases the likelihood that it will occur. Short-term goals (action plans) and timetables should be recorded on the client's goal-setting form along with long-term goals (see Personal Goals and Activity Plans).
3. **Identify potential obstacles, and plan strategies for overcoming them.** Inevitably there will be obstacles and setbacks in carrying out one's activity goals, so it is important to try to predict in advance what these might be and plan possible strategies for dealing with them. Most people can predict what the most likely obstacles will be for them. It may be bad weather, interruptions from friends or relatives, or simply a lack of self-discipline. If weather frequently interrupts walking plans, perhaps backup plans can be made for going to the mall or to a fitness center on bad-

weather days. If procrastination or lack of self-discipline is a major barrier, it may help to schedule exercise sessions with a partner. The extra social pressure and expectations from partners, as well as the added enjoyment of exercising together, can help people stay on target with their plans. Projected obstacles and the strategies for overcoming them should be spelled out clearly as part of the goal-setting process (see Personal Goals and Activity Plans).

Monitoring Progress and Reevaluation

Another important step in changing physical activity behavior is monitoring progress and reevaluating. To promote program adherence, you might suggest to your clients that they discuss their goals and plans with a friend, with a family member, or with you, their instructor, and then ask this person to sign the goals contract as a witness to their plans. Participants might even ask their witnesses to question them about their progress at least once a week.

For some people, it can also be motivating to keep a daily or weekly exercise log to track and record their progress toward their long-term goals (such as losing 10 pounds) and to monitor their adherence to weekly activity plans. In fact, it is important to recognize that increasing one's activity level is a worthy accomplishment by itself and often leads to other unexpected rewards (apart from the original goal), such as feeling and looking better, having improved self-confidence or self-efficacy, sleeping better, and having a higher level of energy and ambition. Interestingly, these supplemental, more intrinsic rewards are often most responsible for keeping people active long after their original goal has been met (i.e., the 10 pounds have been lost and the granddaughter's wedding is over).

The challenging part, however, is that it takes time (sometimes many weeks) for a person to reach the stage at which the intrinsic rewards associated with physical activity and fitness (e.g., the joy of being active, the sense of well-being, and the feeling of accomplishment) become sufficient motivation for maintaining an exercise program. In the meantime, you should encourage your clients, especially your new clients, to continue with the goal-setting activities described earlier and

also to consider keeping an exercise log of their activities. Figure 5.4 contains an example of a fairly generic weekly activity chart that should be adaptable for most people. Although people's exercise goals and plans may vary considerably, most will be best served by planning activities in the general categories indicated.

Activity Record

Week of

Name

Record number of minutes per day:	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Lifestyle activity: Indicate any moderately strenuous housework, yardwork, recreation, sports, and so on. If you are low-active, add structured exercise (see below).		Cleaned patio, some yardwork (30 min)			Raked leaves (30 min)		Cleaned house (vacuumed, mirrors, baths, and so on) (1.5 hr)
Structured exercise (aerobic): Brisk walking, jogging, aerobic exercise, cycling, treadmill, and so on. (Need at least 20 min, 3-5 times a week.)		Brisk walk (30 min)		Brisk walk (30 min)		Rode bicycle at senior center (20 min)	
Structured exercise (strengthening): You can use elastic tubing, hand weights, weight machines, or calisthenics. (Work upper and lower muscles at least twice a week.)			20 min (upper and lower body)		20 min (upper and lower body)		
Daily totals: Are you getting 30 to 40 minutes of moderate exercise on most days?		60 min	20 min	30 min	50 min	20 min	90 min
Also important: Flexibility and stretching activities You should stretch all muscle and joint areas 2 or 3 times a week, preferably daily.		Lot of stretching after walking				Flexibility and relaxation exercises before bedtime	
Agility and balance activities: These are especially important for those experiencing loss of balance.							

FIGURE 5.4 Sample activity record

The participants' progress toward their goals should also be monitored through periodic reevaluation of fitness parameters and other relevant indicators. Readministering the SFT at regular intervals at least twice a year provides an indication of progress with respect to changes in strength, endurance, flexibility, mobility, balance, and body weight. Changes in other indicators such as blood pressure, blood sugar readings

(if the person has diabetes), golf scores, or self-diagnosed energy levels and feelings of well-being also provide evidence of progress toward goals.

Reports generated by the Senior Fitness Test Software 2.0 can display results from multiple test sessions for an individual to track progress over time. You can also use the software to generate group reports in order to evaluate the needs and progress of an entire group or a specific sex or age range within the group. To learn more, visit <http://sft.humankinetics.com>.

“The SFT assessments help me show progress and set future goals to improve my whole body. I’m always eager for my next 6-month assessment to see which areas I’ve improved upon and which ones still need work.”

Juanita Nichols, age 80, Crestwood Village West, Indianapolis

As clients are setting their goals and planning their programs, they may ask you for guidance in helping them determine what specific activities will best help them reach their goals. Chapter 6 addresses a number of relevant considerations in helping your clients improve their performance, plus it provides specific exercise recommendations.

Summary

The performance tables and charts specifically developed for the SFT can assist professionals in the interpretation of test scores. Data from a nationwide study of 7,000 older adults provide the basis for both norm-referenced and criterion-referenced performance tables as follows:

1. Percentile norm tables provide 5-year age-group percentile ranks for women and men separately on each of the test items, allowing participants to compare their scores with those of others of the same age and sex (see the percentile tables for the Senior Fitness

Test).

2. Two tables provide a simple version of the normative performance scores, giving only the normal ranges of scores (middle 50 percent) for women and men on each test item (see [tables 5.3](#) and [5.4](#)).
3. A table shows recommended fitness standards for older adults that are associated with maintaining adequate functional mobility throughout the life span (see [table 5.5](#)).
4. SFT performance charts provide a graphic display of data showing scoring zones that indicate above average, normal, below average, and low functioning, as well as recommended standards (fitness goals) associated with maintaining functional mobility and independence until late in life (see [figures 5.2](#) and [5.3](#)).

In interpreting test scores, test users are reminded that the performance standards are based on volunteer independent-living older adults throughout the United States who tend to be healthier and more active than the population at large over the age of 60.

Administering the SFT can be an effective way of motivating clients to increase their activity participation. For many, just the process of being evaluated is motivating and can cause positive changes in behavior. Test results can also be used as the basis for individual goal setting and program planning. For some, motivation can be further enhanced by keeping an activity log tracking both program adherence and physical progress. Finally, reevaluation (readministering the SFT at regular intervals) is important in maintaining clients' motivation and interest in their level of physical activity and fitness.

In the next chapter we discuss exercise guidelines for older adults and recommend specific exercises that can help improve performance on each of the SFT items.

Chapter 6

Exercise Recommendations for Older Adults

Improving Senior Fitness Test Scores

Contributing author: Jeana French Miller



In recommending specific exercises to help your clients improve their functional fitness (defined as having the physical capacity to perform normal everyday activities safely and independently without undue fatigue), several factors should be taken into account. In addition to considering their current fitness needs as indicated by their SFT scores, as well as any personal exercise objectives they have (such as losing

weight or managing a health condition), it is important to consider clients' exercise and activity preferences. We often hear that the best exercise to improve performance is *the one that people will do*, suggesting you also should help your clients determine which types of exercise options are most likely to work for them—that is, the ones that best fit their personalities and meet other personal and environmental needs (e.g., access to facilities, financial considerations, transportation, social support). Various exercise options can include exercising alone versus in groups, exercising at home versus going to an exercise class or a fitness center, and engaging in structured exercise activities versus incorporating exercise into daily routines. Other options involve developing an active hobby; doing more yardwork or housework; or getting exercise through various recreational, social, dance, or sport activities. In helping clients plan effective exercise programs, it is also important to consider the general exercise guidelines and recommendations that have been developed by professional experts.

In this chapter, we first review recently published physical activity and exercise guidelines that apply to older adults. We then provide specific recommendations for increasing lifestyle physical activity and for performing structured exercises that can help improve SFT scores. Specific topics include

- physical activity and exercise guidelines for older adults;
- lifestyle exercise;
- guidelines for structured exercise to improve SFT scores; and
- exercises to improve strength, flexibility, agility, and balance.

Physical Activity and Exercise Guidelines for Older Adults

A number of recent publications such as the Physical Activity Guidelines for Americans (U.S. Department of Health and Human Services, 2008), the Canadian Physical Activity Guidelines (Canadian Society for Exercise Physiology, 2011), and statements from the American College of Sports Medicine, the American Heart Association (ACSM, 2009; Nelson et al., 2007), the UK Department of Health, Physical Activity, Health Improvement and Protection (2011), and the World Health

Organization (2010) provide updated guidelines describing recommended levels of physical activity for people of all ages, including older adults, based on strong evidence linking physical activity with improved health and reduced disease, disability, and all-cause mortality. These nationally and internationally recognized guidelines generally suggest that all adults engage in the following: (1) at least 150 minutes of moderate-level (or 75 minutes of vigorous-intensity) aerobic exercise spread over at least 5 days of the week; (2) muscle-strengthening activities that work all major muscle groups (legs, hips, back, abdomen, chest, shoulder, and arms), performed at least 2 days a week; and (3) balance exercises as needed. In addition, although most guidelines do not make specific recommendations regarding the amount of flexibility exercise needed, flexibility is mentioned as being important for maintaining functional mobility and as something that can be improved through range of motion and stretching exercises. In particular, it is recommended that people take an extra 10 minutes each day when performing aerobic or strength exercises to stretch all major muscle and tendon groups (Nelson et al., 2007).

“Although no amount of physical activity can stop the biological aging process, there is evidence that regular exercise can minimize the physiological effects of an otherwise sedentary lifestyle and increase active life expectancy by limiting the development and progression of chronic and disabling conditions.”

ACSM Position Stand (ACSM, 2009)

Moderate aerobic exercise (defined as that which burns at least 150 calories of energy per day, or 1,000 calories a week) would mean (for most older adults) something equivalent to 30 to 40 minutes of brisk walking per day. The sidebar lists examples of other activities providing moderate amounts of exercise. For even greater benefits, the 2008 Physical Activity Guidelines for Americans recommend that older adults increase moderate-intensity aerobic activity to 300 minutes (5 hours) or perform 150 minutes (2.5 hours) of vigorous-intensity aerobic exercise spread throughout the week, especially if the goal is weight reduction.



Examples of Activities That Provide Moderate-Level Exercise (Using 150 Calories of Energy)

Washing and waxing a car for 45 to 60 minutes
Washing windows or floors for 45 to 60 minutes
Gardening for 30 to 45 minutes
Pushing self in a wheelchair for 30 to 40 minutes
Walking 1.75 miles (2.8 km) in 35 minutes (20 minutes/mile)
Walking 2 miles (3.2 km) in 30 minutes (15 minutes/mile)
Raking leaves for 30 minutes
Pushing a stroller 1.5 miles (2.4 km) in 30 minutes
Bicycling 5 miles (8 km) in 30 minutes
Bicycling 4 miles (6.4 km) in 15 minutes
Doing water aerobics for 30 minutes
Swimming laps for 20 minutes
Running 1.5 miles (2.4 km) in 15 minutes (10 minutes/mile)
Stair walking for 15 minutes
Dancing socially for 30 minutes
Shoveling snow for 15 minutes

For the majority of older adults who have not been active on a regular basis and have already experienced some degree of age-related or disease-related physical decline, effective activity programs typically require more prescriptive and targeted exercise than what has been recommended in the more generic physical activity guidelines. Recognizing that there are physical capacity thresholds for performing such functions as walking, stair climbing, and rising from a chair, it is important that older adults engage in exercise programs that not only promote health and reduce risk for disease but also address special areas of physical weakness that may cause them to be at risk for losing their physical independence. Table 5.5 and figures 5.2 and 5.3 provide information on the fitness standards (scores) needed for maintaining mobility and physical independence that can address specific areas of weakness.

Again, in recommending specific exercises for people, the only ones that will be effective are *the ones they will do!* With that in mind, we recommend explaining to your clients that they can improve their fitness level (and test scores) in three ways: (1) by incorporating additional physical activity into their normal daily routines (sometimes referred to as lifestyle exercise); (2) by scheduling time each week to engage in structured types of exercise—that is, exercises designed to address specific fitness components, such as strength or endurance; and (3) by engaging in a combination of lifestyle and structured exercises. Each of these methods has advantages. For some, the advantage of lifestyle exercise is that it also has other purposes (e.g., walking the dog, doing yardwork) and doesn't seem like *real* exercise. The advantage of structured exercise is that it focuses on specific aspects of fitness and is especially effective in addressing any special needs or weaknesses that may have been identified during the client's fitness assessments. The next sections include additional suggestions relative to each of these categories—lifestyle exercise and structured exercise.

Lifestyle Exercise

Most people could significantly increase their activity level and improve their fitness simply by building more activity into their daily lives. Every day there are numerous opportunities to become more active just by altering normal routines and habits—for example, by taking the stairs instead of the elevator, by walking more and driving less, and by doing more housework and yardwork. We especially suggest that you try instilling in your older clients the idea that moving more and sitting less are good for them. Considering that the “use it or lose it” phenomenon becomes increasingly real as we age, the best thing we can do to ensure continued functional ability is to stay as active as possible. The following are just a few examples of how people could significantly increase their daily energy expenditure:

- Taking the stairs instead of an elevator or escalator
- Walking the dog more often
- Walking to the store instead of driving
- Bicycling to a friend's house
- Cleaning the garage or car

- Doing more housework and yardwork
- Playing with grandchildren
- Joining a hiking club or dance group
- Volunteering for active projects
- Parking farther away
- Planning active vacations (e.g., fishing, walking or bicycle tours)
- Walking and using a pull golf cart for clubs instead of riding a cart
- Taking up an active hobby (e.g., dancing, gardening)
- Picking up litter in public places (e.g., parks, beaches)

These lifestyle activities can help maintain functional fitness and may even result in an *increased* level of fitness if there is a substantial increase in activity expenditure over what a person has been accustomed to. Additional benefits, however, can be gained from engaging in structured exercises that specifically address each of the major underlying physical fitness parameters (aerobic endurance, strength, flexibility, and balance) associated with functional independence. The next section provides specific exercises to improve the underlying physical fitness parameters for each of the SFT items.

Guidelines for Structured Exercises to Improve SFT Scores

In addition to incorporating as much activity as possible into their daily routines (i.e., limiting sitting and inactivity as much as possible), most older adults could benefit from setting aside time each day for structured exercise, thus ensuring a well-rounded program of activities that addresses all aspects of physical fitness. Structured or focused exercise is especially important for clients who received low scores on the SFT and need to improve their fitness in one or two specific categories. This section provides information on exercise recommendations to improve test scores for SFT items, each of which is associated with a major component of physical fitness. Specific physical activity and exercise training guidelines and precautions associated with various musculoskeletal, cardiovascular, pulmonary, and metabolic disorders and frail populations are beyond the scope of this chapter. Therefore, we recommend other publications such as *Physical Activity Instruction of Older Adults* (Jones & Rose, 2005), *ACSM's Guidelines for Exercise*

Testing and Prescription (2009), and *ACSM's Exercise Management for Persons With Chronic Diseases and Disabilities* (2009). In addition, it is always advisable to remind clients of the signs of overexertion (discussed in chapter 4) and when exercise should be stopped or is not advisable. You can also refer to the recommended adaptations for special populations when administering the SFT, as discussed in chapter 4.

Improving Aerobic Endurance

Aerobic exercise improves the function of the heart, lungs, and blood vessels and helps people feel more energetic. It is helpful to remind clients that maintaining an adequate level of aerobic endurance is critical for physical independence and for reducing risk of cardiovascular disease, diabetes, obesity, and cognitive problems. In the SFT, a person's level of aerobic endurance is measured using either the 6-minute walk test or the 2-minute step test (depending on the setting). Your clients can best improve their aerobic endurance scores by engaging in large-muscle rhythmic activities such as walking, jogging, hiking, stair climbing, and cycling. For most older adults, walking is an ideal form of aerobic exercise, not only because it contributes to one's aerobic fitness but also because of its importance in everyday functioning. Of course practicing the test items is a form of exercise and is also a great way to improve scores. For people whose aerobic endurance scores were exceptionally low on the SFT, we suggest they work on their lower-body strength (see next section) before, or at the same time as, working on their aerobic endurance. For people with balance problems, it is often best to focus on balance and agility training first and lead up to aerobic training within a few weeks. However, if exercise equipment that is safe for people with balance problems (such as a stationary bicycle) is available, aerobic endurance training and balance training can be done concurrently.

The goal of aerobic conditioning is to gradually increase exercise duration, frequency, and intensity to the point where one can maintain the activity for 30 to 40 minutes at a time, preferably at least five times a week, at a moderate level of intensity. For most, a moderate intensity level causes a noticeable increase in breathing rate and heart rate and usually results in some amount of perspiration. Some use the "talk test" to define moderate-intensity exercise, meaning that when exercising at a

moderate level, people should still be able to talk but would not be able to sing. However, we recommend that you use a rating of perceived exertion (RPE) scale with your participants to help determine intensity level during aerobic exercise. On the RPE scale in [table 6.1](#), moderate intensity is indicated by the number 5. Generally, we recommend an RPE between 5 and 6 for most older adults—that is, exercise that would be evaluated as being in the moderate to somewhat hard range. For more frail participants, it is recommended that exercise intensity start at very easy or somewhat easy (3 or 4 on the RPE scale), with intensity level gradually increasing over time. To use the RPE scale effectively, most older adults will require some training and practice.

TABLE 6.1 Rating of Perceived Exertion (RPE) Scale

Rating	Description
1	No effort
2	Little effort
3	Very easy
4	Somewhat easy
5	Moderate
6	Somewhat hard (starting to feel it)
7	Hard
8	Very hard (making an effort to keep up)
9	Very, very hard
10	Maximum effort (can't go any farther)

Adapted from NSCA 2012.

As with all types of exercise, you should always advise your clients to start slow and work up gradually to the desired level; this is especially true for sedentary people and those with chronic conditions. Studies show that aerobic exercise accumulated in short bouts (10 minutes at a time) had similar effects as one continuous bout (30 minutes at one time) with regard to aerobic fitness and weight loss in women (Schmidt, Biwer, & Kalscheuer, 2001). The minimum amount (duration) of aerobic exercise needed to meet the recommended guidelines varies depending on the intensity level. Refer to the list earlier in this chapter for examples of activities that provide the recommended amounts of daily moderate-level physical activity (i.e., activities that use approximately 150 calories of energy per day or 1,000 calories per week, assuming daily participation). Increased participation in these types of aerobic activities should result in improved performance on the SFT aerobic endurance tests—the 6-minute walk and the 2-minute step test.

Aerobic Conditioning Guidelines

- Instruct the participant to start slowly and gradually work up to at least 150 minutes (2.5 hours) each week of moderate-intensity (somewhat hard) aerobic exercise. This is approximately 30 minutes per day if spread over a period of 5 days, but it is preferable to exercise every day of the week.

- For the less fit, encourage 5- to 10-minute bouts of light- to moderate-intensity exercise.
- The more fit can mix it up with a combination of moderate- and vigorous-intensity physical activity, increasing duration with the goal of 300 minutes (5 hours) spread over most or all days of the week.

Aerobic Conditioning Precautions

- Adjust intensity, duration, and frequency based on the participant's fitness level, medical conditions, and physical symptoms (e.g., pain, fatigue, stiffness). Although mild fatigue and pain can be expected, exercise should never leave the participant feeling extremely fatigued or in extreme pain the following day.
- There should be an adequate warm-up and cool-down.
- Outdoor exercise should be avoided in extremes of heat or cold and in icy or smoggy conditions.
- Avoid unsupported exercise for clients with balance problems.
- Stop exercise for any signs of overexertion (refer to chapter 4).

Improving Strength

As discussed in chapters 2 and 3, maintaining an adequate amount of lower- and upper-body strength is necessary for executing a variety of common tasks associated with physical independence such as climbing stairs, walking distances, getting out of a chair or the bathtub, standing up from the floor, lifting and lowering objects, and reducing risk for falls. Reminding participants about the health-related benefits of strength training is also important—such as reducing risk of obesity, bone loss, low-back pain, osteoarthritis, cardiovascular disease, and diabetes. In the SFT, lower-body strength is assessed using a 30-second chair stand; upper-body strength is measured using the arm curl. Any form of exercise that stresses a person's muscles, including many common types of housework and yardwork activities, will help maintain strength. However, if your client scored low on either or both of the 30-second chair stand and arm curl test items and wishes to *increase* his strength, a particular regimen of *progressive resistance exercises* will need to be followed.

Briefly, strength is increased by gradually increasing the resistance placed on a muscle (i.e., by applying what is called the overload principle). Overloading a muscle means making it do more than it is accustomed to doing. This can be accomplished using free weights (similar to the dumbbells used to test arm strength in the SFT), elastic exercise bands, Velcro strap-on weights, exercise machines that are designed for specific muscle groups, or a person's own body weight and gravity. A suitable resistance for stimulating strength depends on the participant's health and fitness status. Generally, according to national recommendations previously mentioned, a beginner should start out with a load of 50 percent of one-repetition maximum (1RM), gradually increasing to approximately 70 to 80 percent of 1RM. A load of 70 to 80 percent of 1RM should cause the lifter to reach fatigue within 8 to 12 repetitions, with *fatigue* meaning the muscles cannot perform another repetition using proper form. Using a leg press as an example, lower-body strength can be increased by selecting a resistance (load) that can be pressed at least 8 times, but no more than 12 times, before the muscle is too fatigued to continue. Then, as muscle strength improves (as it becomes possible to press the selected weight more than 12 times before fatigue), the resistance should be increased, thus causing the muscle to again be overloaded (i.e., to do more than it was accustomed to). This process is repeated throughout the strength development program—as strength improves to the point where the new resistance can be performed more than 12 times without fatigue, it is again replaced with a heavier resistance.

When recommending strengthening exercises that do not use traditional weight machines, such as the standing squat, which uses body weight and position as a form of resistance, the aforementioned guidelines do not directly apply. Although the goal is to complete 8 to 12 repetitions to fatigue, some participants will at first be able to complete only 1 or 2 repetitions. If this is the case, gradually increase the repetitions (not the resistance) until participants are able to complete the exercise between 8 and 12 times. Also, when performing nonequipment exercises (e.g., calisthenics), increasing the load (resistance) is achieved by increasing the challenge, such as changing body position, gripping up on (shortening) a resistance band, using a thicker resistance band, or adding dumbbells. Refer to the sections To Increase the Challenge throughout the chapter for more exercise examples.

Strength training guidelines for older adults recommend performing at least one set of 8 to 12 repetitions, to the point of fatigue, for each of the major muscle groups twice a week. Important muscle groups for older adults are those needed for lower-body functioning (hip extensors, knee extensors, and ankle plantar flexors and dorsiflexors), for upper-body functioning (biceps, triceps, shoulders, and back extensors), and for trunk and core stability (abdominals and lower back). Strengthening exercises can also be performed more than two times a week, with at least 48 hours between sessions.

Strength Conditioning Guidelines

- Always include a short warm-up (including dynamic flexibility exercises) before doing strength exercises to raise body temperature and get more blood to the extremities.
- Progress slowly, and cautiously increase both the range of motion and the amount of weight for people with chronic conditions or who are frail or less fit.
- The goal is to perform at least one set of 8 to 12 repetitions, to the point of fatigue, for each of the major muscle groups twice a week.
- Rest time between exercises depends on exercise type and resistance level. Participants should rest for 1 to 2 minutes after most exercises and 2 to 3 minutes after multiple-joint exercises using heavy resistance.
- Place exercises using larger muscle groups (e.g., leg press, bench press, seated row, standing squat, chest press, and wall push-up) at the beginning of the workout; place single-joint and isolated muscle actions (e.g., triceps extensions, biceps curls, knee extensions, and leg curls) toward the end of the session.
- Both concentric (shortening) and eccentric (lengthening) muscle actions are recommended.
- Increase weight (progression) by about 2 to 10 percent when the desired number of repetitions is met (larger muscle groups generally can handle a greater increase in load).
- If using a resistance band, switch to another band that provides more resistance after the participant can do 8 to 12 repetitions.
- Exercises should be conducted through the full range of motion.
- Remind participants to breathe throughout the movement (generally exhale during the exertion phase and inhale during the release phase).

Strength Conditioning Precautions

- Participants should talk with their doctors if they are unsure about doing a particular exercise, especially if they have had joint surgeries.
- Use only resistance levels that allow the participant to keep proper body alignment and form during the exercise.
- For the less conditioned and people with chronic pain disorders, stretch the same muscle group after each resistance exercise.
- Remind participants to avoid hyperextending or locking the joints.
- Remind participants to avoid jerking or thrusting the weight.
- Participants should always stay within their pain-free range.
- Allow 48 hours between moderate- to high-intensity workouts.

Improving Flexibility

Evidence suggests that reduced flexibility can result in impaired mobility and functional limitation (Holland, Tanaka, Shigematsu, & Nakagaichi, 2002; Nelson et al., 2007). As discussed in chapters 2 and 3, flexibility (the range of motion possible at one or multiple joints) is important for maintaining good posture, reducing the risk for injuries and back problems, reducing pain and stiffness, and performing tasks of daily living (e.g., putting on socks and shoes, inspecting and washing feet, kneeling down to pick up objects from the floor, putting on overhead garments, combing hair). Although most physical activity guidelines provide little information about the exact type and amount of flexibility exercise needed, all agree on the importance of flexibility and that it be part of a well-rounded physical activity program. In the SFT, upper-body flexibility is assessed using the back scratch test; lower-body flexibility is measured using the chair sit-and-reach.

When older adults perform aerobic or strength training activities, they should take an extra 10 minutes to stretch the major muscle and tendon groups, holding each stretch for 10 to 30 seconds and repeating it three to five times. Also, before participating in aerobic or strengthening exercise, it is important to first warm up the muscles and joints through walking or light calisthenics and then do some dynamic flexibility exercises (moving a joint through a given range of motion but not holding the joint in an end position) to help prevent injuries to the body tissue. Static flexibility

exercises (moving the joint through a single movement plane until a given end point is reached and then held for several seconds) should be included as part of the cool-down period after exercise sessions to help with muscle relaxation and removal of unwanted waste products, thus reducing muscle soreness and stiffness. More comprehensive guidelines for flexibility training of older adults can be found in the book *Physical Activity Instruction of Older Adults* (Jones & Rose, 2005). A well-rounded exercise program should include flexibility exercises for all muscle and joint areas—ankle, knee, hip, back, shoulder, trunk, and neck. We also recommend that older adults incorporate stretching movements into their activities of daily living (e.g., using a full range of movement to vacuum, sweep, reach for things, or wash the car).

Flexibility Exercise Guidelines

- Include flexibility exercises for stretching major body parts.
- Add at least 10 minutes of stretching to cardiorespiratory and strength training routines; include dynamic stretching during the warm-up and static during the cool-down.
- Have participant do a warm-up of 5 to 10 minutes before performing static stretches.
- Participants should maintain normal breathing throughout each stretch and focus attention on the muscle being stretched, trying to limit movement in other body parts.
- Emphasize joints that have obvious range limitations or stiffness.
- For static stretching, participants should move slowly into the end position (a feeling of gentle tension, not pain).
- Joints should always be slightly bent (not locked) while stretching.
- For static stretching, each exercise should be held for 10 to 30 seconds and should be repeated three to five times (trying to stretch farther each time).
- Ideally, you want your clients to increase holding of the static stretch to at least 20 seconds to receive optimal benefits.

Flexibility Exercise Precautions

- Understand the contraindicated exercises (e.g., high-impact activities, heavy resistance, ballistic movements) for older adults with recent injuries, surgeries (e.g., hip, back, shoulder), joint replacements, and chronic conditions (e.g., osteoporosis, spinal

stenosis).

- Emphasize good body alignment (e.g., avoid hyperextension or locking of joints).
- Remind participants to breathe normally even when holding a stretch.
- Make sure participants avoid jerking and bouncing into a static stretch.

Improving Agility and Dynamic Balance

A combination of agility (involving speed and coordination) and dynamic balance (maintaining postural stability while moving) is important for a number of common mobility tasks such as walking, negotiating curbs, and climbing stairs as well as for making quick movements to avoid hazards in the environment, getting on and off a bus in a timely manner, crossing the street before the light turns red, and getting up quickly to answer a phone or a doorbell. Having adequate levels of agility and dynamic balance is also important for reducing risk for falls and loss of independence. In the SFT, agility and dynamic balance are measured using the 8-foot up-and-go test.

Since unintentional falls among older adults are a major public health problem in the United States, balance was mentioned as an important aspect of exercise programs for at-risk older adults in all the recently published physical activity guidelines. More specifically, the Physical Activity Guidelines for Americans (U.S. Department of Health and Human Services, 2008) suggests that older adults at risk for falls should perform balance and moderate-intensity muscle-strengthening activities for 90 minutes a week, plus moderate-intensity walking for about 1 hour a week, preferably spreading the activity over 3 or more days.

Unlike improving aerobic endurance, strength, and flexibility, improving agility and balance requires a multidimensional approach that targets the multiple systems contributing to postural control and speed (e.g., sensory, motor, and cognitive). Findings from several studies show that balance is best improved, and the risk of falling reduced, if older adults participate in specific balance and agility activities along with aerobic, strength, and flexibility exercises (Rose, 2010).

“Since I have been exercising, I have noticed positive changes in balance, flexibility, and endurance. The annual Senior Fitness Tests allow me to track my improvements. . . . My overall physical conditioning has improved, as well as a knee injury that has steadily gotten better.”

George Schussler, age 75, Maple Knoll Wellness Center, Cincinnati, Ohio

The best exercises to improve agility and balance involve a variety of motor tasks performed under varying types of sensory and cognitive environments. Examples of such activities, which can range from simple to complex, include calf raises, balancing on one foot with eyes open and with eyes closed, tandem (heel-to-toe) balancing, walking on unstable surfaces such as foam pads or rocker boards, walking at different speeds using a variety of gait patterns, and other more complex activities that include quick weight transfers and crossover stepping (e.g., the grapevine move in a dance routine). For more capable older adults, participation in complex exercise routines to music and in various sports activities (e.g., tennis and badminton) that involve balance and coordination are excellent ways to maintain agility and balance.

Agility and Balance Training Guidelines

- Agility and balance activities should be included as part of a well-rounded exercise program.
- Older adults at risk for falls should perform balance and moderate-intensity muscle-strengthening activities for 90 minutes a week, plus moderate-intensity walking for about 1 hour a week, preferably spread out over 3 or more days.
- Balance and agility are best improved through exercises that involve multiple systems (e.g., somatosensory, visual, vestibular, musculoskeletal) and cognitive challenges.

Agility and Balance Training Precautions

- If working with a frail client or one with poor balance, make sure the person is closely supervised at all times.
- Use caution with fast movements, including quick turns and changes in position. This may increase risk for dizziness and falls.

- To increase stability and safety during an exercise, provide participant something to hold on to (e.g., sturdy chair, wall, cane, walker).
- Progress the participant to the next level of exercise difficulty when the preceding challenge can be done safely.
- Eliminate any unnecessary clutter that clients might trip over.
- Make sure the participant is wearing stable, flat shoes.

Exercises to Improve Strength, Flexibility, Agility, and Balance

This section provides examples of low-risk nonmachine exercises (reproducible for use as handouts) that can help clients and participants improve lower- and upper-body strength (30-second chair stand and arm curl tests), lower- and upper-body flexibility (chair sit-and-reach and back scratch tests), and agility and dynamic balance (8-foot up-and-go test). In addition, we highly recommend participation in group exercise classes (e.g., Pilates, yoga, tai chi, aerobic dance) or personalized programs at fitness facilities that use exercise equipment to improve functional fitness.

The Senior Fitness Test Software 2.0 includes printable versions of the exercise instruction sheets in this section.

For flexibility, agility, and dynamic balance exercises, remind clients that warming up the muscles for at least 10 minutes before exercising is essential. For clients who need additional exercises to improve agility and dynamic balance, *FallProof!*, a comprehensive balance and mobility training program (Rose, 2010), provides evidence-based targeted exercises to reduce fall risk.

EXERCISES TO IMPROVE LOWER-BODY STRENGTH

Standing Squat



Exercise Instructions

- Stand in front of a sturdy chair, with feet shoulder-width apart. The back of the chair will be against a wall.
- As you slowly bend at the knees and hips to lower your buttocks into the chair, bring both arms forward to help with balance and control.
- Pause and settle your weight in the chair.
- Return to the standing position by pressing through the heels and straightening the knees and hips.
- Perform 8 to 12 times or until fatigue.

Tips

- Remember to breathe throughout the movement.
- Keep eyes looking forward to a vertical target (e.g., corner of a wall, door frame).
- Push off your thighs if you need to.

To Increase the Challenge

- Don't sit completely before doing the next stand.
- Hold hand weights.
- Increase the number of repetitions or sets.

Safety Precautions

- Make sure the chair is stable, with its back firmly against the wall.
- Never perform exercise to a point of pain.

Single-Leg Lunge



a



b

Exercise Instructions

1. Stand behind a sturdy chair, with the right foot in front of the left foot (*a*). Place your hands on the chair for stability.
2. Keeping a slight bend in both knees, bring your left knee toward

the ground, stopping when your right knee almost goes over the right toes (*b*).

3. Slowly return to the starting position. Perform 8 to 12 times or until fatigue.
4. Switch legs so that the left leg is in front, and repeat.

Tips

- Make sure the forward knee does not go beyond the toes of the forward foot.
- Make sure the upper body is not leaning forward or backward.
- Remember to breathe throughout the movement.

To Increase the Challenge

- Try to perform the exercise without holding on to the chair.
- Increase the number of repetitions or sets.

Safety Precautions

- Hold on to a wall or chair for stability.
- Keep the knee just behind the forward toes.
- Make sure the chair is sturdy and stable and if possible placed on a nonslip surface (such as carpet).

Hip Extension



Exercise Instructions

1. Stand behind a sturdy chair or next to a wall, and place your hands on the chair or wall for stability. Place the feet shoulder-width apart.

2. Lift one leg up directly behind your body; keep the supporting leg straight, with the knee slightly bent, pause, and then slowly return leg to starting position.
3. Repeat exercise with opposite leg, alternating between the two for 8 to 12 repetitions.

Tips

- Avoid leaning forward or arching your back.
- Squeeze the buttocks as you lift the heel of your foot backward.
- Remember to breathe throughout the movement.

To Increase the Challenge

- Tighten the abdominal muscles when lifting your leg.
- Hold the lift longer.
- Add an ankle weight.
- Perform 8 to 12 repetitions on the same leg instead of alternating legs.

Safety Precautions

- Keep the supporting knee slightly bent during the exercise.
- If using a chair, make sure it is sturdy and stable.

Leg Curl



Exercise Instructions

1. Stand behind a sturdy chair or next to a wall, and place your hands on the chair or wall for stability. Place the feet slightly apart.
2. Slowly lift your right heel toward your buttocks, pause, and then

- slowly return your right foot to the ground.
3. Repeat with the left foot, alternating between the two for 8 to 12 repetitions.

Tips

- Maintain proper posture, with ears above the shoulders and shoulders above the hips.
- Remember to breathe throughout the movement. Ideally, exhale as you lift your heel and inhale as you extend your leg.

To Increase the Challenge

- Tighten the abdominal muscles and buttocks when raising the leg.
- Perform 8 to 12 repetitions in a row on the same leg instead of alternating.
- Add ankle weights.

Safety Precautions

- Perform this exercise slowly and controlled.
- Make sure the chair is sturdy and stable and if possible placed on a nonslip surface (such as carpet).

EXERCISES TO IMPROVE UPPER-BODY STRENGTH

Chest Press



a



b

Exercise Instructions

1. While seated, place the resistance band behind your upper back just under the armpits, and grab the ends of the band with your hands (*a*).

2. Press your arms forward and parallel with the floor until the elbows are straight (*b*).
3. Slowly return to the starting position. Complete 8 to 12 repetitions.

Tips

- Make sure the band is inside the elbows.
- Exhale as you extend the arms, and inhale as you bring them back.

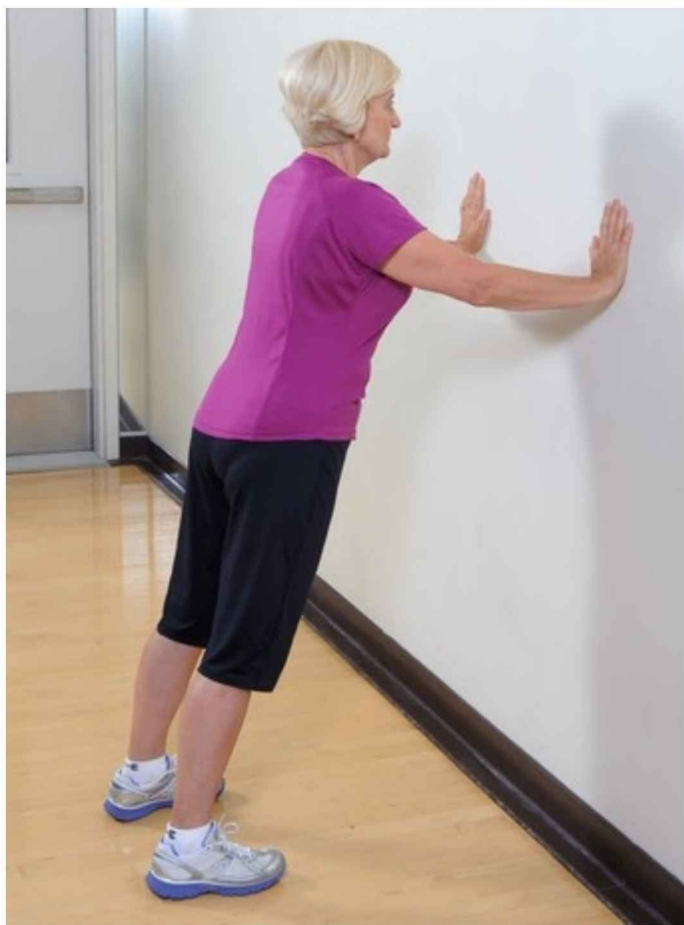
To Increase the Challenge

- Adjust the grip closer to your body to increase resistance on the band.
- Change to a heavier resistance band.
- Perform the exercise while standing.
- Increase the number of repetitions.

Safety Precautions

- Perform the exercise slowly and controlled.
- Check to make sure the resistance band is not fraying or damaged.

Wall Push-Up



a



b

Exercise Instructions

1. Face a wall, standing a little farther than arm's length away, feet shoulder-width apart.
2. Extend your arms, lean forward, and place your palms flat against

- the wall, shoulder-width apart (*a*).
3. With the arms fully extended, slowly breathe in and bend the elbows, lowering your body toward the wall (*b*). Keep your feet flat on the floor and the body straight.
 4. Exhale and slowly return the arms to the fully extended position. Complete 8 to 12 repetitions.

Tips

- Your body weight should be lifted by the arms; don't be tempted to use abdominal muscles or lower-body muscles to pull yourself up.

To Increase the Challenge

- Move your feet slightly farther away from the wall. Move to knee-style push-ups on the floor once your initial strength has increased.
- Increase the number of repetitions or sets.

Safety Precautions

- Avoid overextending the arms or locking the elbows.
- Avoid slippery surfaces.
- Wear sturdy, nonslip shoes.

Biceps Curl



Exercise Instructions

1. While seated, lift your right foot, wrap one end of the resistance band around it, and place the foot on the floor.
2. Take the other end of the band in your right hand, with the arm

- extended toward the ground.
3. Slowly bend at the elbow, bringing the palm up toward the shoulder.
 4. Slowly return to the starting position. Complete 8 to 12 repetitions.
 5. Switch the band to your left side, and repeat the exercise using your left arm.

Tips

- Keep the elbows close to the body.
- Exhale as you bend the elbow; inhale as you extend the elbow.
- The exercise can be performed in a standing position if the band is long enough.

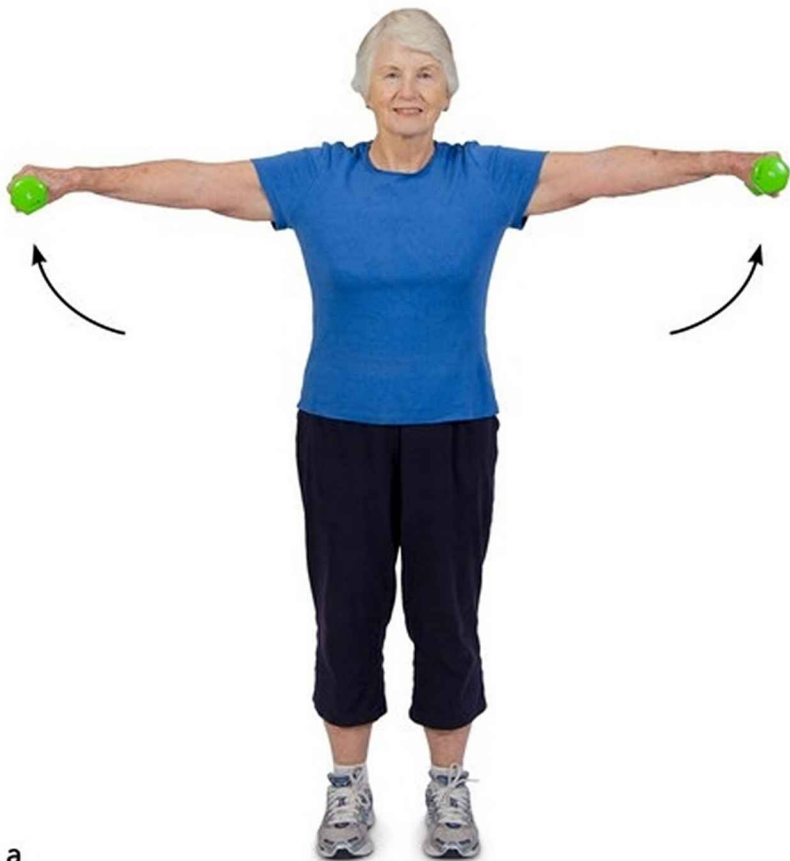
To Increase the Challenge

- Grab lower on the band to increase the resistance.
- Change to a thicker (more resistance) band.
- Increase the number of repetitions.

Safety Precautions

- Make sure the band is wrapped securely around the foot.
- Bend only at the elbow, not the wrist.
- Check to make sure the resistance band is not frayed or damaged.
- Using a resistance band with a handle is helpful, especially for people with arthritis.

Lateral Arm Raise



a



Exercise Instructions

1. Stand upright, with feet shoulder-width apart, knees slightly bent, and arms at the sides.
2. Hold a dumbbell in each hand beside the thighs, with palms turned

toward the body.

3. With straight arms, slowly lift to shoulder height (arms parallel to the floor) (*a*), pause, and then slowly return arms to the starting position. This exercise can also be done in a seated position using a resistance band (*b*).
4. Select a hand weight that allows completion of 8 to 12 repetitions to fatigue.

Tips

- Keep your back straight, with head and eyes forward.
- Do not lean forward.
- Do not bend the wrists or elbows.
- Inhale as you lift, and exhale as you lower.
- Lift the dumbbells slowly and under control rather than swinging them up.

To Increase the Challenge

- Use a heavier weight or thicker band if you can do more than 12 repetitions using good form.

Safety Precautions

- Do not hyperextend or lock your elbows.
- Do not raise the dumbbells higher than the shoulders.
- Avoid tensing neck muscles.

EXERCISES TO IMPROVE LOWER-BODY FLEXIBILITY

Hamstring Stretch



a



b



c

Exercise Instructions

1. Lie on your back on an exercise mat or firm surface.
2. Slide the right foot back toward the hip so that foot is flat on the floor and the knee is pointing toward the ceiling; keep the left leg straight and flat along the floor (*a*).
3. Reach down with both hands and gently grab the back of the right thigh to pull the knee toward the chest (*b*) until you feel gentle tension, not pain.
4. Hold the position for 10 to 30 seconds, and then slowly return the leg to the starting position.
5. Repeat with the opposite leg.
6. Alternating between the two legs, repeat the stretch three to five times on each side.

Tip

- Exhale as you pull the knee toward the chest, and relax the shoulders.

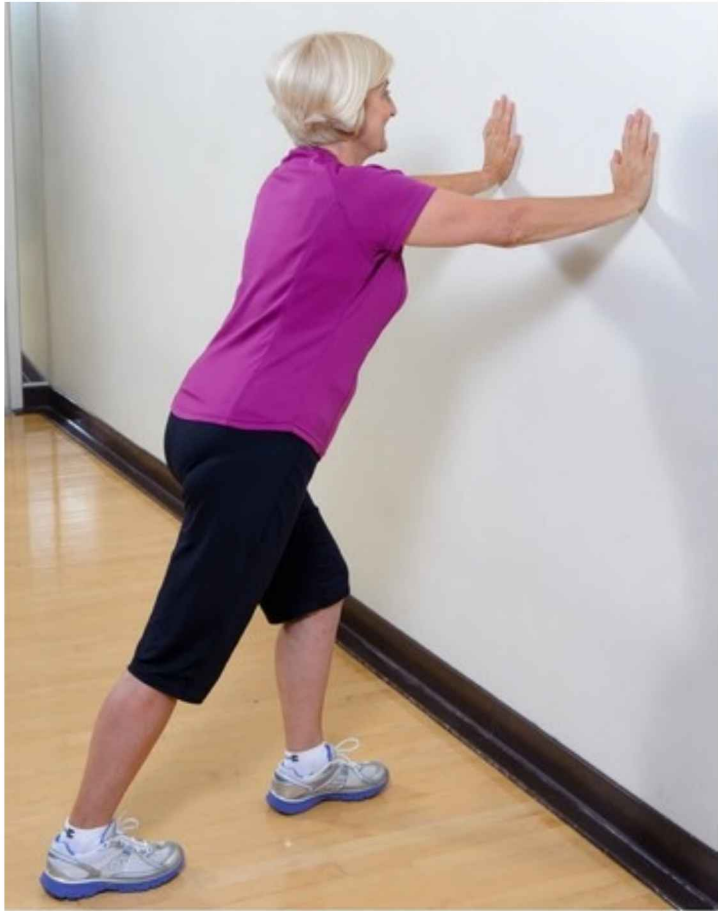
To Increase the Challenge

- Gradually hold the stretch longer, with the goal being 30 seconds.
- To increase tension, slowly extend the leg up and gently pull the leg toward your face (*c*).

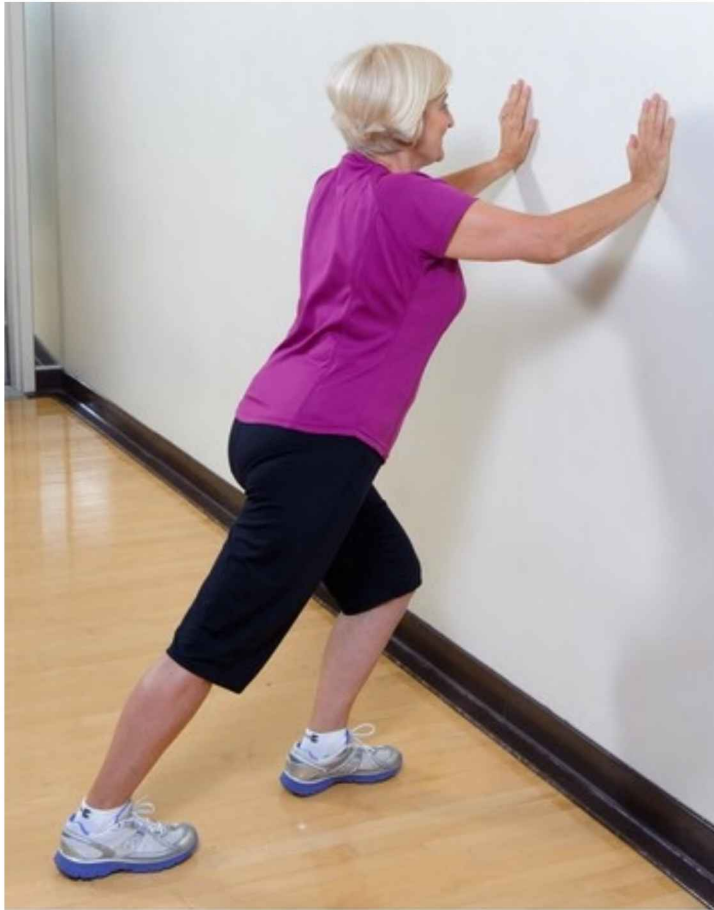
Safety Precautions

- Do a general warm-up for at least 10 minutes before doing flexibility exercises.
- Breathe continuously throughout the exercise.
- Never bounce the knee or move to a point of pain.

Standing Calf Stretch



a



b

Exercise Instructions

1. Stand facing a wall slightly farther than arm's length away, with feet shoulder-width apart.
2. Step forward with the left leg, and bend the left knee. Put your

palms flat against the wall at shoulder height and shoulder-width apart (*a*).

3. Keeping both feet on the floor, bend the right knee slightly until you feel a stretch in your right calf muscle (*b*).
4. Hold the position for 10 to 30 seconds, and then slowly return to the starting position.
5. Repeat with the opposite leg.
6. Alternating between the two legs, repeat the stretch three to five times on each side.

Tip

- Be sure the toes of both feet are pointing directly forward and that the heels are in line with the toes.

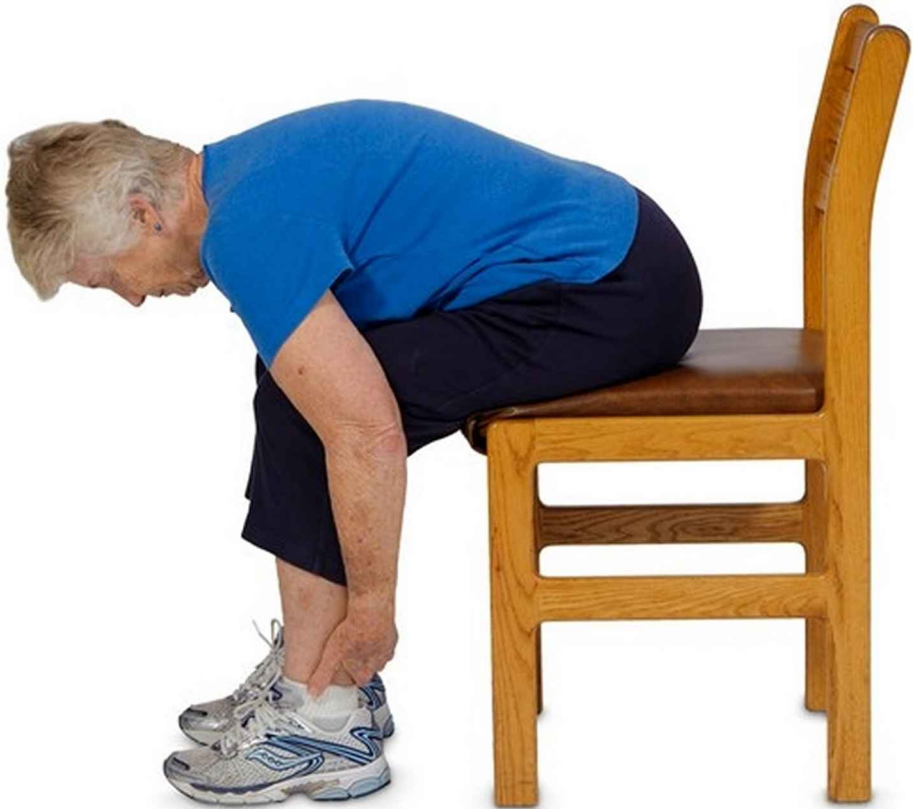
To Increase the Challenge

- Increase tension by bending the front knee more or moving the back foot farther from the wall.
- Gradually hold the stretch longer, with the goal being 30 seconds.

Safety Precautions

- Do a general warm-up for at least 10 minutes before doing flexibility exercises.
- Avoid hyperextension or locking of the knee and elbow joints.
- Continue breathing throughout the exercise.

Seated Back Extension



Exercise Instructions

1. Sit securely toward the front of a sturdy armless chair, with feet flat on the floor and shoulder-width apart.
2. Place your hands under your thighs.
3. On the exhale, slowly bend forward from the hips. Keep your back

and neck straight.

4. Slightly relax your neck and lower your chin, and slowly bend farther forward.
5. Allow your hands to slide down toward the heels or to a point of tension. Hold the position for 10 to 30 seconds.
6. Slowly inhale and straighten up by pulling your abdominal muscles toward the spine.
7. Repeat the stretch three to five times.

Tip

- When straightening, slowly uncurl each part of the spine, starting from the pelvis and working your way up.

To Increase the Challenge

- Gradually increase the stretch so you touch the floor.

Safety Precautions

- Do a general warm-up for at least 10 minutes before doing flexibility exercises.
- To reduce pressure on the abdomen, separate the feet more.
- People with osteoporosis should not do this stretch.
- Avoid dropping the head too far forward, because this may cause dizziness.

Hip Stretch



Exercise Instructions

1. Sitting in a chair with your feet shoulder-width apart, cross the left leg over the right leg so that the left ankle is resting on the right knee.
2. Gently lean your trunk forward.
3. Hold the stretch for 10 to 30 seconds, and repeat on the opposite leg.
4. Alternating between the two legs, repeat the stretch three to five times on each side.

Tip

- Exhale into the stretch.

To Increase the Challenge

- Reach under the crossed leg, and lift the leg and foot toward the chest.
- Increase the repetitions.

Safety Precautions

- Do a general warm-up for at least 10 minutes before doing flexibility exercises.
- People with knee or hip replacements should not do this stretch.
- Do not perform the stretch if it causes any pain.

EXERCISES TO IMPROVE UPPER-BODY FLEXIBILITY

Shoulder and Upper-Arm Stretch



a

A man with grey hair, wearing a blue short-sleeved shirt and dark blue trousers, is standing with his back to the camera. He is holding a white towel behind his back with both hands. His right arm is raised and bent, with his hand reaching up to his head, pulling the towel over his shoulder. His left arm is bent, with his hand reaching down to his waist, also holding the towel. The towel is draped over his back, hanging down between his legs.

b

Exercise Instructions

1. Stand with feet shoulder-width apart.
2. Hold one end of a towel in your right hand.
3. Raise and bend your right arm to drape the towel down your back.

Keep the right arm in this position as you continue holding on to the towel.

4. Reach behind your lower back, and grasp the towel with your left hand (*a*).
5. To stretch the right shoulder, pull the towel down with the left hand (*b*) until you feel comfortable tension, not pain.
6. Hold for 10 to 30 seconds and then relax.
7. Alternating between the arms, repeat the stretch three to five times on each side.

Tips

- Exhale and relax into the stretch as you pull the towel down.
- Continue breathing throughout the exercise.

To Increase the Challenge

- Stretch farther and hold longer.

Safety Precautions

- Do a general warm-up for at least 10 minutes before doing flexibility exercises.
- Avoid tensing the neck.
- Do not perform the exercise if it causes any pain.
- Do not do this exercise if you have had a shoulder replacement.

Chest Stretch





b

Exercise Instructions

1. Standing with feet shoulder-width apart, grasp your hands behind your back.
2. Slowly squeeze the shoulder blades together (*a*, *b*) until you feel

- gentle tension in your chest, shoulders, and arms.
3. Hold for 10 to 30 seconds and then relax.
 4. Repeat the stretch three to five times.

Tips

- Continue breathing throughout the movement.
- Pull your shoulders slightly down as you pull them back.
- Avoid bending forward or backward—keep the trunk vertical.

To Increase the Challenge

- When squeezing the shoulder blades together, slowly lift the arms.
- Stretch farther and hold longer.

Safety Precautions

- Do a general warm-up for at least 10 minutes before doing flexibility exercises.
- Avoid tensing the neck.
- Do not perform the exercise if it causes any pain.

Wall Upper-Body Stretch



a



b

Exercise Instructions

1. Stand slightly farther than arm's length from a wall, feet shoulder-width apart.
2. Lean forward and put your palms flat against the wall at shoulder

- height and shoulder-width apart.
3. Keeping your back straight, slowly walk your hands up the wall (*a*) until your arms are above your head (*b*).
 4. Hold your arms overhead for 10 to 30 seconds.
 5. Slowly walk your hands back down and then relax.
 6. Repeat the stretch three to five times.

Tip

- Keep your feet flat on the floor when leaning forward.

To Increase the Challenge

- Stand farther away from the wall, and lean farther into the stretch.

Safety Precautions

- Do a general warm-up for at least 10 minutes before doing flexibility exercises.
- Avoid tensing the neck.
- Wear sturdy, nonslip shoes.
- Do not perform the exercise if it causes any pain.



Exercise Instructions

1. Stand with feet shoulder-width apart and hands extended down at the sides.
2. Gently pull the left arm down, reaching toward the floor with the

- left hand, and tilt the head to the right.
3. Hold the position for 10 to 30 seconds while continuing to breathe.
 4. Repeat the action, gently pulling the right arm down toward the floor, tilting the head to the left.
 5. Alternating between the two sides, repeat the stretch three to five times on each side.

Tip

- Relax the neck, and exhale into the stretch.

Safety Precautions

- Do a general warm-up for at least 10 minutes before doing flexibility exercises.
- Do not perform the exercise if it causes any pain.
- Be sure to breathe while holding the stretch.

EXERCISES TO IMPROVE AGILITY AND DYNAMIC BALANCE

Calf Raise



Exercise Instructions

1. Stand behind a sturdy chair or next to a wall, and place your hands on the chair or wall for stability. Place the feet shoulder-width apart.

2. Rise up onto your toes, and slowly return to the starting position.
3. Complete 8 to 12 repetitions.

Tips

- Maintain proper posture.
- Push up onto the toes as high as possible without pain.

To Increase the Challenge

- Try to perform the exercise without holding on to the chair or wall.
- Increase the number of repetitions.
- Attach ankle weights.

Safety Precautions

- Do a general warm-up for at least 10 minutes before agility and balance exercises.
- Use a sturdy chair or wall to increase stability and reduce risk of falls.

Seated Sit-Ups





Exercise Instructions

1. Sit securely toward the front of a sturdy armless chair; lean back against the backrest, crossing your arms across the chest (*a*).
2. Slowly move forward from the hips, tightening the abdominal

- muscles.
3. Pause when seated upright (*b*), and return slowly to the starting position.
 4. Perform the exercise 5 to 10 times, slowly and controlled.

Tips

- Pretend there is a grape in your belly button, and squeeze it while moving forward.
- Breathe out while moving forward, and breathe in while returning to the original position.

To Increase the Challenge

- Increase the number of repetitions.
- Perform the exercise more slowly.
- Place your hands behind your head.

Safety Precautions

- Do a general warm-up for at least 10 minutes before agility and balance exercises.
- Do not perform the exercise if you have lower-back pain.



a



b

Exercise Instructions

1. Stand in front of a chair or other supporting surface such as a kitchen counter, with feet shoulder-width apart.
2. Place your hands on the supporting surface to assist with

maintaining balance.

3. Shift your weight from left to right by softening the right knee and taking the body weight over to the right leg (*a*).
4. Notice the right leg becomes heavy and the left leg is light as more of the body weight is supported by the right leg.
5. Hold the position for 5 seconds, and then return to center; repeat in the opposite direction (*b*).
6. Alternating legs, perform the exercise eight times on each side.

Tips

- Keep your eyes forward on a vertical target.
- Keep your shoulders directly above the hips during the shifts.

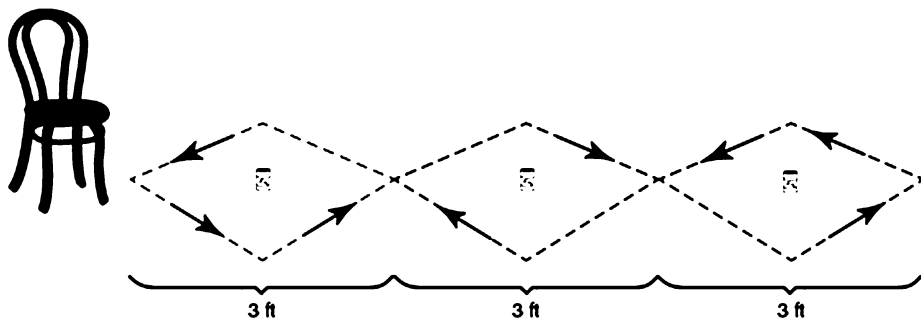
To Increase the Challenge

- Increase the number of repetitions.
- Lift the non-weight-bearing foot off of the floor.
- Close your eyes while performing the exercise (but be sure to open them if you become off balance).

Safety Precautions

- Do a general warm-up for at least 10 minutes before agility and balance exercises. Perform this activity only while holding on to something stable.
- Never push yourself beyond what you consider safe for you.

Zigzag



Exercise Instructions

1. Set up four to six markers (e.g., soup cans, cups) about 3 feet (.9 m) apart in a straight line.
2. Place a sturdy chair at the beginning of the row to mark the starting line. Sit in the chair to start.
3. Get up quickly from the chair, weave (zigzag) through the markers in a left to right or right to left pattern to the end of the course and back, and then sit down and rest for a few seconds.
4. Repeat the exercise three to five times.

Tip

- Start slowly, and work within a safe speed.

To Increase the Challenge

- Increase the speed, repetitions, or number of markers.

Safety Precautions

- Do a general warm-up for at least 10 minutes before agility and balance exercises.
- Place a sturdy chair against the wall for the starting line.
- Wear sturdy, nonslip shoes.
- Never push yourself beyond what you consider safe for you.

Summary

Recognize that the best type of exercise is the kind that people will do. Clients can increase their activity level and their fitness in three ways: (1) by increasing the amount of physical activity they get in their normal daily routines, (2) by participating in a structured exercise program, or (3) by doing a combination of both. Although both types of exercise have their advantages and should be encouraged, structured exercise protocols may be needed to address specific physical weaknesses identified by the SFT scores, particularly for improving aerobic endurance, muscular strength, flexibility, balance, and agility. This chapter provides some structured exercises for improving strength, flexibility, agility, and dynamic balance. Because a thorough discussion of exercise programming for various populations is beyond the scope of this book, readers should also consult the many available additional resources for helping plan older adult exercise programs.

Informed Consent and Assumption of Liability

You are being invited to participate in testing to evaluate your physical fitness. Your participation is entirely voluntary. If you agree to participate, you will be asked to perform a series of assessments designed to evaluate your mobility, upper- and lower-body strength, aerobic endurance, flexibility, agility, and balance. These assessments involve activities such as walking, standing, lifting, stepping, and stretching. The risk of engaging in these activities is similar to the risk of engaging in all moderate exercise and may possibly result in muscular fatigue and soreness; sprains and soft tissue injury; skeletal injury; dizziness and fainting; and the risk of cardiac arrest, stroke, and even death.

If any of the following apply, you should *not* participate in testing without written permission of your physician:

1. Your doctor has advised you not to exercise because of your medical condition(s).
2. You have had congestive heart failure.
3. You are currently experiencing joint pain, chest pain, or dizziness or have exertional angina (chest tightness, pressure, pain, heaviness) during exercise.
4. You have uncontrolled high blood pressure (160/100 or above).

During the assessment you will be asked to perform within your physical comfort zone and never to push to a point of overexertion or beyond what you feel is safe. You will be instructed to notify the person monitoring your assessment if you feel any discomfort or experience any unusual physical symptoms such as shortness of breath, dizziness, tightness or pain in the chest, irregular heartbeat, numbness, loss of balance, nausea, or blurred vision. If you are accidentally injured during testing, the test administrators will be unable to provide treatment to you other than basic first aid. You will be required to seek treatment from your own physician, which must be paid for by you or your insurance company.

You may discontinue participation in testing whenever you wish by asking to do so. By signing this form, you acknowledge the following:

1. I have read the full content of this document. I have been informed of the purpose of the testing and of the physical risks I may encounter.
2. I agree to monitor my own physical condition during tests I am asked to perform, and I agree to stop my participation and inform the person administering the assessment if I feel uncomfortable or experience any unusual symptoms.
3. I assume full responsibility for all risk of bodily injury and death as a result of participating in testing. Should I suffer an injury or become ill during testing, I understand that I must seek treatment from my own physician and that I or my insurance will have to pay for this treatment.

My signature below indicates that I have had an opportunity to ask and have answered any questions I may have had, and that I freely consent to participate in the physical assessment.

Signature: _____ Date:

Print name: _____

...ity (strength, endurance, flexibility, balance, and agility). The test battery was developed and validated through research at California State University at Fullerton and elsewhere.

...will be administered by trained personnel, and procedures for any medical emergency are in place. Participants will be instructed to do the best they can within their comfort zone and never to push themselves to exertion or beyond what they think is safe for them. Technicians have been instructed to discontinue the test if at any time participants show signs of dizziness, pain, nausea, or undue fatigue. Following are the test items:

- 1. Chair stand test (number of stands from a chair in 30 seconds)
- 2. Curl test (number of curls in 30 seconds; 5-pound weight for women, 8-pound weight for men)
- 3. Step test (number of steps in place completed in 2 minutes)
- 4. 6-minute walk test (number of yards walked in 6 minutes—person can rest when necessary)
- 5. Either the 2-minute step test OR the 6-minute walk test will be performed, not both
- 6. Sit-and-reach test (distance one can reach forward toward toes)
- 7. Back scratch test (how far hands can reach behind the back)
- 8. Chair sit-and-go test (time required to get up from a chair, walk 8 feet, and return to the chair)

If, for any medical or other reasons, you believe your patient would not benefit from participation in the fitness testing by your patient would not benefit from participation in the fitness testing, please check the appropriate box on this form. By completing the following form, you are not assuming any responsibility for the test battery.

If you have any questions about the fitness testing, please call _____.

I know of no reason why my patient should not participate.
I believe my patient can participate, but I urge caution because _____

My patient should not engage in the following test items: _____

I do not recommend that my patient NOT participate in testing.

Signature: _____ Date: _____

Physician: _____ Phone #: _____

Participant Instructions Before Assessment

Place: _____

Date: _____

Time: _____

Although the physical risks associated with the testing are minimal, the following reminders are important in ensuring your safety and helping you score the best you can.

1. Avoid strenuous physical activity for 1 or 2 days before the assessment.
2. Avoid excessive alcohol use for 24 hours before testing.
3. Eat a light meal 1 hour before testing.
4. Wear clothing and shoes appropriate for participating in physical activity.
5. Bring a hat and sunglasses for walking outside and reading glasses (if needed) for completing forms.
6. Bring the Informed Consent and Assumption of Liability form and the Medical Clearance form, if required.
7. Inform the test administer of any medical conditions or medications that could affect your performance.

As part of your testing, you will be asked to perform the aerobic endurance test checked below:

_____ 2-minute step test to see how many times you can step (march) in place in 2 minutes

_____ 6-minute walk test around a flat course to determine the distance you can cover in that time

After you have determined that it is safe for you to participate in the tests (see Informed Consent and Assumption of Liability form), you should practice the test you will be taking at least once before test day—that is, time yourself either walking for 6 minutes or stepping (marching) in place for 2 minutes. This will help you determine the pace that will work best for you on test day.

Accident Report Form

Date of injury/illness _____ Time _____ a.m. _____ p.m.
City _____ State _____ Zip code _____
Name _____
Address _____
City _____ State _____ Zip code _____
Sex _____ Age _____ Area code & phone # _____

Check suspected cause of injury or illness:

Fall _____ Overexposure _____ Hyperflexion _____ Blunt trauma _____
Overexertion _____ Hyperextension _____ Other (specify) _____

Describe the nature of injury/illness in detail. (Did it happen before, during, or after activity? How?)

Check site of injury/illness:

None _____ Chest/ribs _____ Hand _____ Neck/throat _____ Wrist _____ Abdomen _____
_____ Ear _____ Head _____ Nose _____ Ankle _____ Elbow _____ Shoulder/collarbone _____
Arm _____ Face _____ Knee _____ Stomach _____ Back (upper) _____ Fingers/thumb _____
Leg _____ Teeth _____ Back (lower) _____ Foot _____ Mouth _____ Toes _____
Other (specify) _____

Check sign and/or symptom:

Abrasion _____ Discoloration _____ Loss of function _____ Pain _____
Visual involvement _____ Confusion _____ Fracture _____ Lung involvement _____
Shock _____ Cuts _____ Heart involvement _____ Nausea/vomiting _____ Sprain/strain _____
Dislocation _____ Internal injury _____ Numbness _____ Swelling _____ Other (specify) _____

What was done for the injured?

By whom?

Was the injured person sent to the hospital? Yes _____ No _____
Was the injured person sent to a physician? Yes _____ No _____
To whom was the injured person released? Self _____ Relative _____ Other (specify) _____

Signature of person filling out form _____

Date _____

Witness signature _____

Date _____

BMI Conversion Chart

Weight (lb)	Height																		
	49	51	53	55	57	59	61	63	65	67	69	71	73	75	77	79	81	83	
66	19	18	16	15	14	13	12	12	11	10	10	9	9	8	8	8	7	7	
70	20	19	18	16	15	14	13	13	12	11	10	10	9	9	8	8	8	7	
75	22	20	19	17	16	15	14	13	12	11	10	10	9	9	8	8	8	7	
79	23	21	20	18	17	16	15	14	13	12	12	11	11	10	9	9	9	8	
84	24	22	21	19	18	17	16	15	14	13	12	12	11	11	10	10	9	9	
88	26	24	22	20	19	18	17	16	15	14	13	12	12	11	11	10	10	9	
92	27	25	23	21	20	19	17	16	15	15	14	13	12	12	11	11	10	10	
97	28	26	24	22	21	20	18	17	16	15	14	14	13	12	12	11	10	10	
101	29	27	25	23	22	20	19	18	17	16	15	14	13	13	12	12	11	10	
106	31	28	26	24	23	21	20	19	18	17	16	15	14	13	13	12	11	11	
110	32	30	27	26	24	22	21	20	18	17	16	15	15	14	13	13	11	11	
114	33	31	29	27	25	23	22	20	19	18	17	16	15	14	14	13	12	12	
119	35	32	30	28	26	24	22	21	20	19	18	17	16	15	14	14	13	12	
123	36	33	31	29	27	25	23	22	21	19	18	17	16	16	15	14	13	13	
128	37	34	32	30	28	26	24	23	21	20	19	18	17	16	15	15	14	13	
132	38	36	33	31	29	27	25	23	22	21	20	19	18	17	16	15	15	14	
136	40	37	34	32	29	28	26	24	23	21	20	19	18	17	16	16	15	14	
141	41	38	35	33	30	28	27	25	24	22	21	20	19	18	17	16	16	15	
145	42	39	36	34	31	29	27	26	24	23	22	20	19	18	17	17	16	15	
150	44	40	37	35	32	30	28	27	25	24	22	21	20	19	18	17	16	15	
154	45	41	38	34	33	31	29	27	26	24	23	22	20	19	18	18	17	16	
158	46	43	40	37	34	32	30	28	26	25	24	22	21	20	19	18	17	16	
163	47	44	41	38	35	33	31	29	27	26	24	23	22	20	19	19	18	17	
167	49	45	42	39	36	34	32	30	28	26	25	23	22	21	20	19	18	17	
172	50	46	43	40	37	35	32	30	29	27	25	24	23	22	21	20	19	18	
176	51	47	44	41	38	34	33	31	29	28	26	25	23	22	21	20	19	18	
180	52	49	45	42	39	36	34	32	30	28	27	25	24	23	22	21	20	19	
185	54	50	46	43	40	37	35	33	31	29	27	26	25	23	22	21	20	19	
189	55	51	47	44	41	38	36	34	32	30	28	27	25	24	23	22	20	20	
194	56	52	48	45	42	39	37	34	32	30	29	27	26	24	23	22	21	20	
198	58	53	49	46	43	40	37	35	33	31	29	28	26	25	24	23	21	20	
202	59	54	50	47	44	41	38	36	34	32	30	28	27	25	24	23	22	21	
207	60	56	52	48	45	42	39	37	35	33	31	29	27	26	25	24	22	21	
211	61	57	53	49	46	43	40	38	35	33	31	30	28	27	25	24	23	22	
216	63	58	54	50	47	44	41	38	36	34	32	30	29	27	26	25	23	22	
220	64	59	55	51	48	44	42	39	37	35	33	31	29	28	26	25	24	23	
224	65	60	56	52	49	45	42	40	37	35	33	31	30	28	27	26	24	23	
229	67	62	57	53	49	46	43	41	38	36	34	32	30	29	27	26	25	24	
233	68	63	58	54	50	47	44	41	39	37	35	33	31	29	28	27	25	24	
238	69	64	59	55	51	48	45	42	40	37	35	33	32	30	28	27	26	24	
242	71	65	60	56	52	49	46	43	40	38	36	34	32	30	29	28	26	25	
246	72	66	61	58	53	50	47	44	41	39	37	35	33	31	29	28	27	25	
251	73	67	63	58	54	51	47	45	42	39	37	35	33	32	30	29	28	26	
255	74	69	64	59	55	52	48	45	43	40	38	36	34	32	31	29	28	26	
260	76	70	65	60	56	52	49	46	43	41	39	36	34	33	31	30	28	27	
264	77	71	66	61	57	53	50	47	44	42	39	37	35	33	32	30	29	27	
268	78	72	67	62	58	54	51	48	45	42	40	38	36	34	32	31	29	28	
273	79	73	68	63	59	55	52	48	46	43	40	38	36	34	33	31	30	28	
277	81	75	69	64	60	56	52	49	46	44	41	39	37	35	33	32	30	29	
282	82	76	70	65	61	57	53	50	47	44	41	40	37	35	34	32	30	29	

Body mass index (BMI) is determined by locating the point at which the horizontal row indicating a person's weight intersects with the vertical column indicating the person's height. BMI values of 19 to 25 generally are considered to be in the healthy range. Values above or below the shaded area may be associated with increased risk for disease and loss of mobility in later years (American College of Sports Medicine, 2010; Losonczy et al., 1995; Ross & Janssen, 2007).

Age-Group Percentile Norms

Chair Stand Test

Arm Curl Test

6-Minute Walk Test

2-Minute Step Test

Chair Sit-and-Reach Test

Back Scratch Test

8-Foot Up-and-Go Test

CHAIR STAND Test (Women)

Percentile Rank	60-64	65-69	70-74	75-79	80-84	85-89	90-94
95	21	19	19	19	18	17	16
90	20	18	18	17	17	15	15
85	19	17	17	16	16	14	13
80	18	16	16	16	15	14	12
75	17	16	15	15	14	13	11
70	17	15	15	14	13	12	11
65	16	15	14	14	13	12	10
60	16	14	14	13	12	11	9
55	15	14	13	13	12	11	9
50	15	14	13	12	11	10	8
45	14	13	12	12	11	10	7
40	14	13	12	12	10	9	7
35	13	12	11	11	10	9	6
30	12	12	11	11	9	8	5
25	12	11	10	10	9	8	4
20	11	11	10	9	8	7	4
15	10	10	9	9	7	6	3
10	9	9	8	8	6	5	1
5	8	8	7	6	4	4	0

CHAIR STAND Test (Men)

Percentile Rank	60-64	65-69	70-74	75-79	80-84	85-89	90-94
95	23	23	21	21	19	19	16
90	22	21	20	20	17	17	15
85	21	20	19	18	16	16	14
80	20	19	18	18	16	15	13
75	19	18	17	17	15	14	12
70	19	18	17	16	14	13	12
65	18	17	16	16	14	13	11
60	17	16	16	15	13	12	11
55	17	16	15	15	13	12	10
50	16	15	14	14	12	11	10
45	16	15	14	13	12	11	9
40	15	14	13	13	11	10	9
35	15	13	13	12	11	9	8
30	14	13	12	12	10	9	8
25	14	12	12	11	10	8	7
20	13	11	11	10	9	7	7
15	12	11	10	10	8	6	6
10	11	9	9	8	7	5	5
5	9	8	8	7	6	4	3

ARM CURL Test (Women)

Percentile Rank	65-69	65-69	70-74	70-74	80-84	65-69	70-74
95	24	22	22	21	20	18	17
90	22	21	20	20	18	17	16
85	21	20	19	19	17	16	15
80	20	19	18	18	16	15	14
75	19	18	17	17	16	15	13
70	18	17	17	16	15	14	13
65	18	17	16	16	15	14	12
60	17	16	16	15	14	13	12
55	17	16	15	15	14	13	11
50	16	15	14	14	13	12	11
45	16	15	14	13	12	12	10
40	15	14	13	13	12	11	10
35	14	14	13	12	11	11	9
30	14	13	12	12	11	10	9
25	13	12	12	11	10	10	8
20	12	12	11	10	10	9	8
15	11	11	10	9	9	8	7
10	10	10	9	8	8	7	6
5	9	8	8	7	6	6	5

ARM CURL Test (Men)

Percentile Rank	60-64	65-69	80-84	75-79	60-64	85-89	90-94
95	27	27	26	24	23	21	18
90	25	25	24	22	22	19	16
85	24	24	23	21	20	18	16
80	23	23	22	20	20	17	15
75	22	21	21	19	19	17	14
70	21	21	20	19	18	16	14
65	21	20	19	18	18	15	13
60	20	20	19	17	17	15	13
55	20	19	18	17	17	14	12
50	19	18	17	16	16	14	12
45	18	18	17	16	15	13	12
40	18	17	16	15	15	13	11
35	17	16	15	14	14	12	11
30	17	16	15	14	14	11	10
25	16	15	14	13	13	11	10
20	15	14	13	12	12	10	9
15	14	13	12	11	12	9	8
10	13	12	11	10	10	8	8
5	11	10	9	9	9	7	6

6-MINUTE WALK Test (Women)

Percentile Rank	60-64	65-69	70-74	75-79	80-84	85-89	90-94
95	741	734	709	696	654	638	564
90	711	697	673	655	612	591	518
85	690	673	650	628	584	560	488
80	674	653	630	605	560	534	463
75	659	636	614	585	540	512	441
70	647	621	599	568	523	493	423
65	636	607	586	553	508	476	406
60	624	593	572	538	491	458	388
55	614	581	561	524	477	443	373
50	603	568	548	509	462	426	357
45	592	555	535	494	447	409	341
40	582	543	524	480	433	394	326
35	570	529	510	465	416	376	308
30	559	515	497	450	401	359	291
25	547	500	482	433	384	340	273
20	532	483	466	413	364	318	251
15	516	463	446	390	340	292	226
10	495	439	423	363	312	261	196
5	465	402	387	322	270	214	150

6-MINUTE WALK Test (Men)

Percentile Rank	60-64	65-69	70-74	75-79	80-84	85-89	90-94
95	825	800	779	762	721	710	646
90	792	763	743	716	678	659	592
85	770	738	718	686	649	625	557
80	751	718	698	661	625	596	527
75	736	700	680	639	604	572	502
70	722	685	665	621	586	551	480
65	710	671	652	604	571	532	461
60	697	657	638	586	554	512	440
55	686	644	625	571	540	495	422
50	674	631	612	555	524	477	403
45	662	618	599	539	508	459	384
40	651	605	586	524	494	442	366
35	638	591	572	506	477	422	345
30	626	577	559	489	462	403	326
25	612	562	544	471	444	382	304
20	597	544	526	449	423	358	279
15	578	524	506	424	399	329	249
10	556	499	481	394	370	295	214
5	523	462	445	348	327	244	160

2-MINUTE STEP Test (Women)

Percentile Rank	60-64	65-69	70-74	75-79	80-84	85-89	90-94
95	130	133	125	123	113	106	92
90	122	123	116	115	104	98	85
85	116	117	110	109	99	93	80
80	111	112	105	104	94	88	76
75	107	107	101	100	90	85	72
70	103	104	97	96	87	81	69
65	100	100	94	93	84	79	66
60	97	96	90	90	81	76	63
55	94	93	87	87	78	73	61
50	91	90	84	84	75	70	58
45	88	87	81	81	72	67	55
40	85	84	78	78	69	64	53
35	82	80	74	75	66	61	50
30	79	76	71	72	63	59	47
25	75	73	67	68	60	55	44
20	71	68	63	64	56	52	40
15	66	63	58	59	51	47	36
10	60	57	52	53	46	42	31
5	52	47	43	45	37	39	24

2-MINUTE STEP Test (Men)

Percentile Rank	60-64	65-69	70-74	75-79	80-84	85-89	90-94
95	135	139	133	135	126	114	112
90	128	130	124	126	118	106	102
85	123	125	119	119	112	100	96
80	119	120	114	114	107	95	91
75	115	116	110	109	103	91	86
70	112	113	107	105	99	87	83
65	109	110	104	102	96	84	79
60	106	107	101	98	93	81	76
55	104	104	98	95	90	78	72
50	101	101	95	91	87	75	69
45	98	98	92	87	84	72	66
40	96	95	89	84	81	69	62
35	93	92	86	80	78	66	59
30	90	89	83	77	75	63	55
25	87	86	80	73	71	59	52
20	83	82	76	68	67	55	47
15	79	77	71	63	62	50	42
10	74	72	66	56	56	44	36
5	67	67	67	47	48	36	26

CHAIR SIT-AND-REACH Test (Women)

Percentile Rank	60-64	65-69	70-74	75-79	80-84	85-89	90-94
95	8.7	7.9	7.5	7.4	6.6	6.0	4.9
90	7.2	6.6	6.1	6.1	5.2	4.6	3.4
85	6.3	5.7	5.2	5.2	4.3	3.7	2.5
80	5.5	5.0	4.5	4.4	3.6	3.0	1.7
75	4.8	4.4	3.9	3.7	3.0	2.4	1.0
70	4.2	3.9	3.3	3.2	2.4	1.8	0.4
65	3.7	3.4	2.8	2.7	1.9	1.3	-0.1
60	3.1	2.9	2.3	2.1	1.4	0.8	-0.7
55	2.6	2.5	1.9	1.7	1.0	0.4	-1.2
50	2.1	2.0	1.4	1.2	0.5	-0.1	-1.7
45	1.6	1.5	0.9	0.7	0.0	-0.6	-2.2
40	1.1	1.1	0.5	0.2	-0.4	-1.0	-2.7
35	0.5	0.6	0.0	-0.3	-0.9	-1.5	-3.3
30	0.0	0.1	-0.5	-0.8	-1.4	-2.0	-3.8
25	-0.6	-0.4	-1.1	-1.3	-2.0	-2.6	-4.4
20	-1.3	-1.0	-1.7	-2.0	-2.6	-3.2	-5.1
15	-2.1	-1.7	-2.4	-2.8	-3.3	-3.9	-5.9
10	-3.0	-2.6	-3.3	-3.7	-4.2	-4.8	-6.8
5	-4.0	-3.9	-4.7	-5.0	-5.0	-6.3	-7.9

CHAIR SIT-AND-REACH Test (Men)

Percentile Rank :	60-64	65-69	70-74	75-79	80-84	85-89	90-94
95	8.5	7.5	7.5	6.6	6.2	4.5	3.5
90	6.7	5.9	5.8	4.9	4.4	3.0	1.9
85	5.6	4.8	4.7	3.8	3.2	2.0	0.9
80	4.6	3.9	3.8	2.8	2.2	1.1	0.0
75	3.8	3.1	3.0	2.0	1.4	0.4	-0.7
70	3.1	2.4	2.4	1.3	0.6	-0.2	-1.4
65	2.5	1.8	1.8	0.7	0.0	-0.8	-1.9
60	1.8	1.1	1.1	0.1	-0.8	-1.3	-2.5
55	1.2	0.6	0.6	-0.5	-1.4	-1.9	-3.0
50	0.6	0.0	0.0	-1.1	-2.0	-2.4	-3.6
45	0.0	-0.6	-0.6	-1.7	-2.6	-2.9	-4.2
40	-0.6	-1.1	-1.2	-2.3	-3.2	-3.5	-4.7
35	-1.3	-1.8	-1.8	-2.9	-4.0	-4.0	-5.3
30	-1.9	-2.4	-2.4	-3.5	-4.6	-4.6	-5.8
25	-2.6	-3.1	-3.1	-4.2	-5.3	-5.2	-6.5
20	-3.4	-3.9	-3.9	-5.0	-6.2	-5.9	-7.2
15	-4.4	-4.8	-4.8	-6.0	-7.2	-6.8	-8.1
10	-5.5	-5.9	-5.9	-7.1	-8.4	-7.8	-9.1
5	-7.3	-7.5	-7.6	-8.8	-10.2	-9.3	-10.7

BACK SCRATCH Test (Women)

Percentile Rank	60-64	65-69	70-74	75-79	80-84	85-89	90-94
95	5.0	4.9	4.5	4.5	4.3	3.5	3.9
90	3.8	3.5	3.2	3.1	2.8	1.9	2.2
85	2.9	2.6	2.3	2.2	1.8	0.8	0.9
80	2.2	1.9	1.5	1.3	0.9	-0.1	-0.1
75	1.6	1.3	0.8	0.6	0.2	-0.9	-1.0
70	1.1	0.7	0.3	0.0	-0.4	-1.6	-1.8
65	0.7	0.2	-0.2	-0.5	-1.0	-2.1	-2.5
60	0.2	-0.3	-0.8	-1.1	-1.6	-2.8	-3.2
55	-0.2	-0.7	-1.2	-1.6	-2.1	-3.3	-3.8
50	-0.7	-1.2	-1.7	-2.1	-2.6	-3.9	-4.5
45	-1.2	-1.7	-2.2	-2.6	-3.1	-4.5	-5.2
40	-1.6	-2.1	-2.6	-3.1	-3.7	-5.0	-5.8
35	-2.1	-2.6	-3.2	-3.7	-4.2	-5.7	-6.5
30	-2.5	-3.1	-3.7	-4.2	-4.8	-6.2	-7.2
25	-3.0	-3.7	-4.2	-4.8	-5.4	-6.9	-8.0
20	-3.6	-4.3	-4.9	-5.5	-6.1	-7.7	-8.9
15	-4.3	-5.0	-5.7	-6.4	-7.0	-8.6	-9.9
10	-5.2	-5.9	-6.6	-7.3	-8.0	-9.7	-11.2
5	-6.4	-7.3	-7.9	-8.8	-9.5	-11.3	-13.0

BACK SCRATCH Test (Men)

Percentile Rank	60-64	65-69	70-74	75-79	80-84	85-89	90-94
95	4.5	3.9	3.5	2.8	3.2	1.7	.7
90	2.7	2.2	1.8	0.9	1.2	-0.1	-1.1
85	1.6	1.0	0.6	-0.3	-0.1	-1.2	-2.2
80	0.6	0.0	-0.4	-1.3	-1.2	-2.2	-3.2
75	-0.2	-0.8	-1.2	-2.2	-2.1	-3.0	-4.0
70	-0.9	-1.6	-2.0	-2.9	-2.9	-3.7	-4.7
65	-1.5	-2.2	-2.6	-3.6	-3.6	-4.3	-5.3
60	-2.2	-2.9	-3.3	-4.3	-4.3	-5.0	-6.0
55	-2.8	-3.5	-3.9	-4.9	-5.0	-5.6	-6.6
50	-3.4	-4.1	-4.5	-5.6	-5.7	-6.2	-7.2
45	-4.0	-4.7	-5.1	-6.3	-6.4	-6.8	-7.8
40	-4.6	-5.3	-5.7	-6.9	-7.1	-7.4	-8.4
35	-5.3	-6.0	-6.4	-7.6	-7.8	-8.1	-9.1
30	-5.9	-6.6	-7.0	-8.3	-8.5	-8.7	-9.7
25	-6.6	-7.4	-7.8	-9.0	-9.3	-9.4	-10.4
20	-7.4	-8.2	-8.6	-9.9	-10.2	-10.2	-11.2
15	-8.4	-9.2	-9.6	-10.9	-11.3	-11.2	-12.2
10	-9.5	-10.4	-10.8	-12.1	-12.6	-12.3	-13.3
5	-11.3	-12.1	-12.5	-14.0	-14.6	-14.1	-15.1

8-FOOT UP-AND-GO Test (Women)

Percentile Rank	60-64	65-69	70-74	75-79	80-84	85-89	90-94
95	3.2	3.6	3.8	4.0	4.0	4.5	5.0
90	3.7	4.1	4.0	4.3	4.4	4.7	5.3
85	4.0	4.4	4.3	4.6	4.9	5.3	6.1
80	4.2	4.6	4.7	5.0	5.4	5.8	6.7
75	4.4	4.8	4.9	5.2	5.7	6.2	7.3
70	4.6	5.0	5.2	5.5	6.1	6.6	7.7
65	4.7	5.1	5.4	5.7	6.3	6.9	8.2
60	4.9	5.3	5.6	5.9	6.7	7.3	8.6
55	5.0	5.4	5.8	6.1	6.9	7.6	9.0
50	5.2	5.6	6.0	6.3	7.2	7.9	9.4
45	5.4	5.8	6.2	6.5	7.5	8.2	9.8
40	5.5	5.9	6.4	6.7	7.8	8.5	10.2
35	5.7	6.1	6.6	6.9	8.1	8.9	10.6
30	5.8	6.2	6.8	7.1	8.3	9.2	11.1
25	6.0	6.4	7.1	7.4	8.7	9.6	11.5
20	6.2	6.6	7.3	7.6	9.0	10.0	12.1
15	6.4	6.8	7.7	8.0	9.5	10.5	12.7
10	6.7	7.1	8.0	8.3	10.0	11.1	13.5
5	7.2	7.6	8.6	8.9	10.8	12.0	14.6

8-FOOT UP-AND-GO Test (Men)

Percentile Rank	60-64	65-69	70-74	75-79	80-84	85-89	90-94
95	3.0	3.1	3.2	3.3	4.0	4.0	4.3
90	3.0	3.6	3.6	3.5	4.1	4.3	4.5
85	3.3	3.9	3.9	3.9	4.5	4.5	5.1
80	3.6	4.1	4.2	4.3	4.9	5.0	5.7
75	3.8	4.3	4.4	4.6	5.2	5.5	6.2
70	4.0	4.5	4.6	4.9	5.5	5.8	6.6
65	4.2	4.6	4.8	5.2	5.7	6.2	7.0
60	4.4	4.8	5.0	5.4	6.0	6.5	7.4
55	4.5	4.9	5.1	5.7	6.2	6.9	7.7
50	4.7	5.1	5.3	5.9	6.4	7.2	8.1
45	4.9	5.3	5.5	6.1	6.6	7.5	8.5
40	5.0	5.4	5.6	6.4	6.9	7.9	8.8
35	5.2	5.6	5.8	6.6	7.1	8.2	9.2
30	5.4	5.7	6.0	6.9	7.3	8.6	9.6
25	5.6	5.9	6.2	7.2	7.6	8.9	10.0
20	5.8	6.1	6.4	7.5	7.9	9.4	10.5
15	6.1	6.3	6.7	7.9	8.3	9.9	11.1
10	6.4	6.6	7.0	8.3	8.7	10.5	11.8
5	6.8	7.1	7.4	9.0	9.4	11.5	12.9

Personal Goals and Activity Plans

Step 1. Describe at least one **long-term goal** you would like to achieve that could result from increased exercise. Goals can be specific, such as improving SFT walking scores from 500 to 600 yards in 6 minutes, or losing 10 pounds before a granddaughter's wedding next summer. Goals can also be more general, such as increasing participation in moderate physical activity to the recommended 30 to 40 minutes per day on most days of the week.

Long-term goal or major exercise objective: _____

Step 2. Describe realistic **short-term goals**, expressed as a 2-week activity plan that will move you toward your goals. Be specific relative to your planned activities and proposed schedule.

Activity

Days/Times

_____	_____
_____	_____
_____	_____
_____	_____

Step 3. Describe any potential obstacles that might keep you from following your plans and possible strategies for overcoming them.

Obstacle

Strategy for Overcoming

_____	_____
_____	_____
_____	_____
_____	_____
_____	_____

Step 4. Formalize your commitment to this plan by signing on the line below. Discuss your plans with a friend, a family member, or your exercise leader. Ask him or her to sign off as a witness and be willing to discuss your progress at least once a week.

Your signature: _____

Date: _____

Witness signature: _____ Date: _____

Measurement Conversions

General Conversions

To convert measurements from English to metric, use these conversion charts. Multiply the English measurement by the appropriate conversion factor.

Weight

To convert	Multiply by
Pounds to kilograms	0.454

Length

To convert	Multiply by
Inches to centimeters	2.54
Feet to meters	0.305
Yards to meters	0.9144
Miles to kilometers	1.609

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1 kg = 2.205 pounds

1 pound = 16 ounces = 453.6 g = 0.454 kg

1 inch = 0.0254 m = 2.54 cm = 25.4 mm

1 foot = 12 inches = 0.3048 m = 30.48 cm = 304.8 mm

1 yard = 3 feet = 0.9144 m

1 mile = 5,280 feet = 1,760 yards = 1,609.35 m = 1.61 km

1 cm = 0.3937 inches

1 m = 100 cm = 1,000 mm = 39.37 inches = 3.28 feet = 1.09 yards

1 km = 0.62 miles

Conversions Specific to Each of the SFT Protocols

30-Second Chair Stand Test

Chair seat height of 17 inches = 43 cm

30-Second Arm Curl Test

5-pound dumbbell for women = 2.3 kg

8-pound dumbbell for men = 3.6 kg

Height and Weight

For 60-inch tape measure, use 150 cm tape measure.

For ease of measurement, position end 50 cm up from the floor.

2-Minute Step Test

String or cord about 30 inches long = 76 cm

Chair Sit-and-Reach Test

Chair seat height of 17 inches = 43 cm

18-inch ruler = 45 cm

Back Scratch Test

18-inch ruler = 45 cm

8-Foot Up-and-Go Test

Chair seat height of 17 inches = 43 cm

8 feet = 2.4 m

6-Minute Walk Test

50-yard course = 45.7 m

Note: For ease of use with metric system, the course can be set up in meters (5 m x 20 m), with the final score converted to yards for comparison with U.S. norms.

References

- Administration on Aging. (2010). *A profile of older Americans: 2010*. A report of the Administration on Aging, U.S. Department of Health and Human Services. Washington, DC.
- Agarwal, S., & Kiely, P.D.W. (2006). Two simple, reliable and valid tests of proximal muscle function, and their application to the management of idiopathic inflammatory myositis. *Rheumatology*, 45, 874-879.
- Alexander, N.B., Schultz, A.B., & Warwick, D.N. (1991). Rising from a chair: Effects of age and functional ability on performance biomechanics. *Journal of Gerontology: Medical Sciences*, 46, M91-M98.
- American Academy of Orthopaedic Surgeons. (1966). *Joint motion: Method of measuring and recording*. Edinburgh: Livingstone.
- American College of Sports Medicine. (2009). ACSM position stand: Exercise and physical activity for older adults. *Medicine and Science in Sports and Exercise*, 41(7), 1510-1530.
- American College of Sports Medicine. (2010). *ACSM's Guidelines for exercise testing and prescription* (8th ed.). Philadelphia: Lippincott, Williams & Wilkins.
- American Psychological Association. (1985). *Standards for educational and psychological tests*. Washington, DC: American Psychological Association.
- Arnold, A.M., Newman, A.B., Cushman, M., Ding, J., & Kritchevsky, S. (2010). Body weight dynamics and their association with physical function and mortality in older adults: The cardiovascular health study. *Journal of Gerontology*, 65A, 63-70.
- Baruth, M., Wilcox, S., Wegley, S., Buchner, D.M., Ory, M.G., Phillips, A., . . . Bazzare, T.L. (2010). Changes in physical functioning in the Active Living Everyday Program of the Active for Life initiative. *International Journal of Behavioral Medicine*. [Online]. doi:10.1007/s12529-010-9108-7.
- Baumgartner, T.A., Jackson, A.S., Mahar, M.T., & Rowe, D.A. (2007). *Measurement for evaluation in physical education and exercise science* (8th ed.). Boston: McGraw-Hill.
- Beck, A.M., Damkjaer, K., & Beyer, N. (2008). Multifaceted nutritional intervention among nursing-home residents has a positive influence on nutrition and function. *Nutrition*, 24(11-12), 1073-1080.
- Bohannon, R.W. (2002). Quantitative testing of muscle strength: Issues and practical options for the geriatric population. *Topics in Geriatric Rehabilitation*, 18(2), 1-17.
- Booth, F.W., Gordon, S.E., Carlson, C.J., & Hamilton, M.T. (2000). Waging war on modern chronic diseases: Primary prevention through exercise biology. *Journal of Applied Physiology*, 88, 774-787.
- Borg, G. (1998). *Borg's perceived exertion and pain scales*. Champaign, IL: Human Kinetics.

- Bouchard, D.R., Beliaeff, S., Dionne, I.J., & Brochu, M. (2007). Fat mass but not fat-free mass is related to physical capacity in well-functioning older individuals: Nutrition as a determinant of successful aging (NuAge)—the Quebec Longitudinal Study. *Journal of Gerontology*, 62, 1382-1388.
- Bouchard, D.R., & Janssen, I. (2010). Dynapenic-obesity and physical function in older adults. *Journal of Gerontology*, 65A, 71-77.
- Bravo, G., Gauthier, P., Roy, P., Tessier, D., Gaulin, P., Dubois, M., & Peloquin, L. (1994). The functional fitness assessment battery: Reliability and validity data for elderly women. *Journal of Aging and Physical Activity*, 2, 67-79.
- Brouha, L. (1943). A step test: A simple method of measuring physical fitness for muscular work in young men. *Research Quarterly*, 14, 31-36.
- Brown, M., & Rose, D.J. (2005). Flexibility training. In C.J. Jones & D.J. Rose (Eds.), *Physical activity instruction for older adults* (pp. 155-174). Champaign, IL: Human Kinetics.
- Brown, M., Sinacore, D.R., Ehsani, A.A., Binder, E.F., Holloszy, J.O., & Kohrt, W.M. (2000). Low-intensity exercise as a modifier of physical frailty in older adults. *Archives of Physical Medicine and Rehabilitation*, 81, 960-965.
- Burger, H., & Marincek, C. (2001). Functional testing of elderly subjects after lower limb amputation. *Prosthetics and Orthotics International*, 25(2), 102-107.
- Canadian Society for Exercise Physiology. (2011). Canadian Physical Activity Guidelines. www.csep.ca/guidelines.
- Cailliet, R. (1988). *Low back pain syndrome*. Philadelphia: Davis.
- Cavani, V., Mier, C.M., Musto, A.A., & Tummers, N. (2002). Effects of a 6-week resistance-training program on functional fitness of older adults. *Journal of Aging and Physical Activity*, 10(4), 443-452.
- Centers for Disease Control and Prevention and the Merck Company Foundation. (2007). *The state of aging and health in America 2007*. Whitehouse Station, NJ: The Merck Company Foundation.
- Centers for Disease Control and Prevention. (2011). Costs of falls among older adults. www.cdc.gov/HomeandRecreationalSafety/Falls/fallcost.html.
- Chakravarty, K., & Webley, M. (1993). Shoulder joint movement and its relationship to disability in the elderly. *Journal of Rheumatology*, 20, 1359-1361.
- Chandler, J.M., & Hadley, E.C. (1996). Exercise to improve physiologic and functional performance in old age. *Clinics in Geriatric Medicine*, 12, 761-784.
- Cohen, J.J., Sveen, J.D., Walker, J.M., & Brummel-Smith, K. (1987). Establishing criteria for community ambulation. *Topics in Geriatric Rehabilitation*, 3(1), 71-77.
- Cotten, D.J. (1971). A modified step test for group cardiovascular testing. *Research Quarterly*, 42, 91-95.
- Courneya, K.S., Booth, C.M., Gill, S., O'Brien, P., Vardy, J., Friedenreich, C.M., . . . Meyer, R.M. (2008). The colon health and life-long exercise change trial: A randomized trial of the National Cancer Institute of Canada Clinical Trials Group.

- Cress, M.E., Petrella, J.K., Moore, T.L., & Schenkman, M.L. (2005). Continuous-scale physical functional performance test: Validity, reliability, and sensitivity of data for the short version. *Physical Therapy*, 85(4), 323-335.
- Csuka, M., & McCarty, D.J. (1985). Simple method for measurement of lower extremity muscle strength. *American Journal of Medicine*, 78, 77-81.
- Cureton, K.J., & Warren, G.L. (1990). Criterion-referenced standards for youth health-related fitness tests: A tutorial. *Research Quarterly for Exercise and Sport*, 61(1), 7-19.
- Damush, T.M., Perkins, A., & Miller, K. (2005). The implementation of an oncologist referred, exercise self-management program for older breast cancer survivors. *Psycho-Oncology*, 15(10), 884-890.
- Demark-Wahnefried, W., Morey, M.C., Clipp, E.C., Pieper, C.F., Snyder, D.C., Sloane, R., & Cohen, H.J. (2003). Leading the way in exercise and diet (Project LEAD): Intervening to improve function among older breast and prostate cancer survivors. *Controlled Clinical Trials*, 24, 206-223.
- DiBrezzo, R., Shadden, B.B., Raybon, B.H., & Powers, M. (2005). Exercise intervention designed to improve strength and dynamic balance among community-dwelling older adults. *Journal of Aging and Physical Activity*, 13(2), 198-209.
- Dobek, J.C., White, K.N., & Gunter, K.B. (2007). The effect of a novel ADL-based training program on performance of activities of daily living and physical fitness. *Journal of Aging and Physical Activity*, 15(1), 13-25.
- Doherty, T.J. (2003). Invited review: Aging and sarcopenia. *Journal of Applied Physiology*, 95, 1717-1727.
- Dugas, E. W. (1996). The development and validation of a 2-minute step test to estimate aerobic endurance in older adults. Unpublished Master's Thesis, California State University, Fullerton, Fullerton, CA.
- Federal Interagency Forum on Aging-Related Statistics. (2010). Older Americans 2010: Key indicators of well-being. Federal Interagency Forum on Aging-Related Statistics, Washington D.C.: U.S. Government Printing Office.
- Fenstermaker, K.L., Plowman, S.A., & Looney, M.A. (1992). Validation of the Rockport Fitness Walking Test in females 65 years and older. *Research Quarterly for Exercise and Sport*, 63, 322-327.
- Fiatarone Singh, M.A. (2002). Exercise to prevent and treat functional disability. *Clinics in Geriatric Medicine*, 18, 431-462.
- Fleg, J.L., Morrell, C.H., Bos, A.G., Brant, L.J., Talbot, L.A., Wright, J.G., & Lakatta, E.G. (2005). Accelerated longitudinal decline of aerobic capacity in healthy older adults. *Circulation*, 112, 674-682. doi:10.1161/CIRCULATIONAHA.105.545459
- Giuliani, C.A., Gruber-Baldini, A.L., Park, N.S., Schrodt, L.A., Rokoske, F., Sloane, P.D., & Zimmerman, S. (2008). Physical performance characteristics of assisted living residents and risk for adverse health outcomes. *The Gerontologist*, 48, 203-

- Golding, L., Myers, C., & Sinning, W. (1989). *Y's way to physical fitness* (3rd ed.). Champaign, IL: Human Kinetics.
- Gross, J., Fetto, J., & Rosen, E. (1996). *Musculoskeletal examination*. Cambridge: Blackwell Science.
- Guralnik, J.M., Ferrucci, L., Pieper, C.F., Leveille, S.G., Markides, K.S., Ostir, G.V., . . . Wallace, R.B. (2000). Lower extremity function and subsequent disability: Consistency across studies, predictive models, and value of gait speed alone compared with the Short Physical Performance Battery. *Journal of Gerontology*, 55A, M221-M231.
- Guralnik, J.M., Simonsick, E.M., Ferrucci, L., Glynn, R.J., Berkman, L.F., Blazer, D.G., Scherr, P.A., & Wallace, R.B. (1994). A short physical performance battery assessing lower extremity function: Association with self-reported disability and prediction of mortality and nursing home admission. *Journal of Gerontology*, 49, M85-M94.
- Hernandez, J.P., & Franke, W.D. (2005). Effects of a 6-mo endurance-training program on venous compliance and maximal lower body negative pressure in older men and women. *Journal of Applied Physiology*, 99, 1070-1077.
- Holland, G.J., Tanaka, K., Shigematsu, R., & Nakagaichi, M. (2002). Flexibility and physical functions of older adults: A review. *Journal of Aging and Physical Activity*, 10, 169-206.
- Hooker, S.P., & Cirill, L.A. (2006). Evaluation of community coalitions ability to create safe, effective exercise classes for older adults. *Evaluation and Program Planning*, 29, 242-250.
- Hruda, K.V., Hicks, A.L., & McCartney, N. (2003). Training for muscle power in older adults: Effects on functional abilities. *Canadian Journal of Applied Physiology*, 28(2), 178-189.
- Hughes, V.A., Frontera, W.R., Wood, M., Evans, W.J., Dallal, G.E., Roubenoff, R., & Fiatarone Singh, M.A. (2001). Longitudinal muscle strength changes in older adults: Influence of muscle mass, physical activity, and health. *Journal of Gerontology*, 56A(5), B209-B217.
- Jackson, A.S., Sui, X., Hebert, J.R., Church, T.S., & Blair, S.N. (2009). Role of lifestyle and aging on the longitudinal change in cardiorespiratory fitness. *Archives of Internal Medicine*, 169(19), 1781-1787.
- Jackson, A.W., & Baker, A.A. (1986). The relationship of the sit and reach test to criterion measures of hamstring and back flexibility in young females. *Research Quarterly for Exercise and Sport*, 57, 183-186.
- Jackson, A.W., & Langford, N.J. (1989). The criterion-related validity of the sit-and-reach test: Replication and extension of previous findings. *Research Quarterly for Exercise and Sport*, 60, 384-387.
- James, T.W. (1999). *The 30-second arm curl test as an indicator of upper body*

- strength in older adults. Unpublished master's thesis, California State University, Fullerton.
- Johnston, J. (1999). *Validation of a 2-minute step-in-place test relative to treadmill performance in older adults*. Unpublished master's thesis, California State University, Fullerton.
- Jones, C.J., Rikli, R.E., & Beam, W.C. (1999). A 30-s chair-stand test as a measure of lower body strength in community-residing older adults. *Research Quarterly for Exercise and Sport*, 70, 113-119.
- Jones, C.J., Rikli, R.E., Max, J., & Noffal, G. (1998). The reliability and validity of a chair sit-and-reach test as a measure of hamstring flexibility in older adults. *Research Quarterly for Exercise and Sport*, 69, 338-343.
- Jones, C.J., & Rose, D.J. (Eds.). (2005). *Physical activity instruction of older adults*. Champaign, IL: Human Kinetics.
- Jones, C.J., Rutledge, D.N., & Aquino, J. (2010). Predictors of physical performance and functional ability in people 50+ with and without fibromyalgia. *Journal of Aging and Physical Activity*, 18, 353-368.
- Judge, J.O. (2003). Balance training to maintain mobility and prevent disability. *American Journal of Preventive Medicine*, 25, 150-156.
- Kline, G.M., Porcari, J.P., Hintermeister, R., Freedson, P.S., Ward, A., McCarron, R.F., Ross, J., & Rippe, J.M. (1987). Estimation of $\dot{V}O_2$ max from a one-mile track walk, gender, age, and body weight. *Medicine and Science in Sports and Exercise*, 19, 253-259.
- Lambers, S., Van Laethem, C., Van Acker, K., & Calders, P. (2008). Influence of combined exercise training on indices of obesity, diabetes and cardiovascular risk in type 2 diabetes patients. *Clinical Rehabilitation*, 22(6), 483-492.
- Lerner-Frankiel, M.B., Vargas, S., Brown, M., Krusell, L., & Schoneberger, W. (1986). Functional community ambulation: What are your criteria? *Clinical Management in Physical Therapy*, 6(2), 12-15.
- Liu, J., Baiqing, L., & Shnider, R. (2010). Effects of tai chi training on improving physical function in patients with coronary heart diseases. *Journal of Exercise Science and Fitness*, 8(2), 78-84.
- Looney, M.A. (1989). Criterion-referenced measurement: Reliability. In M. J. Safrit & T. M. Woods (Eds.), *Measurement concepts in physical education and exercise science* (pp. 137-152). Champaign, IL: Human Kinetics.
- Losonczy, K.G., Harris, T.B., Cornoni-Huntley, J., Simonsick, E.M., Wallace, R.B., Cook, N.R., Ostfeld, A.M., & Blazer, D.G. (1995). Does weight loss from middle age to old age explain the inverse weight mortality relation in old age? *American Journal of Epidemiology*, 141, 312-321.
- Macaluso, A., & De Vito, G. (2004). Muscle strength, power and adaptations to resistance training in older people. *European Journal of Applied Physiology*, 91, 450-472.

- MacRae, P.G., Lacourse, M., & Moldavon, R. (1992). Physical performance measures that predict faller status in community-dwelling older adults. *Journal of Occupational and Sports Physical Therapy*, 16, 123-128.
- Magee, D.J. (1992). *Orthopedic physical assessment*. Philadelphia: Saunders.
- Mahar, M.R., & Rowe, D.A. (2008). Practical guidelines for valid and reliable youth fitness testing. *Measurement in Physical Education and Exercise Science*, 12(3), 126-145.
- McArdle, W.D., Katch, F.I., Pechar, G.S., Jacobson, L., & Ruck, S. (1972). Reliability and interrelationships between maximal oxygen intake, physical work capacity and step-test scores in college women. *Medicine and Science in Sports*, 4, 182-186.
- McMurdo, M.E., & Rennie, L. (1993). A controlled trial of exercise by residents of old people's homes. *Age and Aging*, 22, 11-15.
- Miotto, J.M., Chodzko-Zajko, W.J., Reich, J.L., & Supler, M.M. (1999). Reliability and validity of the Fullerton Functional Fitness Test: An independent replication study. *Journal of Aging and Physical Activity*, 7, 339-353.
- Morey, M.C., Pieper, C.F., & Cornoni-Huntley, J. (1998). Physical fitness and functional limitations in community-dwelling older adults. *Medicine and Science in Sports and Exercise*, 30, 715-723.
- Morrow, J.R., Jr., Jackson, A.W., Disch, J.G., & Mood, D.P. (2011). *Measurement and evaluation in human performance* (4th ed.). Champaign, IL: Human Kinetics.
- Nagi, S.Z. (1965). Some conceptual issues in disability and rehabilitation. In M.B. Sussman (Ed.), *Sociology and rehabilitation* (pp. 100-113). Washington, DC: American Sociological Association.
- Nagi, S.Z. (1991). Disability concepts revisited: Implication for prevention. In A.M. Pope & A.R. Tarlov (Eds.), *Disability in America: Toward a national agenda for prevention* (pp. 309-327). Washington, DC: National Academy Press.
- National Center for Health Statistics. (1991). *National Health Interview Survey 1991 (10): Vital Health Statistics*. Washington, DC: National Center for Health Statistics.
- National Institutes of Health. (2012). Research for a new age. NIH Publication No. 93-1129.
- www.healthandage.com/html/min/nih/content/booklets/research_new_age/page3.htm.
- Nelson, M.E., Rejeski, W.J., Blair, S.N., Duncan, P.W., Judge, J.O., King, . . . Castaneda-Sceppa, C. (2007). Physical activity and public health in older adults: Recommendation from the American College of Sports Medicine and the American Heart Association. *Medicine and Science in Sports and Exercise*, 39(8), 1435-1445.
- Osness, W.H., Adrian, M., Clark, B., Hoeger, W., Rabb, D., & Wiswell, R. (1996). *Functional fitness assessment for adults over 60 years*. Dubuque, IA: Kendall/Hunt.
- Paterson, D.H., Jones, G.R., & Rice, C.L. (2007). Ageing and physical activity: Evidence to develop exercise recommendations for older adults. *Applied Physiology, Nutrition, and Metabolism*, 32(Suppl. 2E), S69-S108.
- Paterson, D.H., & Warburton, D.E.R. (2010). Physical activity and functional

- limitations in older adults: A systematic review related to Canada's Physical Activity Guidelines. *International Journal of Behavioral Nutrition and Physical Activity*, 7, 1-22.
- Patterson, P., Wiksten, D.L., Ray, L., Flanders, C., & Sanphy, D. (1996). The validity and reliability of the back saver sit-and-reach test in middle school girls and boys. *Research Quarterly for Exercise and Sport*, 64, 448-451.
- Pearson, J.A., Burkhardt, E., Pifalo, W.B., Palaggo-Toy, T., & Krohn, K. (2005). A lifestyle modification intervention for the treatment of osteoporosis. *American Journal of Health Promotion*, 20(1), 28-33.
- Peterson, M.J., Crowley, G.M., Sullivan, R.J., Morey, M.C. (2004). Physical function in sedentary and exercising older veterans as compared to national norms. *Journal of Rehabilitation Research and Development*, 41(5), 653-658.
- Pettee Gabriel, K.K., Rankin, R.L., Lee, C., Charlton, M.E., Swan, P.D., & Ainsworth, B.E. (2010). Test-retest reliability and validity of the 400-meter walk test in healthy, middle-aged women. *Journal of Physical Activity and Health*, 7, 649-657.
- Physical Activity Guidelines Advisory Committee. (2008). Physical Activity Guidelines Advisory Committee report. U.S. Department of Health and Human Services. www.health.gov/PAGuidelines/committeereport.aspx.
- Podsiadlo, D., & Richardson, S. (1991). The timed "up and go": A test of basic functional mobility for frail elderly persons. *Journal of the American Geriatric Society*, 39, 142-148.
- Rikli, R.E., & Jones, C.J. (1998). The reliability and validity of a 6-minute walk test as a measure of physical endurance in older adults. *Journal of Aging and Physical Activity*, 6, 363-375.
- Rikli, R.E., & Jones, C.J. (1999a). Development and validation of a functional fitness test for community-residing older adults. *Journal of Aging and Physical Activity*, 1, 129-161.
- Rikli, R.E., & Jones, C.J. (1999b). Functional fitness normative scores for community-residing adults, ages 60-94. *Journal of Aging and Physical Activity*, 1, 162-181.
- Rikli, R.E., & Jones, C.J. (2000). Physical activity level, fitness, and functional ability of community-residing older adults. *Medicine and Science in Sports and Exercise*, 28, S153.
- Rikli, R.E., & Jones, C.J. (2012). Development and validation of criterion-referenced clinically relevant fitness standards for maintaining physical independence in later years. *The Gerontologist*, 0,1-13 DOI:10.1093/geront/gns071.
- Rose, D. (2010). *FallProof!* (2nd ed.). Champaign, IL: Human Kinetics.
- Rose, D.J., Jones, C.J., & Lucchese, N. (2002). Predicting the probability of falls in community-residing older adults using the 8-foot up-and-go: A new measure of functional mobility. *Journal of Aging and Physical Activity*, 10, 466-475.
- Rosow, I., & Breslau, N. (1966). A Guttman Health Scale for the aged. *Journal of*

- Gerontology, 21, 556-559.
- Ross, R., & Janssen, I. (2007). Physical activity, fitness, and obesity. In C. Bouchard, S.N. Blair, & W. Haskell (Eds.), *Physical activity and health* (pp.174-191). Champaign, IL: Human Kinetics.
- Safrit, M.J., & Wood, T.M. (1995). *Introduction to measurement in physical education and exercise science*. St. Louis: Mosby-Year Book.
- Sakai, T., Tanaka, K., & Holland, G.J. (2002). Functional and locomotive characteristics of stroke survivors in Japanese community-based rehabilitation. *American Journal of Physical Medicine and Rehabilitation*, 81(9), 675-683.
- Santana-Sosa, E., Barriopedro, M.I., López-Mojares, L.M., Pérez, M., & Lucia, A. (2008). Exercise training is beneficial for Alzheimer's patients. *International Journal of Sports Medicine*, 29(10), 845-850.
- Schmidt, W.D., Biwer, C.J., & Kalscheuer, L.K. (2001). Effects of long versus short bout exercise on fitness and weight loss in overweight females. *Journal of the American College of Nutrition*, 20(5), 494-501.
- Seeman, T.E., Berkman, L.F., Charpentier, P.A., Blazer, D.G., Alpert, M.A., & Tinetti, M.E. (1995). Behavioral and psychosocial predictors of physical performance: MacArthur Studies of Successful Aging. *Journal of Gerontology*, 50, M177-M183.
- Shephard, R.J. (1997). *Aging, physical activity, and health*. Champaign, IL: Human Kinetics.
- Simonsick, E.M., Fan, E., & Fleg, J.L. (2006). Estimating cardiorespiratory fitness in well-functioning older adults: Treadmill validation of the long distance corridor walk. *Journal of the American Geriatrics Society*, 54, 127-132.
- Siu, A.L., Reuben, D.B., & Hays, R.D. (1990). Hierarchical measures of physical function in ambulatory geriatrics. *Journal of the American Geriatrics Society*, 38, 1113-1119.
- Sperling, L. (1980). Evaluation of upper extremity function in 70-year old men and women. *Scandinavian Journal of Rehabilitative Medicine*, 12, 139-144.
- Spiriduso, W.W., Francis, K.L., & MacRae, P.G. (2005). *Physical dimensions of aging*. (2nd ed.). Champaign, IL: Human Kinetics.
- Starkey, C., & Ryan, J.L. (1996). *Evaluation of orthopedic and athletic injuries*. Philadelphia: Davis.
- Sternfeld, B., Ngo, L., Satariano, W.A., & Tager, I.B. (2002). Association of body composition with physical performance and self-reported functional limitation in elderly men and women. *American Journal of Epidemiology*, 156, 110-121.
- Stump, T., Clark, D.O., Johnson, R.J., & Wolinsky, F.D. (1997). The structure of health status among Hispanic, African American, and white older adults. *The Journals of Gerontology*, 52B(Special Issue), 49-60.
- Takeshima, N., Rogers, N.L., Rogers, M.E., Islam, M.M., Koizumi, D., & Lee, S. (2007). Functional fitness gain varies in older adults depending on exercise mode. *Medicine and Science in Sports and Exercise*, 39(11), 2036-2043.

- Tinetti, M.E., Speechley, M., & Ginter, S.F. (1988). Risk factors for falls among elderly persons living in the community. *New England Journal of Medicine*, 319, 1701-1707.
- Tinetti, M.E., Williams, T.F., & Mayewski, R. (1986). Fall risk index for elderly patients based on number of chronic conditions. *American Journal of Medicine*, 80, 429-434.
- UK Department of Health, Physical Activity, Health Improvement and Protection. (2011). Start Active, Stay Active: A report on physical activity from the four home countries' chief medical officers. London.
www.nutrinfo.com/archivos/ebooks/act_fca_uk.pdf.
- U.S. Department of Health and Human Services. (2008). Physical Activity Guidelines for Americans. www.health.gov/paguidelines.
- U.S. Department of Health and Human Services. (2011). Healthy People 2020. Office of Disease Prevention and Health Promotion. Washington, DC.
www.healthypeople.gov/2020/topicsobjectives2020/objectiveslist.aspx?topicId=31.
- U.S. Department of Health and Human Services. (2006). Medicare Current Beneficiary Survey (MCBS), Centers for Medicare and Medicaid Services.
<http://www.cms.hhs.gov/MCBS>.
- U.S. Department of Housing and Urban Development. (1999). *Housing our elders*. Washington, DC: Office of Policy Development and Research.
- Vandervoort, A.A. (2002). Aging of the human neuromuscular system. *Muscle and Nerve*, 25, 17-25.
- Wang, A.W., Gilbey, H.J., & Ackland, T.R. (2002). Perioperative exercise programs improve early return of ambulatory function after total hip arthroplasty: A randomized, controlled trial. *American Journal of Physical Medicine and Rehabilitation*, 81(11), 801-806.
- Warburton, D.E.R., Gledhill, N., & Quinney, A. (2001). The effects of changes in musculoskeletal fitness and health. *Canadian Journal of Applied Physiology*, 26(2), 161-216.
- Warren, B.J., Dotson, R.G., Nieman, D.C., & Butterworth, D.E. (1993). Validation of a 1-mile walk test in elderly women. *Journal of Aging and Physical Activity*, 1, 13-21.
- Welk, G.J., & Meredith, M.D. (Eds.). (2008). *Fitnessgram/Activitygram reference guide*. Dallas, TX: Cooper Institute.
- Wiacek, M., & Hagner, W. (2008). The history and economic impact on the functional fitness of elderly in the south-eastern region of Poland: A comparison with US citizens. *Archives of Gerontology and Geriatrics*, 46(2), 221-226.
- Wilk, M., Kocur, P., Rozanska, A., Przywarska, I., Dylewicz, P., Owczarski, T. . . Borowicz-Bienkowska, S. (2005). Assessment of the selected physiological effects of Nordic walking performed as a part of a physical exercise program during the second phase of rehabilitation after a myocardial infarction. *Medical Rehabilitation*,

9(2), 20-25.

Winnick, J.P., & Short, F.X. (1999). *The Brockport Physical Fitness Test manual*. Champaign, IL: Human Kinetics.

Woodward, T.W., & Best, T.M. (2000). The painful shoulder: Part I. Clinical evaluation. *American Family Physician*, 61(10), 3079-3088.

World Health Organization. (2010). Global recommendations on physical activity for health. Geneva: World Health Organization.

http://whqlibdoc.who.int/publications/2010/9789241599979_eng.pdf.

Yamauchi, T., Islam, M.M., Koizumi, D., Rogers, M.E., Rogers, N.L., & Takeshima, N. (2005). Effect of home-based well-rounded exercise in community-dwelling older adults. *Journal of Sports Science and Medicine*, 4(4), 563-571.

Yan, T., Wilber, K.H., Aguirre, R., & Trejo, L. (2009). Do sedentary older adults benefit from community-based exercise? Results from the active start program. *The Gerontologist*, 49, 847-855.

About the Authors



Roberta E. Rikli, PhD, is professor of kinesiology and dean emeritus of

the college of health and human development at California State University in Fullerton and was cofounder of the LifeSpan Wellness Program at Fullerton. For the past two decades she has done extensive work in physical performance assessment with a particular focus on senior fitness. She has published numerous scientific papers on her work and has made over 100 presentations at conferences and workshops in the United States, Canada, Germany, France, Finland, Scotland, Brazil, China, Korea, and Japan.

Dr. Rikli has served on the editorial boards of three scientific journals and is a regular reviewer for several others. She has held leadership positions in professional organizations, including the International Society for Aging and Physical Activity; the American Alliance for Health, Physical Education, Recreation and Dance (AAHPERD); the American Kinesiology Association; and the National Academy of Kinesiology.

Dr. Rikli lives in Orange, California, and enjoys playing golf, hiking, and playing tennis.



C. Jessie Jones, PhD, is professor and chair of the department of health science at California State University in Fullerton. She is director of the Fibromyalgia and Pain Management Center at Fullerton and cofounder of the Center for Successful Aging.

Dr. Jones is internationally known for her research, program design,

curriculum development, and instructor training in the field of exercise science and aging. She has taught senior fitness classes and conducted training workshops for senior fitness instructors for over 25 years. Her work has been covered in numerous publications and presented at conferences worldwide. Her professional memberships include the Gerontological Society of America, the American Public Health Association, and the American Academy for Pain Management.

Dr. Jones lives in Fullerton, California, where she enjoys dancing, drumming, hiking, playing golf, and jogging with her dog.